

A Robust Statistic for Monetary Non-Neutrality in Menu Cost Economies

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[PRELIMINARY]

Abstract

The first two entries of the IRF of firm-level prices to firm-level cost shocks accurately approximate the generalized Phillips curve Jacobian in a wide class of pricing models. This result is exact in Calvo and in the Gertler-Leahy menu cost model, is numerically accurate for a broad set of CalvoPlus parameterizations and in models with micro real rigidities, shocks of different persistence, leptokurtic shocks, and measurement error. Our approach approximates the hazard rate of the Calvo model closest to the unknown true menu-cost Jacobian, as defined by Auclert et al. (2024). This virtual Calvo hazard may be arbitrarily different from the frequency of price changes computed in a dataset of firm-level prices, providing a diagnostic of the extent of selection effects.

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1. Introduction

A large research program in monetary economics aims to infer the extent of monetary non-neutrality from unconditional moments of the distribution of price changes. The best-known example is the frequency of price changes: in the textbook Calvo (1983) model it is a sufficient statistic for monetary non-neutrality. This program extends to the more complex case of state-dependent models: there, the frequency of price change is uninformative about the degree of non-neutrality (e.g., Caplin and Spulber 1987), but—under certain conditions—the ratio of kurtosis to frequency of price changes is a sufficient statistic (Alvarez, Le Bihan, and Lippi 2016).

A key challenge in this literature is model misspecification. These unconditional moments are sufficient statistics when the model is correctly specified, but small changes in the environment can break them. Both statistics fail in the presence of micro real rigidities, a key mechanism to reconcile the modest price rigidity in micro data with large monetary non-neutrality in aggregate data (Ball and Romer 1990). The frequency is uninformative in the Rotemberg (1982) and Gertler and Leahy (2008) models, even though both share the Calvo Phillips curve. The kurtosis ratio depends on the distributional assumptions of Alvarez, Le Bihan, and Lippi (2016) and Dotsey and Wolman (2020) show that relaxing them changes the answer.

This paper takes a different route. Instead of unconditional moments of the distribution of price changes, we focus on one conditional moment: the response of a firm’s price to an observed idiosyncratic shock to its cost, estimated over horizons by a local projection. Our main result is that two such local projection coefficients, at impact and one period ahead, form a *robust statistic* for the generalized Phillips curve, the first-order response of the path of inflation to the path of real marginal cost.¹ Assembling the generalized Phillips curve from these two coefficients as if the economy were Calvo recovers the true one exactly when prices are set as in Calvo (1983), as shown in Herreño, Pinardon-Touati, and Thie (2026), but also as in Rotemberg (1982) and as in Gertler and Leahy (2008), where no Calvo lottery exists. In the canonical menu cost model, the assembly is not exact but is numerically tight and extremely robust across a wide range of menu costs, free-adjustment probabilities, shock volatilities, real rigidities, and measurement error in prices.

¹Throughout this paper, Phillips curve refers to the marginal cost-based Phillips curve. Obtaining the output-based Phillips curve additionally requires a mapping from the output gap to real marginal costs.

These conditional moments are highly robust across models because they measure the object the Phillips curve is made of. The response of the price level to an aggregate nominal cost shock is an average of the responses of individual prices to that shock, and we show that it is also the average of firms' responses to their own idiosyncratic nominal cost shock. A local projection of price changes on an observed cost shifter therefore directly measures the correct object. By contrast, an unconditional moment of the distribution of price changes is an outcome: the average of what the pricing policy produces from whatever shocks the economy contains. Recovering a response from an outcome requires a model of the policy and of the shocks.² In addition, only two local projection coefficients suffice because the generalized Phillips curve of menu cost models is close to having the geometric structure of the Calvo one, as shown by Auclert et al. (2024). The two entries pin down an implied survival rate of price gaps, and the Calvo structure extrapolates it with a high degree of numerical accuracy.

The approach is simple to implement. It requires a panel of firm-level prices and an observed idiosyncratic cost shifter; in our analytical derivations, the shifter is a random walk. We assume the discount factor β is known. The first step is to estimate the local projection of cumulative price changes on the shifter at impact and one period ahead, which yields two coefficients, b_0 and b_1 . The second step is to compute $\phi^{\text{IRF}} = b_1/b_0 - 1$, the survival probability of prices in the Calvo economy that would generate these two coefficients, and the implied Phillips curve slope $\kappa^{\text{IRF}} = b_0(1 - \beta\phi^{\text{IRF}})/\phi^{\text{IRF}}$. The third step is to proceed as if the economy were Calvo: the New Keynesian Phillips curve $\pi_t = \beta\mathbb{E}_t\pi_{t+1} + \kappa^{\text{IRF}} mc_t^r$, iterated forward for any path of mc_t^r , is the estimate of the generalized Phillips curve. No step requires knowing the data-generating process, and the computational cost is negligible.

We start by providing an analytical characterization of our robust statistic across models. The environment is a continuum of firms facing isoelastic demand curves, producing with labor in potentially decreasing returns to scale technologies and facing a mixture of idiosyncratic shocks. Firms minimize the profit loss from mispricing plus the cost of adjusting prices, which is zero when they draw a free adjustment opportunity and positive otherwise. Depending on the adjustment technology and on the distribution of idiosyn-

²For instance, such moments cannot separate strong real rigidity with large cost shocks from weak real rigidity with small ones, although the two imply different Phillips curves.

cratic shocks, the environment nests the Calvo (1983), Golosov and Lucas (2007), Gertler and Leahy (2008), and CalvoPlus (Nakamura and Steinsson 2010) models.

Our first result is an equivalence between the macro response to a macro shock and the average micro response to a micro shock. Consider the sequence-space Jacobian mapping the impulse response of nominal marginal costs to that of the price level, the nominal pass-through matrix. We show that in each of these models, under conditions on the idiosyncratic shocks that we characterize, the average response of firms' prices to their own random walk idiosyncratic cost shock—the micro IRF—identifies the response of the aggregate price level to a permanent unanticipated nominal marginal cost shock horizon by horizon—the row sums of the nominal pass-through matrix.³ Our second result is that, although the generalized Phillips curve a priori depends on the entire nominal pass-through matrix, its first two entries suffice, either exactly or to a numerically tight approximation.

To develop intuition, we show, model by model, what the two coefficients identify. The generalized Phillips curve depends on two forces: the desired pass-through of costs into prices, $\omega \leq 1$, pinned down by real rigidities, and the survival of a cost shock in firms' price gaps, set by the adjustment technology. The impact coefficient b_0 captures both forces, while the ratio $\phi^{\text{IRF}} = b_1/b_0 - 1$ captures the survival rate alone. In the Calvo model, $b_0 = \omega(1 - \alpha)$ and $\phi^{\text{IRF}} = \alpha$, where α is the non-adjustment probability. Because the survival function is geometric, its one-period value summarizes it, and the two coefficients fully determine the Phillips curve. Importantly, both b_0 and b_1 are responses of realized prices and so reflect selection—which firms adjust after a shock and by how much—when present. In the state-dependent Gertler and Leahy (2008) model, for instance, a firm reconsiders its price only when hit by a large idiosyncratic shock, with probability $1 - \eta$, and a firm that reconsiders and keeps its price had the right price on average, so that its expected price absorbs the cost shock whether or not it adjusts. The survival function is geometric, at η , and the Phillips curve has the Calvo structure with η in place of α . In this case, b_0 and ϕ^{IRF} pick up $\omega(1 - \eta)$ and η , although η is not the frequency of price changes, and our statistic is again exact.

In the canonical menu cost model the survival function is not geometric. By the representation of Auclert et al. (2024), the pass-through matrix is a weighted average of the

³The result is related to, but distinct from, Caballero and Engel (2007), who show that the macro response to a macro shock equals the average micro response to the same macro shock. Ours relates responses to two different shocks, which is what makes the aggregate response estimable from a firm-level regression.

pass-through matrices of exactly two time-dependent models, one for the extensive margin, the movements of the adjustment bands, and one for the intensive margin, the movements of the reset price. We show that the micro IRF exactly identifies the weighted average of the two survival functions horizon by horizon, with weights equal to the shares of the impact pass-through due to each margin, and that ϕ^{IRF} is its one-period value. The Calvo assembly extrapolates this one-period value geometrically. The replacement is accurate because, as Auclert et al. (2024) document, the conditional survival functions of the two margins deviate from a common constant in opposite directions so that their average is close to constant.

Whether the approach accurately recovers the generalized Phillips curve of menu cost models is a quantitative question, which we answer in two steps. First, Auclert et al. (2024) show that the generalized Phillips curve of a menu cost model is well approximated by that of a Calvo model with a suitably chosen survival probability. We show that whenever this approximation is accurate, our statistic recovers this survival probability, and therefore the true Phillips curve, with a proportional accuracy. Second, we show numerically that the approximation is highly accurate. It reproduces the full dynamic response to marginal cost shocks in the Golosov and Lucas (2007) and Nakamura and Steinsson (2010) calibrations. To evaluate accuracy across a large number of parameterizations of the menu cost, the probability of free adjustment, and the volatility of idiosyncratic shocks, we then focus on two scalar metrics: the impact entry of the generalized Phillips curve and the Phillips multiplier of Barnichon and Mesters (2021), equal to the ratio of the present discounted value of the inflation response to that of the output response. Across parameterizations, the metrics computed from our assembled Phillips curve move one-for-one with the true ones and explain virtually all of their variance. Our approach does as well as a Calvo model whose survival probability is chosen to best fit the true Phillips curve Jacobian, but does not require knowing it.

The accuracy of the approach extends beyond the canonical model. First, it accommodates decreasing returns to scale and strategic complementarities in price setting (which we model à la Kimball 1995), two forms of micro-level real rigidities, while existing sufficient statistics do not. The intuition is that firm-level pass-through already incorporates them, so that no adjustment is needed. Hence, our statistic performs as well as in the baseline model. Second, it does not require the idiosyncratic shocks affecting firms' desired

prices to be normally distributed or to follow a random walk, two assumptions that existing sufficient statistics maintain. Third, it is unaffected by measurement error in prices.

Compared to approaches relying on unconditional moments of the distribution of price changes, the main cost of our approach is that it requires a plausibly exogenous firm-level cost shifter. We do not believe this to be a major impediment in practice since such pass-through regressions have been estimated in many contexts.⁴

Related literature. This work contributes to three strands of the literature. First, we contribute to the literature that infers monetary non-neutrality from unconditional moments of the distribution of price changes. This agenda has grown alongside the evidence for state-dependent models, in which the frequency of price changes is not a sufficient statistic. In these models, the full size distribution of price changes is informative about monetary non-neutrality (Midrigan 2011; Alvarez and Lippi 2014). In particular, Alvarez, Le Bihan, and Lippi (2016) show that the ratio of kurtosis to frequency is sufficient for the cumulative response of the price level to a small monetary shock.⁵ This result holds around the zero inflation steady-state and requires quadratic losses, Gaussian random-walk idiosyncratic shocks, and the absence of micro-level real rigidities so that an adjusting firm closes its price gap entirely.⁶ The subsequent literature has shown that each of these assumptions is a source of misspecification. With fat-tailed shocks, Karadi and Reiff (2019) match the same frequency, size, and kurtosis of price changes with models that have very different cumulative price responses. With strategic complementarities, Klenow and Willis (2016) reproduce the same frequency, dispersion, and persistence of price changes either with

⁴These analyses have been performed in the United States (Gopinath and Itskhoki 2010; Fajgelbaum et al. 2020; Ganapati, Shapiro, and Walker 2020; Cavallo et al. 2021; Renkin, Montialoux, and Siegenthaler 2022), Belgium (Amiti, Itskhoki, and Konings 2019; Gagliardone et al. 2025), France (Lafrogne-Joussier, Martin, and Méjean 2026), Sweden (Carlsson and Nordström Skans 2012), Finland (Benzarti et al. 2020), Hungary (Karadi and Reiff 2019; Harasztosi and Lindner 2019), and India (Herreño, Pinardon-Touati, and Thie 2026). The cost shifters are exchange-rate movements, tariffs, imported-input prices, energy prices, unit labor costs, exposure to minimum-wage increases, value-added tax changes, and shift-share combinations of input prices.

⁵With the additional assumption of Golosov and Lucas (2007) preferences and given a labor supply elasticity, the response of the price level fully determines the real wage, and hence, labor supply and output.

⁶Alvarez, Lippi, and Oskolkov (2022) show that the result extends to any symmetric adjustment hazard but that the same four assumptions remain necessary. Alvarez, Lippi, and Souganidis (2023) allow for strategic complementarities, summarized by a single parameter θ , the negative of the elasticity of a firm's optimal price to the aggregate price, and find that the cumulative output response becomes the kurtosis ratio scaled by $1/\sqrt{1+\theta}$; the scaling is exact in the Calvo model and numerically close in menu cost models. However, θ is not a moment of the distribution of price changes: it must be calibrated, or estimated from the response of prices to costs and to competitors' prices, that is, from a regression of the kind we study.

no complementarity and small shocks or with a large Kimball superelasticity and large shocks. Dotsey and Wolman (2020) reach the same underidentification conclusion while targeting six moments of the distribution of price changes. Consistent with these results, Hong et al. (2023) find across producer-price sectors that only the frequency is robustly related to measured non-neutrality, and that kurtosis has little predictive power. These moments also depend on how prices are measured: Nakamura and Steinsson (2008), Eichenbaum, Jaimovich, and Rebelo (2011) and Kehoe and Midrigan (2015) show that the measured frequency of price changes is highly sensitive to the treatment of temporary sales and to the use of posted versus reference prices. Relative to this literature, we propose conditional moments, the response of prices to an observed cost shock, whose mapping to the Phillips curve is the same across time- and state-dependent models, does not depend on the distribution of idiosyncratic shocks or on micro-level real rigidities, and is unaffected by variation in prices that is unrelated to the shock.

Second, we contribute to the literature on generalized Phillips curves in Ss models. Auclert et al. (2024) define the generalized Phillips curve as the sequence-space Jacobian from real marginal cost to inflation, and find numerically that the generalized Phillips curve of menu cost models is close to that of a Calvo model with a suitably chosen “synthetic” Calvo survival probability. However, operationalizing this result requires knowing the underlying model to compute the synthetic survival probability that best approximates the menu cost generalized Phillips curve.⁷ Our robust statistic operationalizes the Calvo approximation without knowledge of the underlying model. Our approach builds on an aggregation argument reminiscent of Caballero and Engel (2007). They show that in a stationary Ss model the impact response of the price level to a small nominal common shock is the average of firms’ marginal responses to it. We show that it also equals the average response of firms’ prices to their own idiosyncratic nominal cost shock, the object a cross-sectional regression identifies.

Third, we contribute to the empirical literature that estimates the pass-through of cost shocks into prices at the firm level and uses it to learn about monetary non-neutrality (Gopinath and Itskhoki 2010).⁸ Closest to us, Gagliardone et al. (2025) and Herreño,

⁷Alternatively, a researcher can rely on the Alvarez, Le Bihan, and Lippi (2016) sufficient statistics and the known kurtosis of the Calvo model to infer the synthetic Calvo survival probability, but this approach suffers from the misspecification risk highlighted above. We document this numerically in our results.

⁸A large literature also studies pass-throughs outside of the question of monetary non-neutrality (e.g., Amiti, Itskhoki, and Konings 2019; Sangani 2026).

Pinardon-Touati, and Thie (2026) both estimate pass-through regressions to recover the slope of the marginal-cost Phillips curve in the Calvo framework. In particular, Herreño, Pinardon-Touati, and Thie (2026) show that in the Calvo model the pass-through at two horizons identifies this slope. This paper shows what the same regressions identify in state-dependent models: the Phillips curve exactly in the Gertler–Leahy model and to a tight approximation in the canonical menu cost model, so that the two-coefficient recipe applies without knowing the pricing model.⁹

2. Environment and Generalized Phillips Curve

This section lays out the environment common to the three models we consider and defines the object we want to recover, the generalized Phillips curve. Throughout, lower-case letters denote log deviations from a zero-inflation steady state.

2.1. Environment

Firms and marginal costs. A continuum of firms $i \in [0, 1]$ produce differentiated varieties, aggregated with a CES aggregator with elasticity $\varepsilon > 1$, so that firm i faces the demand $Y_{it} = (P_{it}/P_t)^{-\varepsilon} Y_t$. Each firm produces with the technology $Y_{it} = A_{it} N_{it}^\gamma$, $\gamma \in (0, 1]$, where N_{it} is labor hired at the economy-wide nominal wage W_t and A_{it} is idiosyncratic productivity. In log deviations, the nominal marginal cost of firm i is

$$(1) \quad mc_{it} = mc_t - \frac{a_{it}}{\gamma} + \frac{1-\gamma}{\gamma} (y_{it} - Y_t), \quad mc_t \equiv w_t + \frac{1-\gamma}{\gamma} Y_t,$$

where mc_t is what we call *aggregate nominal marginal cost*. Real per-period profit at the nominal price P_{it} is:

$$(2) \quad \Pi_t(P_{it}) = \left(\frac{P_{it}}{P_t} \right)^{1-\varepsilon} Y_t - \frac{W_t}{P_t} \left[\frac{Y_t}{A_{it}} \left(\frac{P_{it}}{P_t} \right)^{-\varepsilon} \right]^{1/\gamma}.$$

⁹A distinct literature estimates the slope of the Phillips curve from cross-sectional regressions across regions (Hazell et al. 2022; Cerrato and Gitti 2022; Fitzgerald et al. 2024). Our statistic identifies the marginal-cost Phillips curve while this approach typically measures the output or unemployment based Phillips curve, which jointly identifies the slope of the marginal-cost Phillips curve and the response of marginal cost to the output gap.

The flexible-price—or frictionless desired price—maximizes (2) and is equal to a constant markup $\frac{\varepsilon}{\varepsilon-1}$ over the firm's own marginal cost. Solving for this quantity, we obtain

$$(3) \quad p_{it}^* = \omega \left(mc_t - \frac{a_{it}}{\gamma} \right) + (1 - \omega) P_t, \quad \omega \equiv \frac{1}{1 + \varepsilon(1 - \gamma)/\gamma} \in (0, 1],$$

once the constant $\omega \ln \frac{\varepsilon}{\varepsilon-1}$ is dropped. Throughout, “desired price” is a shorthand for (3). With constant returns, $\omega = 1$, and the desired price is nominal marginal cost itself. With decreasing returns a firm that raises its price sells less and so produces at lower marginal cost, which damps the desired response to a cost shock by a factor $\omega < 1$. The coefficient ω acts as an index of *micro-level real rigidity*.¹⁰

Loss function. Let $x_{it} \equiv p_{it} - p_{it}^* = \ln P_{it} - \ln P_{it}^*$ denote the *price gap*. Expanding (2) to second order around the desired price, the per-period cost of a mispriced good is

$$(4) \quad \Pi_t(P_{it}^*) - \Pi_t(P_{it}) = \frac{1}{2} C_{it} x_{it}^2 + O(|x_{it}|^3), \quad C_{it} = \frac{\varepsilon - 1}{\omega} Y_t \left(\frac{P_{it}^*}{P_t} \right)^{1-\varepsilon}$$

Dividing by C_{it} then measures every loss in units of the curvature: the per-period loss becomes $\frac{1}{2} x_{it}^2$, and the cost of adjusting, which depends on the adjustment technology introduced below, is measured in the same unit. Where that cost is a menu cost we take it to be proportional to frictionless revenue, ξC_{it} , so that in curvature units it is a constant ξ common across firms and dates. Using standard second-order arguments, equation (4) leads to the following objective function:

$$(5) \quad \mathbb{E}_0 \sum_{t \geq 0} \beta^t \left[\frac{1}{2} x_{it}^2 + (\text{cost of adjusting}_{it}) \right],$$

Adjustment technologies. We consider two distinct adjustment technologies.

Calvo. This is model (C). With probability $1 - \alpha$, i.i.d. across firms and time, the firm may reset its price at no cost; otherwise its price is frozen.

¹⁰ Every result below uses only the reduced form (3) together with a loss that is quadratic in the gap. Two other sources of micro-level real rigidities deliver the same reduced form: a firm-specific labor market whose wage the firm internalizes, $w_{it} - w_t = \varphi(n_{it} - N_t)$ with φ the inverse Frisch elasticity, which is isomorphic to (1) with $\frac{1-\gamma}{\gamma} = \varphi$; and Kimball demand, for which (3) holds to first order with ω set by the superelasticity. In this latter case, the curvature of the loss then depends on the firm's relative cost, so the reduction to the gap below holds only to second order. We provide numerical results for this case in Section 5.1.

Menu cost with free adjustments. The Gertler-Leahy model (GL) and canonical menu cost model (MC) both use this technology. The firm pays a menu cost $\xi_{it} \in \{0, \xi\}$ to change its price, with a free adjustment ($\xi_{it} = 0$) materializing with probability λ , i.i.d. across firms and time: the CalvoPlus specification of Nakamura and Steinsson (2010), which nests Golosov and Lucas (2007) ($\lambda = 0$) and Calvo ($\xi \rightarrow \infty$).¹¹ Whenever the firm reconsiders its price, the optimal policy has an *Ss* form: it adjusts if and only if its gap lies outside an inaction band $[-\bar{x}, \bar{x}]$ around the reset price (Appendix A.2.2 and B.4.1).

More generally, the menu cost may be drawn from an arbitrary distribution and take more than two values; we refer to this case as the generalized menu cost model (GMC). The main text states results for (MC) and notes in each case how they extend to (GMC), with the full (GMC) results left to Appendix A.3.

Idiosyncratic shocks. We decompose idiosyncratic productivity into two components,

$$(6) \quad -\frac{a_{it}}{\gamma} = z_{it} + \vartheta_{it},$$

so that

$$(7) \quad p_{it}^* = \omega(mc_t + z_{it} + \vartheta_{it}) + (1 - \omega) P_t.$$

The component z_{it} is a generic productivity process. The component ϑ_{it} is a cost shifter that the econometrician observes and uses as a regressor; for instance in Herreño, Pinardon-Touati, and Thie (2026) it is a firm-specific input-cost shock. The split (6) is a statement about what the econometrician observes, not about how the firm behaves. Both components are independent driftless random walks:

$$(8) \quad \vartheta_{it} = \vartheta_{i,t-1} + \epsilon_{it}, \quad \mathbb{E}[\epsilon_{it}] = 0, \quad \text{Var}(\epsilon_{it}) = \sigma_{\vartheta}^2$$

$$(9) \quad z_{it} = z_{i,t-1} + \epsilon_{it}^z, \quad \mathbb{E}[\epsilon_{it}^z] = 0, \quad \text{Var}(\epsilon_{it}^z) = \sigma_z^2.$$

where ϵ_{it} and ϵ_{it}^z are mutually independent, i.i.d. across firms and time, and independent of all adjustment draws at all dates; in particular, both are independent of the firm's pre-determined state (its price, its gap and its adjustment history). The three models differ in

¹¹In Nakamura and Steinsson's specification the menu cost takes a low or a high value each period; we set the low value to zero, so that $\xi_{it} \in \{0, \xi\}$.

the distributional assumptions on these shocks.

Calvo (C). ϵ_{it} and ϵ_{it}^z can have any distribution with the moments above.

Menu Cost (MC). ϵ_{it}^z has a symmetric, single-peaked and continuously differentiable density. For our baseline identification results, ϵ_{it} is Gaussian, and we discuss the non-Gaussian case in Appendix B.5.4.

Gertler-Leahy (GL). The driving shock is a large *turbulence* shock: with probability $1-\eta$, i.i.d. across firms and time, $z_{it} = z_{i,t-1} + \epsilon_{it}^z$ with $\omega\epsilon_{it}^z \sim U[-\varsigma/2, \varsigma/2]$. By contrast, innovations to mc_t , P_t and ϑ_{it} are small: they move the desired price by at most a first-order constant m over the horizon a price is expected to last, and $\varsigma > 4\bar{x} + 2m$, so that the band sits strictly inside the support of the turbulence shock. As explained below, this infrequent, uniform and large turbulence draw is what makes the model tractable.

With these assumptions, the desired price is a random walk in every model. Then, the firm's problem depends on $(mc_t, z_{it}, \vartheta_{it}, P_t)$ only through p_{it}^* . In the stationary cross-section the reset price is then the current desired price and the reset gap is zero.¹²

We state our main results under the random-walk assumption for simplicity, but this assumption can be relaxed. In models (C) and (GL), every analytical result goes through with modified formulas. In model (MC), we do not provide analytical results for this case, but Section 5.4 presents numerical results that show that our method performs equally well. Appendix B.6 details the three cases.

Aggregation. To first order the log price index is $P_t = \int_0^1 p_{it} di$, and since $\int (z_{it} + \vartheta_{it}) di = 0$, $\int p_{it}^* di = \omega mc_t + (1-\omega)P_t$, so that $P_t = mc_t + \omega^{-1} \int x_{it} di$. Inflation is $\pi_t = P_t - P_{t-1}$. Real marginal cost is $mc_t^r \equiv mc_t - P_t$. The rest of the economy (labor supply, the Euler equation, monetary policy) determines mc_t^r and is only needed for the general-equilibrium exercises.

2.2. Pass-Through Jacobians and the Generalized Phillips Curve

We use sequence-space notation: bold symbols denote sequences starting at date 0, column vectors indexed by $t = 0, 1, \dots$ (truncated at a horizon T in computations), $\mathbf{P} = (P_0, P_1, \dots)'$, $\mathbf{mc} = (mc_0, mc_1, \dots)'$, and so on. L and F are the lag and lead operators,

¹²See Section 3.3.1 for (C), Appendix A.2 for (GL), Appendix B.4.1 for (MC).

square matrices of the same dimension with ones on the first subdiagonal and superdiagonal respectively, so that $(L\mathbf{x})_t = x_{t-1}$, $(L\mathbf{x})_0 = 0$ and $(F\mathbf{x})_t = x_{t+1}$; I is the identity matrix and $\mathbf{1} = (1, 1, \dots)'$. For a Jacobian matrix J we write $J[t, s]$ for its entry in row t and column s , both indexed from 0, so that $[J\mathbf{1}]_h = \sum_s J[h, s]$ is the date- h response to a permanent unit shock.

Starting from the steady state, each model $m \in \{C, GL, MC\}$ maps a perfect-foresight sequence of desired prices into a sequence of price levels through the aggregation of individual price-setting responses. Its first-order representation is the *desired-price pass-through Jacobian*

$$(10) \quad J_{P,p^*}^m[t, s] \equiv \frac{\partial P_t}{\partial p_s^*}, \quad \mathbf{P} = J_{P,p^*}^m \mathbf{p}^*,$$

the response of the price level at t to a common shift of every firm's desired price at date s . This is a partial-equilibrium object: it treats the path of desired prices as exogenous and includes no feedback from prices to desired prices. By (7), holding the price level constant, a unit change in mc_s shifts every desired price by ω at date s , so the *nominal marginal cost pass-through Jacobian* is

$$(11) \quad J_{P,mc}^m[t, s] \equiv \left. \frac{\partial P_t}{\partial mc_s} \right|_{\mathbf{P}} = \omega J_{P,p^*}^m[t, s].$$

Using $\mathbf{p}^* = \omega \mathbf{mc} + (1 - \omega)\mathbf{P}$ and $\mathbf{mc}^r = \mathbf{mc} - \mathbf{P}$, the price level solves the fixed point $\mathbf{P} = J_{P,p^*}^m(\omega \mathbf{mc} + (1 - \omega)\mathbf{P}) = J_{P,p^*}^m(\omega \mathbf{mc}^r + \mathbf{P})$, so the *real marginal cost pass-through Jacobian* writes:

$$(12) \quad J_{P,mc^r}^m = (I - J_{P,p^*}^m)^{-1} \omega J_{P,p^*}^m = \omega (I - J_{P,p^*}^m)^{-1} J_{P,p^*}^m.$$

Finally, we define the *generalized Phillips curve Jacobian*, as:

$$(13) \quad J_{\pi,mc^r}^m = (I - L) J_{P,mc^r}^m.$$

Given J_{π,mc^r}^m , inflation is pinned down by the path of real marginal cost, $\boldsymbol{\pi} = J_{\pi,mc^r}^m \mathbf{mc}^r$, and hence by any demand-side block that determines $\mathbf{mc}^r$. Equations (12)–(13) are pure linear algebra given J_{P,p^*}^m and ω , with J_{P,p^*}^m the only model-specific object.

With a slight abuse of terminology, we refer to $J_{\pi,mc}^m[0,0]$, the impact response of inflation to an unanticipated and purely transitory increase in real marginal cost, as the slope of the Phillips curve.

3. A Robust Statistic for the Generalized Phillips Curve

3.1. The Statistic

Consider the following regression in the stationary cross-section of firms, for horizons $h = 0, 1, 2, \dots$ (the statistic uses $h = 0$ and $h = 1$):

$$(14) \quad p_{it+h} - p_{i,t-1} = a_h + b_h \epsilon_{it} + u_{it}$$

b_h is the population local projection of cumulative price changes on the idiosyncratic innovation. Then, define:

$$(15) \quad \phi^{\text{IRF}} \equiv \frac{b_1}{b_0} - 1$$

$$(16) \quad \kappa^{\text{IRF}} \equiv \frac{b_0(1 - \beta\phi^{\text{IRF}})}{\phi^{\text{IRF}}}$$

Let $\Theta_\beta \equiv (I - \beta F)^{-1}$. Our estimate of the generalized Phillips curve is then:

$$(17) \quad J_{\pi,mc}^{\text{IRF}} = \kappa^{\text{IRF}} \Theta_\beta = \kappa^{\text{IRF}} \begin{bmatrix} 1 & \beta & \beta^2 & \beta^3 & \dots \\ 0 & 1 & \beta & \beta^2 & \dots \\ 0 & 0 & 1 & \beta & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

There are two key steps for understanding the identification of the generalized Phillips curve. First, we show that the micro IRF b_h is equal to the response of the aggregate price level to a permanent unit shock to nominal marginal cost, $[J_{P,mc}^m \mathbf{1}]_h$ at every horizon h . Second, we show that the first two entries of $J_{P,mc}^m \mathbf{1}$ —identified by b_0 and b_1 —allow to provide a highly accurate approximation of the generalized Phillips curve $J_{\pi,mc}^m$ in a wide class of models.

3.2. From the Cross-Sectional to the Aggregate Pass-Through

In every model m , the population local projection of cumulative price changes on the idiosyncratic innovation is given by:

$$(18) \quad b_h^m \equiv \frac{\text{Cov}(p_{i,t+h} - p_{i,t-1}, \epsilon_{it})}{\text{Var}(\epsilon_{it})}, \quad h = 0, 1, 2, \dots,$$

We write $b^m = (b_0^m, b_1^m, \dots)'$ and call it the *micro IRF*.

PROPOSITION 1 (Micro–macro bridge). *In each model, under the conditions listed below, the micro IRF equals the response of the aggregate price level to a permanent unit shock to nominal marginal cost:*

$$(19) \quad b_h^m = [J_{P,mc}^m \mathbf{1}]_h \quad \text{for all } h \geq 0.$$

The conditions, beyond the independence assumptions on the innovations maintained after (9), are: none in (C); the small-shock assumptions of Section 2.1 in (GL); and, in (MC) and its extension (GMC), $\epsilon_{it} \sim N(0, \sigma_{\vartheta}^2)$.

Proof (main argument, full proof in Appendix B.5). Fix t and a firm. Let S be its predetermined state, ϵ_{it} its identifying innovation, and $\mathcal{D}$ the other random draws that affect its prices through $t+h$; the cumulative price change is then a deterministic function $f_h(\epsilon_{it}; S, \mathcal{D})$ of the innovation. The proof has two steps.

The first step shows that the aggregate response is the cross-sectional average derivative of the price path with respect to the firm's *own* shifter,

$$(20) \quad [J_{P,mc}^m \mathbf{1}]_h = \frac{d}{d\delta} \mathbb{E}_{S,\epsilon,\mathcal{D}}[f_h(\epsilon_{it} + \delta; S, \mathcal{D})] \Big|_{\delta=0},$$

in every model. Under the random walk, the shift δ raises the desired price by $\omega\delta$ at every $s \geq 0$ and changes nothing else. Since the firm's problem depends on $(mc_t, z_{it}, \vartheta_{it}, P_t)$ only through p_{it}^* in the normalized pricing problem used here, and the price level is held fixed in (11), this is, from the firm's point of view, an unanticipated permanent increase in mc by δ . Averaging over the stationary cross-section gives the response of the aggregate price level to a permanent unit nominal-cost shock with the price level fixed.¹³

¹³Caballero and Engel (2007) offer a related but different result: they relate the macro IRF to a common

The second step shows that the regression coefficient b_h^m equals the right-hand side of (20). The regression weights each firm's price change by its own innovation; the average derivative weights the local responses of f_h by the population density of the innovation. The two coincide: (i) for every response function f_h when the innovation is Gaussian, by Stein's identity — which is why (MC) and (GMC) require the same condition regardless of the menu-cost distribution H , since the argument only uses the shape of f_h , not the two-point structure of ξ_{it} —, or (ii) for any distribution when the response (averaged over the other draws $\mathcal{D}$) is affine in the innovation, which is the case in (C) and (GL). $\square$

This Proposition easily accommodates two extensions. First, if ϵ_{it} is not Gaussian in the menu cost model, $[J_{P,mc}^m \mathbf{1}]_h$ remains identified but the correct estimator becomes an IV regression of the cumulative price change on ϵ_{it} with instrument $s(\epsilon_{it})$ where $s(\epsilon)$ is the score of the shock's density. We present the identification proof in Appendix B.5.4 and provide numerical results in Appendix C.3. Second, even when shocks are not random walks, this analytical identification result carries over to the (C) and (GL) case (with modified formulas), and we provide numerical results for the (MC) case (Appendix B.6).

COROLLARY 1 (Pass-through and the surviving gap). *In every model,*

$$(21) \quad b_h^m = \omega + \frac{\text{Cov}(x_{i,t+h}, \epsilon_{it})}{\sigma_{\vartheta}^2}.$$

PROOF. Write $p_{i,t+h} - p_{i,t-1} = (p_{i,t+h}^* - p_{i,t-1}^*) + x_{i,t+h} - x_{i,t-1}$. With ϑ a random walk and ϵ_{it} independent of all other innovations and of the aggregates, $\text{Cov}(p_{i,t+h}^* - p_{i,t-1}^*, \epsilon_{it}) = \omega \sigma_{\vartheta}^2$ by (7), and $x_{i,t-1}$ is predetermined so $\text{Cov}(x_{i,t-1}, \epsilon_{it}) = 0$. $\square$

Pass-through at horizon h falls short of ω , the pass-through of the shock into the desired price, exactly to the extent that the price gap h periods after the shock still carries the shock. This identity holds in every model and is the device we use to explain the shape of the micro IRF. It also shows what the long-run pass-through measures: with a random-walk shifter, $b_h^m \rightarrow \omega$, the desired-price pass-through coefficient, irrespective of the adjustment technology.

shock to the average of firms' marginal responses to that same shock; we relate it to the average of firms' marginal responses to their own micro shock.

What is not identified. The micro IRF identifies the vector $J_{P,mc}^m \mathbf{1}$ —the response to an unanticipated permanent cost shock with the price level held fixed—and not the matrix $J_{P,mc}^m$: idiosyncratic shocks are never anticipated, so the columns describing responses to anticipated future cost changes are outside the regression’s reach. In addition, without further structure, it does not separately identify ω from J_{P,p^*}^m at any finite horizon. Since the generalized Phillips curve (13) depends on the entire matrix J_{P,p^*}^m , the micro IRF does not identify it in general.

3.3. From the Pass-Through Matrix to the Generalized Phillips Curve

This section shows how only the first two entries of $J_{P,mc}^m \mathbf{1}$ allow to construct a highly robust estimate of the generalized Phillips curve across price setting models. To build intuition, we start from the Calvo model, where the two entries of $J_{P,mc}^m \mathbf{1}$ are sufficient to recover the entire generalized Phillips curve exactly. We then show that the same two entries are exactly sufficient in the (GL) model, which allows to provide intuition for our key result that—even if not exact—our statistic provides an extremely robust estimate of the generalized Phillips curve in the general class of (MC) models.

3.3.1. Calvo and the IRF Statistics

For any non-adjustment probability $\alpha \in (0, 1)$ define

$$\Upsilon(\alpha) \equiv (I - \alpha L)^{-1} = \begin{bmatrix} 1 & 0 & 0 & \cdots \\ \alpha & 1 & 0 & \cdots \\ \alpha^2 & \alpha & 1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}, \quad \Theta(\alpha) \equiv (I - \beta \alpha F)^{-1} = \begin{bmatrix} 1 & \beta \alpha & (\beta \alpha)^2 & \cdots \\ 0 & 1 & \beta \alpha & \cdots \\ 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

A firm that resets at t keeps its price until its next adjustment opportunity, so it minimizes $\mathbb{E}_t \sum_{s \geq 0} (\beta \alpha)^s \frac{1}{2} (p_{it}^{\text{new}} - p_{i,t+s}^*)^2$ and sets

$$(22) \quad p_{it}^{\text{new}} = (1 - \beta \alpha) \mathbb{E}_t \sum_{s \geq 0} (\beta \alpha)^s p_{i,t+s}^*, \quad \text{or} \quad \mathbf{p}_i^{\text{new}} = (1 - \beta \alpha) \Theta(\alpha) \mathbf{p}_i^*,$$

where $\mathbf{p}_i^* = \omega(\mathbf{mc} + \mathbf{z}_i + \mathbf{v}_i) + (1 - \omega)\mathbf{P}$ by (7) and, for the idiosyncratic components, the sequence is the expected path. $\Theta(\alpha)$ maps the sequence of desired prices into the sequence

of reset prices. The expected price of the firm weights reset prices across vintages by the survival probabilities,

$$(23) \quad \bar{p}_{it} = (1 - \alpha) \sum_{j \geq 0} \alpha^j p_{i,t-j}^{\text{new}}, \quad \text{or} \quad \bar{\mathbf{p}}_i = (1 - \alpha) \Upsilon(\alpha) \mathbf{p}_i^{\text{new}}.$$

$\Upsilon(\alpha)$ maps reset prices into the price level by aggregating over surviving vintages. Then,

$$(24) \quad \bar{\mathbf{p}}_i = J_{P,p^*}^C(\alpha) \mathbf{p}_i^*, \quad J_{P,p^*}^C(\alpha) \equiv (1 - \alpha)(1 - \beta\alpha) \Upsilon(\alpha) \Theta(\alpha).$$

Aggregating across firms, the idiosyncratic components wash out and $\mathbf{P} = J_{P,p^*}^C(\alpha)(\omega \mathbf{mc} + (1 - \omega)\mathbf{P})$: the desired-price pass-through of the Calvo model is $J_{P,p^*}^C(\alpha)$, and its nominal marginal cost pass-through Jacobian is $J_{P,mc}^C(\alpha) = \omega J_{P,p^*}^C(\alpha)$. Two properties of $J_{P,p^*}^C(\alpha)$ drive everything that follows.

LEMMA 1. Let $\kappa(\alpha) \equiv (1 - \alpha)(1 - \beta\alpha)/\alpha$. Then

- (i) $[J_{P,p^*}^C(\alpha)\mathbf{1}]_h = 1 - \alpha^{h+1}$;
- (ii) $(I - L)(I - J_{P,p^*}^C(\alpha))^{-1} J_{P,p^*}^C(\alpha) = \kappa(\alpha) \Theta_\beta$, so that by (12)–(13) $J_{\pi,mc^r}^C(\alpha) = \omega \kappa(\alpha) \Theta_\beta$.

Proof: Appendix B.1.1.

Property (ii) is the New Keynesian Phillips curve $\pi_t = \omega \kappa(\alpha) mc_t^r + \beta \pi_{t+1}$ written in sequence space $\boldsymbol{\pi} = \omega \kappa(\alpha) \Theta_\beta \mathbf{mc}^r$.

PROPOSITION 2 (The IRF statistics in the Calvo model). Suppose that in any model m the desired-price pass-through Jacobian has the Calvo structure, $J_{P,p^*}^m = J_{P,p^*}^C(\alpha)$ for some $\alpha \in (0, 1)$, with desired-price pass-through $\omega \in (0, 1]$, and that Proposition 1 applies. Then

- (i) $b_h^m = \omega(1 - \alpha^{h+1})$, so that

$$(25) \quad \phi^{\text{IRF},m} \equiv \frac{b_1^m}{b_0^m} - 1 = \alpha;$$

- (ii) the generalized Phillips curve is $J_{\pi,mc^r}^m = \kappa^{\text{IRF},m} \Theta_\beta$, i.e. $\pi_t = \kappa^{\text{IRF},m} mc_t^r + \beta \pi_{t+1}$, with

$$(26) \quad \kappa^{\text{IRF},m} \equiv \frac{b_0^m (1 - \beta \phi^{\text{IRF},m})}{\phi^{\text{IRF},m}}.$$

Hence, given β , the first two entries of the micro IRF determine the entire generalized Phillips curve.

PROOF. (i) follows from Proposition 1 and Lemma 1(i). (ii) follows from Lemma 1(ii), $J_{\pi,mc}^m = \omega\kappa(\alpha)\Theta_\beta$, with $\omega\kappa(\alpha) = \omega(1-\alpha)(1-\beta\alpha)/\alpha = b_0(1-\beta\alpha)/\alpha$ by (i). $\square$

This works because of two things. b_0 captures the first entry of the passthrough Jacobian. The mapping from the on-impact passthrough to the Phillips curve only depends on α (given β), which is identified by the ratio b_1/b_0 . The two entries of the micro IRF are therefore sufficient to recover the entire generalized Phillips curve.

The micro-to-macro mapping is a fixed-point problem: firms reset their prices taking the aggregate price level as given. The strength of the feedback from prices to desired prices is governed by $1-\alpha$, the degree of price flexibility (and does not depend on ω , since the price level enters the desired price with total weight 1). The micro IRF identifies α because, by Corollary 1, $\omega - b_h = -\text{Cov}(x_{i,t+h}, \epsilon_{it})/\sigma_\epsilon^2 = \omega\alpha^{h+1}$: the shortfall of pass-through from its long-run value ω at horizon h is ω times the probability that the shock still sits in the firm's price gap. This shortfall decays at the geometric rate α , and the ratio $b_1/b_0 = 1 + \alpha$ reads it off because real rigidity scales every entry of the micro IRF by the same ω . If prices are very sticky, many firms cannot adjust in the first period, so a larger remaining mass of firms still wants to adjust between $t = 0$ and $t = 1$. The level b_0 then identifies ω .¹⁴

REMARK 1 (Persistent shifters). With $\vartheta_{it} = \rho\vartheta_{i,t-1} + \epsilon_{it}$ and $J_{P,p^*}^m = J_{P,p^*}^C(\alpha)$, we instead have:

$$b_0 = \omega \frac{(1-\alpha)(1-\beta\alpha)}{1-\beta\alpha\rho}, \quad \frac{b_1}{b_0} = \alpha + \rho,$$

so that $\alpha = b_1/b_0 - \rho$ and $\kappa = \omega\kappa(\alpha) = b_0(1-\beta\alpha\rho)/\alpha$, with ρ estimated from the autocorrelation of the shifter. The random-walk formulas (25)–(26) are the case $\rho = 1$.

We can already note that the Calvo structure of $J_{P,p^*}^C(\alpha)$ is not tied to the Calvo lottery. In the Rotemberg model, where every price changes every period at a convex cost, the desired-price pass-through is exactly $J_{P,p^*}^C(\alpha_R)$ with α_R the stable root of the firm's Euler equation, so Proposition 2 applies verbatim and the statistic is exact while the frequency of price changes is one (Appendix B.2).

¹⁴Given a third entry the system is overidentified, $b_2 = \omega(1-\alpha^3)$, which provides a specification test of the Calvo structure. Note that the micro IRF does not identify β : under a random walk no entry depends on β , and under an AR(1) shifter β enters only through the level factor $(1-\beta\alpha)/(1-\beta\alpha\rho)$ of Remark 1, which is confounded with ω .

3.3.2. Gertler–Leahy

In the GL economy the decision to adjust is state dependent, so there is no structural non-adjustment probability that a Calvo econometrician could target. Nevertheless, the desired-price pass-through Jacobian is exactly Calvo, with the probability $\eta_\lambda \equiv (1 - \lambda)\eta$ that a price is neither hit by turbulence nor freely reconsidered in place of α , and our statistic recovers η_λ (set-up in Appendix A.2, proofs in Appendix B.3).

Reset behavior. A key result is that a firm reconsiders its price only when hit by turbulence or when it draws a free adjustment.¹⁵ A price therefore lasts until the next hit or free draw, an event of per-period probability $1 - \eta_\lambda$. After a hit the firm adjusts if and only if its would-be gap leaves the band $[-\bar{x}, \bar{x}]$, $\bar{x} = \sqrt{2(1 - \eta_\lambda\beta)}\xi$; after a free draw it always adjusts. A firm that adjusts at t faces the same trade-off as a Calvo resetter, since its price survives each period with probability η_λ regardless of its gap: its reset price is the Calvo formula (22) with η_λ in place of α , evaluated along the path with no further hit,

$$(27) \quad p_{it}^{\text{new}} = \omega z_{it} + (1 - \eta_\lambda\beta) \mathbb{E}_t \sum_{s \geq 0} (\eta_\lambda\beta)^s \left[\omega (mc_{t+s} + \vartheta_{i,t+s}) + (1 - \omega) P_{t+s} \right],$$

where z_{it} enters separately because it is frozen until the next hit. Let $\bar{p}_{it}$ and $\bar{p}_{it}^{\text{new}}$ denote the expectations of p_{it} and p_{it}^{new} over the turbulence process (hits and draws), given the paths of mc , P and ϑ_i . Turbulence draws being mean zero,

$$(28) \quad \bar{p}_{it}^{\text{new}} = (1 - \eta_\lambda\beta) \mathbb{E}_t \sum_{s \geq 0} (\eta_\lambda\beta)^s \left[\omega (mc_{t+s} + \vartheta_{i,t+s}) + (1 - \omega) P_{t+s} \right].$$

Aggregation. Consider a firm hit at t without a free draw. Its would-be gap, $y_{it} \equiv p_{i,t-1} - p_{it}^{\text{new}}$, is the gap after its last hit, inside the band, plus the drift of the reset price since then, bounded by m , minus the turbulence draw $\omega \epsilon_{it}^z$, uniform with support width $\varsigma > 4\bar{x} + 2m$: y_{it} is therefore uniform on an interval of width ς containing the band. The firm keeps its

¹⁵This follows from the decision-cost assumption (A4) of Appendix A.2: a firm that is neither hit nor free does not pay the cost and keeps its price. The small-shock assumptions (A1)–(A3) make (A4) consistent, since the gain from reconsidering without a hit is second order while the gain after a hit is not, and keep the band strictly inside the support of the turbulence draw.

price if $|y_{it}| \leq \bar{x}$, probability $2\bar{x}/\varsigma$, and resets otherwise, so

$$(29) \quad \begin{aligned} \mathbb{E}[p_{it} \mid \text{hit}] &= \Pr(\text{adjust} \mid \text{hit}) \mathbb{E}[p_{it} \mid \text{hit}, \text{adjust}] + \Pr(\text{keep} \mid \text{hit}) \mathbb{E}[p_{it} \mid \text{hit}, \text{keep}] \\ &= \left(1 - \frac{2\bar{x}}{\varsigma}\right) p_{it}^{\text{new}} + \frac{2\bar{x}}{\varsigma} \left(p_{it}^{\text{new}} + \mathbb{E}[y_{it} \mid |y_{it}| \leq \bar{x}]\right) = p_{it}^{\text{new}}, \end{aligned}$$

since $\mathbb{E}[y_{it} \mid |y_{it}| \leq \bar{x}] = 0$: non-adjusters are spread symmetrically around the reset price, so the expected price of a hit firm equals its reset price whether or not it adjusts. Free adjusters are at the reset price by construction. Hence

$$(30) \quad \bar{p}_{it} = \underbrace{\lambda \bar{p}_{it}^{\text{new}}}_{\text{free adjusters}} + \underbrace{(1 - \lambda)(1 - \eta) \bar{p}_{it}^{\text{new}}}_{\text{hit, no free draw}} + \underbrace{\eta \lambda \bar{p}_{i,t-1}}_{\text{neither}} = \eta \lambda \bar{p}_{i,t-1} + (1 - \eta \lambda) \bar{p}_{it}^{\text{new}},$$

the Calvo law (23) with hazard $1 - \eta \lambda$ and reset price (28): adjustment is state dependent firm by firm, but in expectation every price behaves as if reset with probability $1 - \eta \lambda$ regardless of its gap. Integrating across firms, $\mathbf{P} = J_{P,p^*}^C(\eta \lambda)[\omega \mathbf{mc} + (1 - \omega) \mathbf{P}]$.

PROPOSITION 3 (Gertler–Leahy). *In the GL economy,*

- (i) *The desired-price pass through Jacobian is Calvo: $J_{P,p^*}^{\text{GL}} = J_{P,p^*}^C(\eta \lambda)$, and hence $J_{\pi,mc^r}^{\text{GL}} = J_{\pi,mc^r}^C(\eta \lambda)$;*
- (ii) *The IRF statistics are exact: $b_h^{\text{GL}} = \omega(1 - \eta \lambda^{h+1})$; $\phi^{\text{IRF,GL}} = \eta \lambda$; and $J_{\pi,mc^r}^{\text{GL}} = \kappa^{\text{IRF,GL}} \Theta_\beta$;*
- (iii) *The virtual hazard $1 - \phi^{\text{IRF,GL}}$ exceeds the actual frequency of adjustment: $\text{freq} = \lambda + (1 - \lambda)(1 - \eta)(1 - 2\bar{x}/\varsigma) < 1 - \eta \lambda$.*

Proof: Appendix B.3.2.¹⁶

Despite the model not being Calvo, the statistics b_0 and $\phi^{\text{IRF,GL}}$ recover the correct structural objects, even if they depend on $\eta \lambda$ as opposed to the actual frequency of adjustment freq . First, $b_0 = \omega(1 - \eta \lambda)$. The shifter innovation moves the desired price by $\omega \epsilon_{it}$ but never triggers a reconsideration; the price moves only on a hit or a free draw, with probability $1 - \eta \lambda$, and in that event the expected price is the reset price whether or not the firm adjusts, by (29), which moves one for one with $\omega \epsilon_{it}$. The expected price change at impact is thus $(1 - \eta \lambda) \omega \epsilon_{it}$. Here, selection means that firms that don't reset had the right price on average, and so the loading of the actual price on the desired price moved by the shock is higher than the actual frequency of adjustment. Second, the statistic $\phi^{\text{IRF,GL}}$ recovers $\eta \lambda$

¹⁶The proof relies on Proposition 1 applying. Note that this is not the case if, instead of the shifter innovation ϵ_{it} , the econometrician regresses on the turbulence innovation ϵ_{it}^z . See discussion in Appendix B.5.5.

and not freq. By Corollary 1, $\omega - b_h = -\text{Cov}(x_{i,t+h}, \epsilon_{it})/\sigma_\theta^2$: the shortfall of pass-through from ω is the part of the shock still sitting in the firm's gap, in expectation over hits and draws. After a hit or a free draw the shock is fully absorbed in expectation, whether or not the firm adjusts; so it survives in the expected gap at horizon h only if neither event has occurred in $h + 1$ periods, probability η_λ^{h+1} . Thus $\omega - b_h = \omega\eta_\lambda^{h+1}$, and $\phi^{\text{IRF, GL}} = \eta_\lambda$.

In the (GL) model, the assumptions ensure that the whole Jacobian is exactly Calvo. However, the insights from this discussion carry over to the general (MC) model: our statistic approximates the true Jacobian even when the economy that generates it is not Calvo, because it measures the survival of the shock in the *expected* price, not the survival rate of realized prices.

3.3.3. Canonical Menu Cost

In the canonical menu cost model, J_{P,p^*}^{MC} does not have the Calvo structure. We proceed in three steps: what b_0 identifies; what the term structure of the micro IRF identifies *exactly*, namely the average survival function of Auclert et al. (2024)'s two-margin representation, of which $\phi^{\text{IRF, MC}}$ is the one-period-ahead value; and why extrapolating this object geometrically, which is all the Calvo structure adds, is accurate.

Policy function. The firm reconsiders its price every period, and its pre-adjustment gap is $x_{i,t-1} - \omega(\epsilon_{it}^z + \epsilon_{it})$. In the steady state the reset gap is $x^* = 0$ and the band is symmetric; the pre-adjustment gap has a stationary density $g(x)$, and the frequency of price adjustment is $\text{freq} = 1 - (1 - \lambda) \int_{-\bar{x}}^{\bar{x}} g(x) dx$.

On-impact pass-through. By Proposition 1 and Appendix B.4.2,

$$(31) \quad b_0^{\text{MC}} = [J_{P,mc}^{\text{MC}} \mathbf{1}]_0 = \omega[\text{freq} + 2(1 - \lambda) g(\bar{x}) \bar{x}].$$

The on-impact pass-through, measured by the regression coefficient b_0^{MC} , encodes the selection effect. Selection is the dependence of the adjustment decision on the gap: unless it gets a free adjustment, a firm adjusts if and only if its gap lies outside the band, so adjusters are the firms with the largest gaps, not a random sample of firms. b_0 contains it mechanically, because the shock displaces every gap and thereby pushes the firms sitting

just inside the edges of the band into adjustment, and the covariance weights these recruited adjusters by their full price change, $\bar{x}$, rather than by the marginal displacement that recruited them. Therefore, the impact pass-through exceeds ω times the frequency (the Calvo-world coefficient).

Term structure of the pass-through matrix. The building block is a time-dependent (TD) model in which a price set at date t is still in place at $t+s$ with probability Φ_s ($\Phi_0 = 1$, Φ_s decreasing):

$$(32) \quad J_{P,p^*}^{\text{TD}}(\Phi) = \frac{1}{(\sum_{s \geq 0} \Phi_s)(\sum_{s \geq 0} \beta^s \Phi_s)} \Upsilon^\Phi \Theta^\Phi,$$

where Υ^Φ is lower-triangular Toeplitz with entries Φ_{t-s} and Θ^Φ is upper-triangular Toeplitz with entries $\beta^{s-t} \Phi_{s-t}$; the Calvo model is the case $\Phi_s = \alpha^s$, for which $J_{P,p^*}^{\text{TD}}(\Phi) = J_{P,p^*}^{\text{C}}(\alpha)$. Auclert et al. (2024) prove that the desired-price pass-through of the canonical menu cost model is a convex combination of two TD Jacobians,

$$(33) \quad J_{P,p^*}^{\text{MC}} = \varrho_e J_{P,p^*}^{\text{TD}}(\Phi^e) + \varrho_i J_{P,p^*}^{\text{TD}}(\Phi^i), \quad \varrho_e + \varrho_i = 1, \quad J_{P,mc}^{\text{MC}} = \omega J_{P,p^*}^{\text{MC}},$$

whose *virtual survival functions* Φ^e and Φ^i encode the extensive margin (movements of the adjustment bands) and the intensive margin (movements of the reset price).¹⁷

LEMMA 2 (TD partial sums). *For any TD model, $[J_{P,p^*}^{\text{TD}}(\Phi)\mathbf{1}]_h = \sum_{s=0}^h \Phi_s / \sum_{s \geq 0} \Phi_s$.*

The permanent-shock IRF of a TD model is the normalized partial sum of its survival function. Applying Lemma 2 to the mixture (33) and using Proposition 1, $b_h = \omega [J_{P,p^*}^{\text{MC}}\mathbf{1}]_h$,

$$(34) \quad b_h^{\text{MC}} = b_0^{\text{MC}} \sum_{s=0}^h \bar{\Phi}_s, \quad \bar{\Phi}_s \equiv m_e \Phi_s^e + m_i \Phi_s^i$$

with weights $m_e = 2(1 - \lambda)g(\bar{x})\bar{x}/(b_0/\omega)$ and $m_i = \text{freq}/(b_0/\omega)$ summing to one.

¹⁷Let $E^s(x)$ denote the steady-state expectation of the gap of a firm s periods ahead, given a gap x today ($E^0(x) = x$; the firm may adjust in between). Then $\Phi_s^e = E^s(\bar{x})/\bar{x}$, the surviving fraction of a marginal gap that starts at the adjustment threshold, and $\Phi_s^i = E^{s'}(0)$, the sensitivity of the expected future gap to the current gap at the reset point; the mixture weights are $\varrho_e = 2(1 - \lambda)g(\bar{x})\bar{x} \sum_s \Phi_s^e$ and $\varrho_i = \text{freq} \sum_s \Phi_s^i$. See Auclert et al. (2024).

Identification from the term structure. The shape of the micro IRF identifies the average virtual survival function $\bar{\Phi}_h$ horizon by horizon:

$$(35) \quad \frac{b_h^{\text{MC}} - b_{h-1}^{\text{MC}}}{b_0^{\text{MC}}} = \bar{\Phi}_h, \quad h \geq 1.$$

In particular, our statistic recovers a structural object of the canonical menu cost model exactly:

$$(36) \quad \phi^{\text{IRF,MC}} = \frac{b_1^{\text{MC}}}{b_0^{\text{MC}}} - 1 = \bar{\Phi}_1,$$

the average one-period survival of a marginal price gap. What the Calvo structure adds is only the geometric extrapolation of $\bar{\Phi}_1$ to later horizons, to which we turn below.

Like $b_0^{\text{MC}}, \phi^{\text{IRF,MC}}$ internalizes selection effects. It is the one-period survival of a marginal *gap*, not of a price spell: a firm that carries the shock past impact has a larger gap and is disproportionately likely to be pushed across a threshold next period, so the shock content of prices mean-reverts faster than price spells end, and $\phi^{\text{IRF,MC}}$ falls short of $1 - \text{freq}$ (Appendix B.4.2). The ratio reads the survival of the shock in the expected price—the object the Jacobian is built from—which selection accelerates; the frequency counts spells, which selection leaves untouched. The gap between the two, $(1 - \text{freq}) - \phi^{\text{IRF,MC}}$, is therefore a model-free measure of the strength of the selection effect, a feature we document numerically below.

This exact identification result carries over to the (GMC) case with an arbitrary distribution of menu costs: the micro IRF still identifies the average one-period survival exactly at horizon one (Appendix A.3).

The Calvo extrapolation. Our IRF statistics assembles the generalized Phillips curve $J_{\pi,mc}^{\text{IRF,MC}} = J_{\pi,mc}^{\text{C}}(\phi^{\text{IRF,MC}}) = \kappa^{\text{IRF,MC}} \Theta_\beta$ with first entry $\kappa^{\text{IRF,MC}} = b_0^{\text{MC}}(1 - \beta\phi^{\text{IRF,MC}})/\phi^{\text{IRF,MC}}$. It takes the identified pair $([J_{P,mc}^{\text{MC}} \mathbf{1}]_0, \bar{\Phi}_1)$ and extrapolates the survival function geometrically, $\bar{\Phi}_s \approx (\phi^{\text{IRF,MC}})^s$. The extrapolation involves two approximations:

- (i) *Along the permanent-shock IRF, horizons $h \geq 2$.* With the levels b_0^{MC} and b_1^{MC} matched, the error is

$$b_0^{\text{MC}} \sum_{s=2}^h (\bar{\Phi}_s - (\phi^{\text{IRF,MC}})^s),$$

the geometric extrapolation set against the true average survival function. It is small because, as Auclert et al. (2024) document numerically, the virtual survival functions of the two margins converge to the same constant and deviate from it in *opposite* directions—the extensive margin from above, the intensive margin from below—so the deviations roughly offset in the average $\bar{\Phi}$, leaving it close to geometric. This approximation is also testable: $\bar{\Phi}_h = (\phi^{\text{IRF,MC}})^h$ at $h \geq 2$ is a geometric-survival restriction, and estimated deviations from it measure the extrapolation error directly.

- (ii) *From the IRF to the matrix.* When each margin’s survival function is close to the *same* geometric—which is what the offsetting virtual survival functions deliver in practice—the shared $\Upsilon^\Phi \Theta^\Phi$ architecture of (32) transfers the proximity to every entry of the Jacobian.

A bound. For an infinite matrix A and a reporting horizon $H \geq 1$, let

$$(37) \quad \|A\|_H \equiv \max_{0 \leq t \leq H} \sum_{s \geq 0} |A[t, s]|$$

denote the maximum absolute row sum over the first $H + 1$ rows; it satisfies $|[A\mathbf{1}]_t| \leq \|A\|_H$ for $t \leq H$.

ASSUMPTION 1. *There exist $\alpha^* \in (0, 1)$ and $\bar{e} \geq 0$ with $\|J_{P,p^*}^{\text{MC}} - J_{P,p^*}^{\text{C}}(\alpha^*)\|_H \leq \bar{e}$ and $\bar{e} < 1 - \alpha^*$.*

The assumption says that *some* Calvo desired-price pass-through Jacobian is close to that of the menu cost model, the numerical finding of Auclert et al. (2024). Define the approximation error $E \equiv J_{P,p^*}^{\text{MC}} - J_{P,p^*}^{\text{C}}(\alpha^*)$. The proposition below then bounds the error of the Calvo Jacobian assembled from $\phi^{\text{IRF,MC}}$.

PROPOSITION 4 (Menu cost). *Under the conditions of Proposition 1 and Assumption 1:*

- (i) *the desired-price pass-through Jacobian satisfies*

$$(38) \quad \|J_{P,p^*}^{\text{MC}} - J_{P,p^*}^{\text{C}}(\phi^{\text{IRF,MC}})\|_H \leq \bar{e} + M_H(I_\alpha) \frac{(2 + \alpha^*)\bar{e}}{1 - \alpha^* - \bar{e}} \equiv B_H,$$

where $I_\alpha \equiv [\min\{\alpha^*, \phi^{\text{IRF,MC}}\}, \max\{\alpha^*, \phi^{\text{IRF,MC}}\}]$ and $M_H(I_\alpha) \equiv \sup_{\alpha \in I_\alpha} \|\partial J_{P,p^*}^{\text{C}}(\alpha)/\partial \alpha\|_H$, a finite number determined by β , H and I_α whenever $I_\alpha \subset (0, 1)$; hence the nominal Jacobians satisfy $\|J_{P,mc}^{\text{MC}} - \omega J_{P,p^*}^{\text{C}}(\phi^{\text{IRF,MC}})\|_H \leq \omega B_H$ with the true ω ;

(ii) on the system truncated at the reporting horizon H , with $\|\cdot\|$ the maximum-absolute-row-sum norm (submultiplicative there) and the truncated $I - J_{P,p^*}^{\text{MC}}$ invertible, the generalized Phillips curves built from (12)–(13) with the true real rigidity ω satisfy

$$(39) \quad \|J_{\pi,mc^r}^{\text{MC}} - J_{\pi,mc^r}^{\text{C}}(\phi^{\text{IRF,MC}})\| \leq 2\omega \|(I - J_{P,p^*}^{\text{C}}(\phi^{\text{IRF,MC}}))^{-1}\| \|(I - J_{P,p^*}^{\text{MC}})^{-1}\| B_H;$$

Proof. See Appendix B.4.3. □

Part (ii) deserves a comment. Every entry of J_{π,mc^r}^m depends on the *whole* pass-through matrix: $(I - J_{P,p^*}^m)^{-1} J_{P,p^*}^m = \sum_{k \geq 1} (J_{P,p^*}^m)^k$, and the entries of $(J_{P,p^*}^m)^k$ sum over round trips through all dates—the wage–price spiral, in which prices move desired prices at every date, through nominal costs $mc = mc^r + P$ with weight ω and directly with weight $1 - \omega$, and those desired prices move prices again. The amplification factor $\|(I - J_{P,p^*}^{\text{C}}(\phi^{\text{IRF,MC}}))^{-1}\| \|(I - J_{P,p^*}^{\text{MC}})^{-1}\|$ in (39) measures the strength of the wage–price spiral: it is large when prices are flexible, and it grows with the reporting horizon, since a permanent movement in real marginal cost feeds inflation indefinitely. At the Phillips-curve level the Calvo approximation is therefore most reliable in sticky-price economies and over moderate horizons.

4. Performance of Robust Statistic Approximation in the Canonical Menu Cost Model

This section presents numerical results on the performance of our method for estimating the degree of monetary non-neutrality in menu cost models.

4.1. Simulation Set-Up

Simulations. From the solution of the stationary problem of the menu cost model, we perform a Monte Carlo simulation to estimate the sufficient statistic. For any given parameterization, the simulation takes as inputs the vector of discretized price gaps x_{grid} , the stationary distribution over price gaps g , the transition matrix of price gaps conditional on no adjustment Q_{not} , the adjustment-probability function $q(x)$, and the target price gap x^* . It then simulates a large cross section of N firms drawn from g , generates a sequence of latent desired prices p_{it}^* using Gaussian shocks ε_{it} , and assembles firm-level pre-adjustment

gaps $x_{it}^{\text{pre}} = p_{i,t-1} - p_{it}^*$. Finally, it simulates adjustment from Bernoulli draws according to $q(x_{it}^{\text{pre}})$: if the firm adjusts, it sets $p_{it} = x^* + p_{it}^*$; if it does not adjust, it keeps $p_{it} = p_{i,t-1}$.

With this simulation at hand, we first discard a burn-in period, and then compute the statistic defined in Section 3.1: we estimate the $h \in \{0, 1\}$ local projection (14) of price changes on marginal-cost shocks, infer the micro IRF-implied survival rate ϕ^{IRF} using (15), and compute the robust-statistic Calvo Jacobian $J_{\pi, mc}^{\text{IRF}}$ as (17). In addition, from the solution of the model, we compute the true generalized Phillips curve $J_{\pi, mc}^{\text{MC}}$.

Parameterizations. The literature has considered different parameterizations of this general model, with two widely-used calibrations in Golosov and Lucas (2007) and Nakamura and Steinsson (2010). We fix a number of parameters, as detailed in Table 1. We replicate the model-based moments targeted by these two papers, conditional on the parameter values fixed in Table 1, and report the resulting calibrations of the other parameters in Table 2.

TABLE 1. Key parameters of the menu cost model

Parameter	notation	value
Discount factor	β	0.99
Elasticity of substitution across varieties	ε	6
Returns to scale	γ	1
Std. of idiosyncratic shocks	σ_{ϑ}	calibrated
Std. of menu cost	σ_k	0.05
Frequency of free price adjustments	λ	calibrated
Average menu cost	μ_k	calibrated

TABLE 2. Alternative parameterizations of the pricing frictions.

Parameter	Golosov and Lucas	Nakamura and Steinsson
μ_k	0.006	0.051
λ	0.000	0.179
σ_{ϑ}	0.046	0.06

Beyond these two standard parametrizations, we exploit the flexibility of the CalvoPlus model and show the performance of our approach across a wide range of calibrations. We solve 113 menu cost calibrations that span the mean menu cost, idiosyncratic marginal-cost volatility, and free-adjustment probability. The mean menu cost ranges logarithmically from 0.002 to 0.08, σ_{ϑ} ranges linearly from 0.02 to 0.12, and λ ranges linearly from 0 to 0.30,

with five values for each parameter. All other parameters are held at their benchmark values. We only keep parameterizations where the quarterly frequency of price changes is at least 14%, keeping 113 out of 125 possible combinations.

Note that because we consider Gaussian shocks as a baseline, the distinction in equations (8)-(9) between the Gaussian identifying shock ϑ and the potentially non-Gaussian unobserved shock z loses content, and we keep ϑ only.

Metrics. We define two main metrics to assess the quality of our numerical approximation of the generalized Phillips curve. First, we consider the first entry of the Phillips curve Jacobian, which is sufficient to capture the on-impact inflationary effects of purely transitory marginal cost changes. Second, we consider the Barnichon and Mesters (2021) Phillips multiplier statistic—the ratio between the present discounted values of the inflation and output impulse responses functions. This metric summarizes the generalized Phillips curve at all horizons and has the advantage of isolating the properties of the supply block of the model in generating inflationary pressures after a demand-induced change in output. It requires to solve the model in general equilibrium, and hence to specify an IS curve and a monetary policy rule. We use standard formulations, detailed in Appendix A.4.

Benchmarks. Across these metrics, we compare our estimate of the generalized Phillips curve assembled from the IRF-implied survival rate $J_{\pi,mc^r}^{\text{IRF}} = J_{\pi,mc^r}^{\text{C}}(\phi^{\text{IRF}})$ to the true generalized Phillips curve J_{π,mc^r}^{MC} . We additionally consider two other Calvo approximations to J_{π,mc^r}^{MC} as benchmarks.

First, we compute the Calvo survival rate ϕ^{FI} that best approximates the entire true Jacobian J_{π,mc^r}^{MC} , as defined by Auclert et al. (2024), and consider the Calvo approximation $J_{\pi,mc^r}^{\text{FI}} = J_{\pi,mc^r}^{\text{C}}(\phi^{\text{FI}})$. Computing ϕ^{FI} requires knowing the true Phillips curve Jacobian, i.e., knowing the structure of the model, we hence call it the full-information benchmark.

Second, we consider the feasible Calvo approximation of the generalized Phillips curve proposed by Auclert et al. (2024) by relying on the Alvarez, Le Bihan, and Lippi (2016) sufficient statistic. Because the Calvo model provides a close approximation to the menu cost model, it must approximately have the same cumulative impulse response of output, and hence, the same ratio of kurtosis to frequency: $\frac{\text{Kur}^{\text{C}}}{\text{freq}^{\text{C}}} = \frac{\text{Kur}^{\text{MC}}}{\text{freq}^{\text{MC}}}$. In addition, in the

discrete-time Calvo model, $\text{Kur}^C = 3(2 - \text{freq}^C)$. Hence, upon measuring the true ratio $\frac{\text{Kur}^{\text{MC}}}{\text{freq}^{\text{MC}}}$ in the data, one can infer a virtual Calvo survival probability ϕ^{Kur} that would yield this ratio in a Calvo world, namely:

$$(40) \quad \phi^{\text{Kur}} = 1 - \frac{1}{\frac{1}{6} \frac{\text{Kur}^{\text{MC}}}{\text{freq}^{\text{MC}}} + \frac{1}{2}},$$

We consider the associated Calvo approximation of the Phillips curve Jacobian $J_{\pi, mc^r}^C(\phi^{\text{Kur}})$ as a second benchmark.

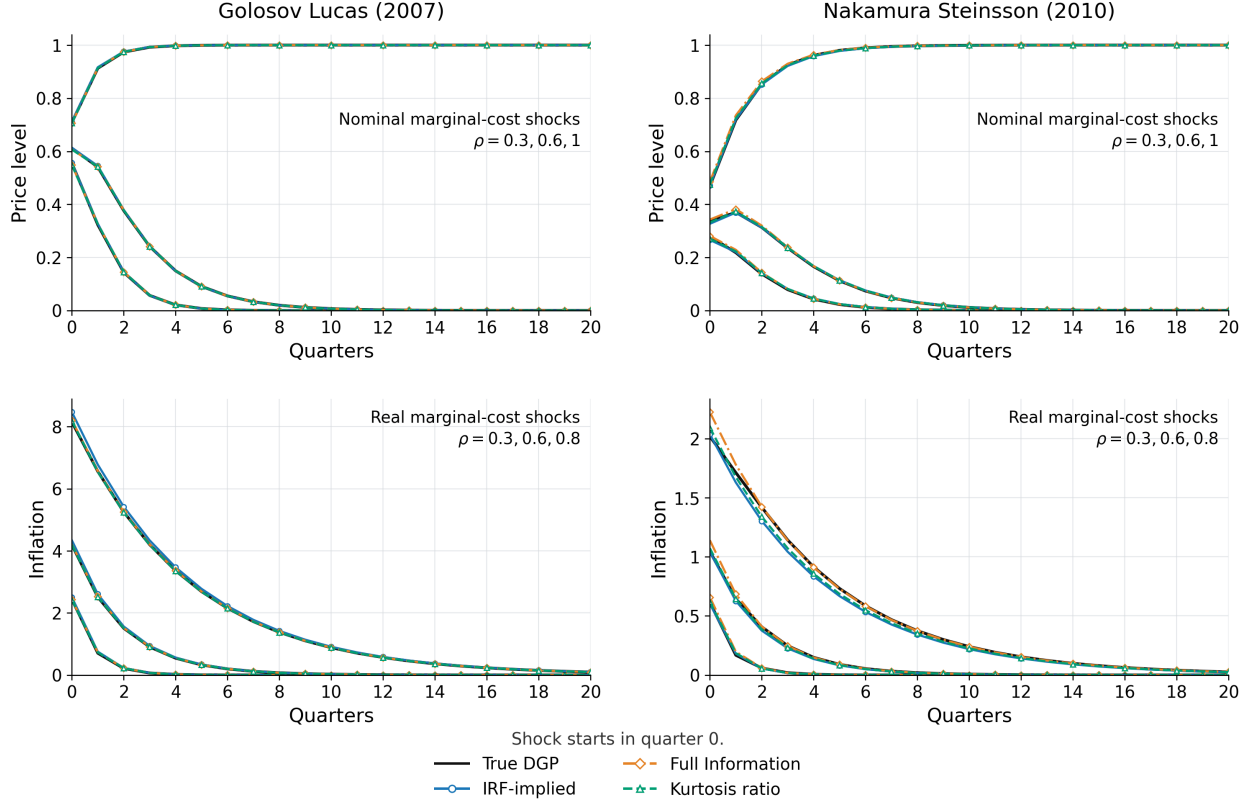
4.2. Results

Approximating the menu cost generalized Phillips curve. Figure 1 documents our key result: our micro IRF-implied approximations of the nominal passthrough and Phillips curve Jacobians generate aggregate impulse responses to arbitrary nominal and real marginal cost shocks that are virtually identical to the true ones. The top row reports the response of the price level to a nominal marginal cost shock, which the pass-through matrix $J_{P, mc}^{\text{MC}}$ determines; the bottom row reports the response of inflation to a real marginal cost shock, which the generalized Phillips curve $J_{\pi, mc^r}^{\text{MC}}$ determines. The left column uses the Golosov–Lucas calibration and the right column the Nakamura–Steinsson calibration. Each panel shows three shocks of different persistence. The black lines are the true responses, computed from $J_{P, mc}^{\text{MC}}$ and $J_{\pi, mc^r}^{\text{MC}}$; the blue lines are the responses predicted by the Calvo approximations built from the micro IRF, $J_{P, mc}^{\text{IRF}}$ and $J_{\pi, mc^r}^{\text{IRF}}$. Across calibrations and persistence levels, the approximated responses follow the true ones closely. Appendix Figures C.3 and C.4 show the same for anticipated shocks.

The close match follows from the fact that the micro IRF-implied Calvo approximation reproduces the entire nominal passthrough and Phillips curve Jacobians very well, and that these matrices suffice for the first-order dynamics of prices. Appendix Figures C.1 and C.2 emphasize this point by showing selected columns of the true and approximated nominal passthrough and Phillips curve Jacobians.

The figure also reports two benchmarks: the full-information Calvo approximation, which selects the survival rate from the entire true Jacobian (orange), and the kurtosis-implied approximation (green). The IRF-implied responses are as close to the true ones as the full-information responses, although the latter has a much higher informational

FIGURE 1. Responses to Nominal and Real Marginal-Cost Innovations



Notes: The left column uses the Golosov–Lucas (2007) calibration and the right column the Nakamura–Steinsson (2010) calibration. The top row reports price-level responses to nominal marginal-cost paths $(1, \rho, \rho^2, \dots)$ for $\rho \in \{0.3, 0.6, 1\}$; $\rho = 1$ is a permanent nominal innovation. The bottom row reports inflation responses to real marginal-cost paths with $\rho \in \{0.3, 0.6, 0.8\}$. Black denotes the true DGP, blue the IRF-implied approximation, orange Full Information, and green the Kurtosis ratio. Within each color, the three paths correspond to the persistence values listed inside the panel.

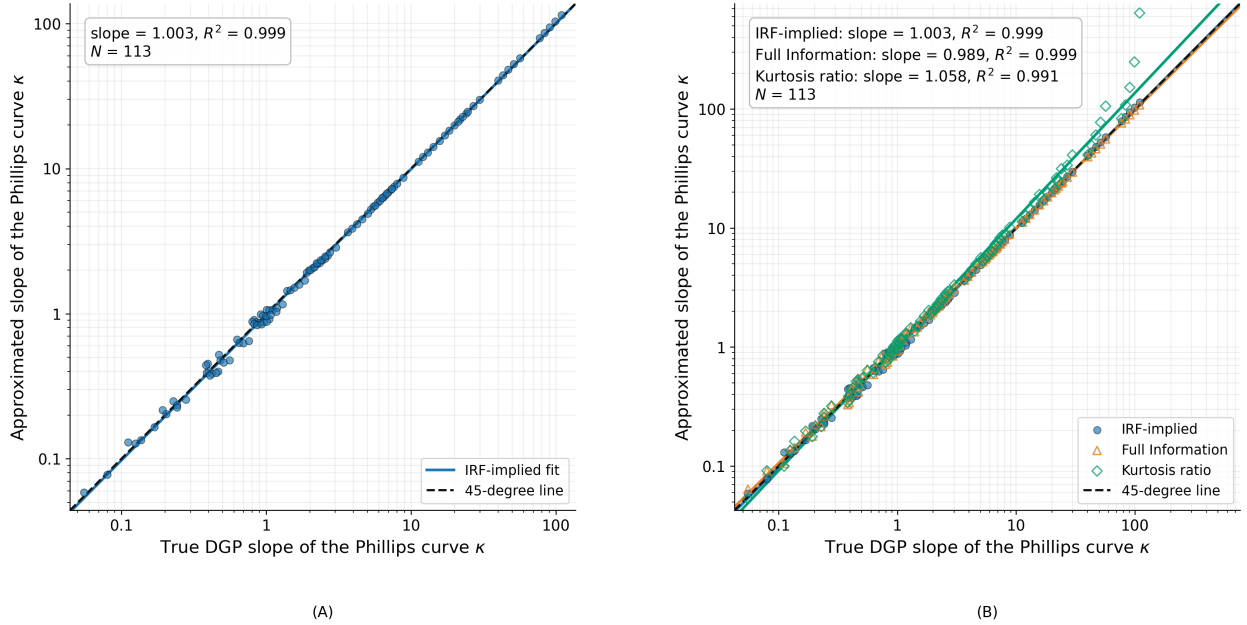
burden. The kurtosis-implied responses are equally close in this exercise, in which $\omega = 1$ and shocks are Gaussian, the assumptions under which the kurtosis ratio is valid; the results below show how they diverge once these assumptions are relaxed.

To move beyond these two parameterizations and show that our approach more generally approximates the true Phillips curve Jacobian, we move to the two summary metrics introduced above: the first entry of the Phillips curve Jacobian (the slope of the Phillips curve), and the Phillips multiplier.

Figure 2(A) shows our approximation of the slope of the Phillips curve $J_{\pi, mc^r}^{\text{IRF}}[0, 0]$ against its true counterpart $J_{\pi, mc^r}^{\text{MC}}[0, 0]$ across calibrations. Our approximation is highly accurate across parameterizations. The slope between the true slope and our approximation is 1.003, and the R^2 of the relation between the two objects is 0.999. The underlying

DGPs used for these statements have a wide range of inflation responses to a purely transitory shocks, from 0.06 to 100. Panel (B) adds to the figure the full-information and kurtosis ratio Calvo approximations. Our approximation performs as well as the full-information one.

FIGURE 2. The Slope of the Phillips Curve: $J_{\pi,mc^r}^{MC}[0, 0]$

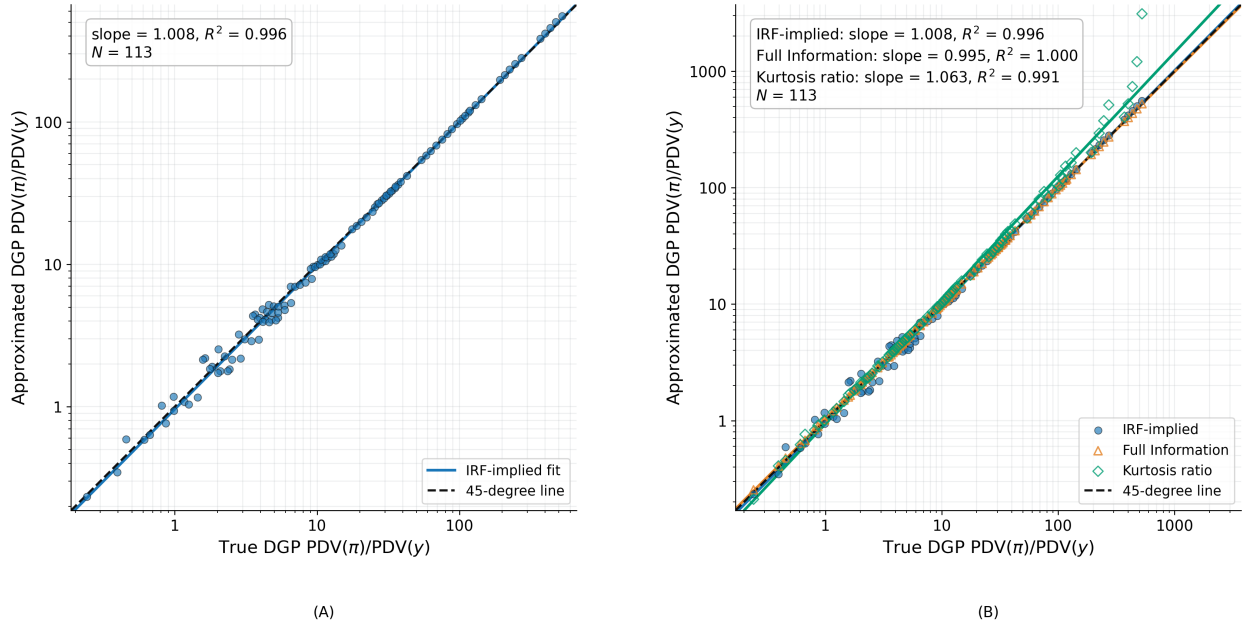


Note: The x-axis reports the first entry of the true generalized Phillips curve entry $J_{\pi,mc^r}^{MC}[0, 0]$, referred to as the slope of the Phillips curve. The y-axis reports the same entry in various approximations of J_{π,mc^r}^{MC} . Panel A reports $J_{\pi,mc^r}^{IRF}[0, 0]$, the micro IRF-implied slope of the Phillips curve. Panel B shows $J_{\pi,mc^r}^{IRF}[0, 0]$ (blue circles) and adds two other Calvo approximations: the full-information approximation $J_{\pi,mc^r}^{FI}[0, 0]$ (orange triangles) and the kurtosis-implied approximation $J_{\pi,mc^r}^{Kur}[0, 0]$ (green diamonds). Each panel uses the same 113 CES calibrations with quarterly adjustment frequencies above 14% (observed range 15.3–79.9%). Both axes are logarithmic. Solid lines are fitted regressions in logs, and the black dashed line denotes exact agreement.

To investigate the performance of our approximation beyond purely transitory shocks, we show results for the Phillips multiplier $PDV(\Pi)/PDV(Y)$. Figure 3(A) shows that the robust statistic approach recovers the correct Phillips multiplier across a wide range of parameterizations. The slope between the true and approximated ratios is 1.008 and an $R^2 = 0.996$. Panel (B) adds the same ratio as implied by the full-information and the kurtosis-implied Calvo approximations. The results are again very similar.

Implied survival rates. A natural skepticism around the results is that our robust statistic is accurate because the calibrations we use imply the model behaves very much like a

FIGURE 3. Phillips Multiplier



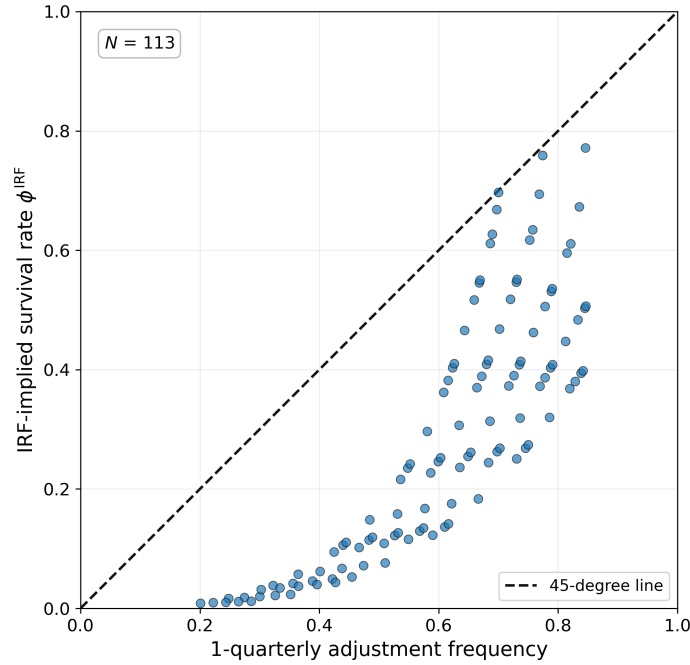
Note: The x-axis reports the Phillips multiplier ($PDV(\pi)/PDV(y)$) computed in a model with the true generalized Phillips curve entry $J_{\pi,mc}^{MC}$. The y-axis reports the same metric for various approximations of $J_{\pi,mc}^{MC}$. Panel A reports the Phillips multiplier obtained using $J_{\pi,mc}^{IRF}$, the micro IRF-implied generalized Phillips curve. Panel B shows the Phillips multiplier implied by $J_{\pi,mc}^{IRF}$ (blue circles) and adds two other Calvo approximations: the full-information approximation using $J_{\pi,mc}^{FI}$ (orange triangles) and the kurtosis-implied approximation using $J_{\pi,mc}^{Kur}$ (green diamonds). Each panel uses the same 113 CES calibrations with quarterly adjustment frequencies above 14% (observed range 15.3–79.9%). Both axes are logarithmic. Solid lines are fitted regressions in logs, and the black dashed line denotes exact agreement.

Calvo model. This is not the case. To illustrate that, we plot our IRF-implied Calvo survival rate and compare it with the unconditional moment $1 - \text{freq}$.

The results are presented in Figure 4. If the models were close to Calvo, the ϕ^{IRF} would recover $1 - \text{freq}$, and all points would lie on top of the 45 degree line. Instead, all points fall below the 45 degree line: the extent of price rigidity that rationalizes the Jacobian is lower than implied by $1 - \text{freq}$. This is because the selection effect implies that, for a given mass of adjusters, prices are more flexible under the menu cost model than under the Calvo model. Some parameterizations are close to the diagonal since the CalvoPlus model nests the Calvo model. The vertical distance between a particular point and the 45 degree line is a model-free measure of the selection effect.

In sum, the goodness of fit of our robust statistic is not by construction, but rather a reflection of the information contained in the dynamic pass-throughs.

FIGURE 4. Micro IRF-implied Survival Rate and the Frequency of Price Changes



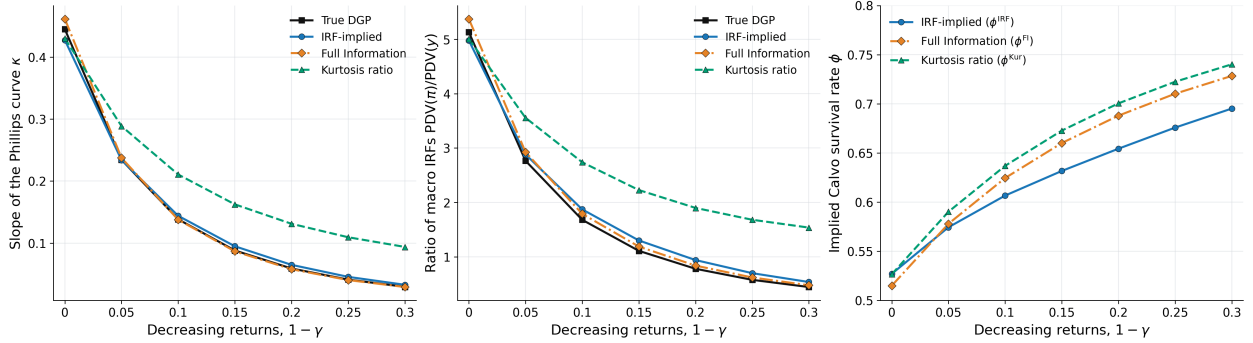
Note: The vertical axis reports the micro IRF-implied survival rate ϕ^{IRF} , inferred from the local projection coefficients b_0 and b_1 . The horizontal axis reports the frequency-based Calvo value, $1 - \text{freq}$, where freq is the unconditional quarterly frequency of price adjustment. Each point is one of 113 CES menu-cost calibrations with $\text{freq} > 0.14$. The dashed line is the Calvo benchmark, under which the two quantities coincide.

Robustness to micro-level real rigidities. Our baseline calibration follows Golosov and Lucas (2007) and Nakamura and Steinsson (2010) and assumes constant returns to scale, $\gamma = 1$. We now show that our robust statistic approach performs equally well with micro-level real rigidities. Figure 5 shows the performance of our robust statistic for approximating the slope of the Phillips curve (left) or the Phillips multiplier (middle), for values of $1 - \gamma$ ranging from 0 (constant returns) to 0.3 (returns to scale equal to 0.7).

Our robust statistic (in blue) tracks the true object (in black) extremely well across calibrations of returns to scale. By contrast, the kurtosis-implied Calvo approximation (dashed green line) deviates from the true object once we depart from constant returns. This is because moments from the unconditional distribution of price changes cannot tell apart strong real rigidity with large cost shocks and weak real rigidity with small cost shocks, while they have different implications for the Phillips curve. The existing sufficient statistics solve this identification problem by making structural assumptions, like constant returns and Gaussian shocks, with the risk for model mis-specification, as exem-

plified here.

FIGURE 5. Performance of the Robust Statistic under Decreasing Returns to Scale



Notes: This figure shows the performance of our approximation of the generalized Phillips curve as returns to scale γ vary. The x-axis is $1 - \gamma$. We report: the slope of the Phillips curve $J_{\pi,mc^r}^m[0, 0]$ (left), the Phillips multiplier (middle), and the Calvo survival rate ϕ (right). Each panel compares the true menu cost model J_{π,mc^r}^{MC} with three Calvo approximations $J_{\pi,mc^r}^C(\phi)$, which use the IRF-implied survival rate ϕ^{IRF} , the full-information survival rate ϕ^{FI} , and the kurtosis-implied survival rate ϕ^{Kur} . The right panel has no true model object. All other parameters are held at the Nakamura and Steinsson (2010) calibration.

5. Extensions

5.1. Beyond CES: The Kimball Extension

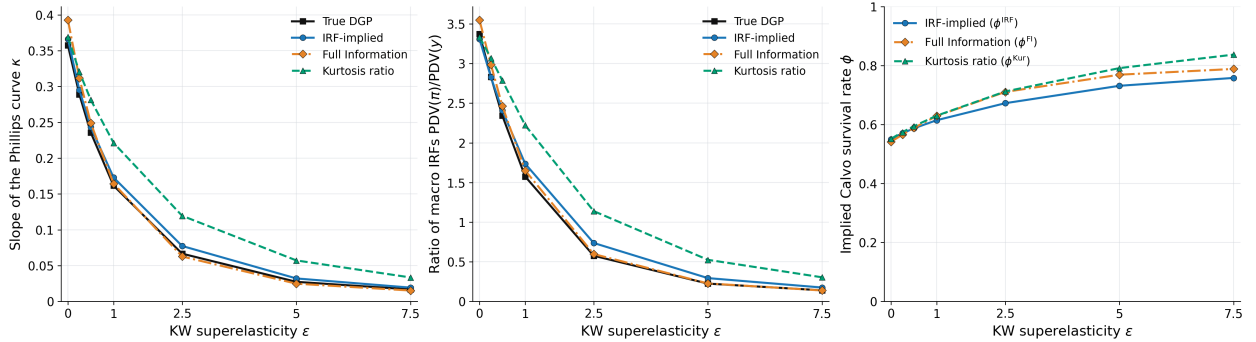
Strategic complementarities change the extent to which firms want to change their prices in the flexible price equilibrium, the incentives that firms have to pay the menu costs, altering the extensive and intensive margins of adjustment, and the size of money non-neutralities. In this section, we document that non-isoelastic demand curves microfounded by a Kimball demand system do not affect the performance of the robust statistic.

As we have shown before, when the true DGP is Calvo, the robust statistic is exact and inclusive of accounting for micro-level real rigidities since the pass-through at every horizon is multiplicative on the flexible price equilibrium desired pass-through ω . Therefore, with strategic complementarities, the formula for ω changes and depends on the superelasticity of demand, but the insight that two entries of the micro IRF suffice to identify the generalized Phillips curve remain.

Figure 6 shows the performance of our approximation when varying the superelasticity of demand away from the CES case. The other parameters are fixed at the Nakamura and Steinsson (2010) parameterization of the menu cost model. The zero superelasticity

corresponds to the CES case we have discussed already. The left panel shows the slope of the Phillips curve that the three Calvo approximations (micro IRF-implied, full information, and kurtosis-implied) infer, as well as the true value. The middle panel shows the same results for the Phillips multiplier. The right panel shows the virtual survival rate inferred by the three approximation methods. Away from the CES case, the IRF-implied approximation continues to track the true slope of the Phillips curve and Phillips multiplier. By contrast, the kurtosis-implied statistic overshoots for both objects and hence infers too much monetary neutrality.

FIGURE 6. Performance of the Robust Statistic under Non-Isoelastic Demand Curves



Notes: This figure shows the performance of our approximation of the generalized Phillips curve as the Klenow–Willis demand superelasticity ε varies. We report: the slope of the Phillips curve $J_{\pi,mc^r}^m[0, 0]$ (left), the Phillips multiplier (middle), and the Calvo survival rate ϕ (right). Each panel compares the true menu cost model J_{π,mc^r}^{MC} with three Calvo approximations $J_{\pi,mc^r}^C(\phi)$, which use the IRF-implied survival rate ϕ^{IRF} , the full-information survival rate ϕ^{FI} , and the kurtosis-implied survival rate ϕ^{Kur} . The right panel has no true model object. All other parameters are held at the Nakamura and Steinsson (2010) calibration.

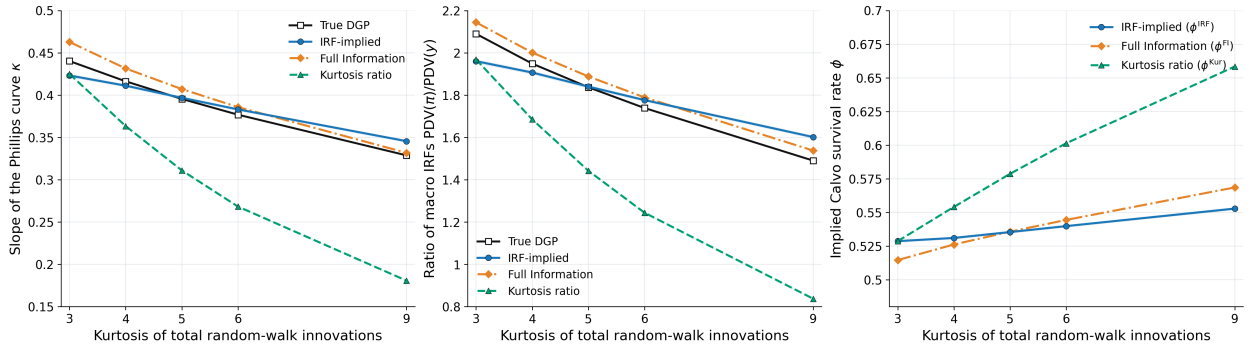
5.2. Leptokurtic Shocks

A large literature has emphasized that the distribution of the idiosyncratic shocks that move firms' desired prices matters for monetary non-neutrality, as fat-tailed shocks weaken the selection effect (Midrigan 2011; Karadi and Reiff 2019). This raises two difficulties. First, the distribution of shocks cannot be recovered from the distribution of price changes without imposing some structure on the policy function that maps one into the other. Second, existing sufficient statistics such as Alvarez, Le Bihan, and Lippi (2016) and its successors require Gaussian shocks. We now show that our method performs similarly well when fat-tailed shocks hit firms' desired prices.

To do so, we reinstate the distinction of equations (9)-(8) between z_{it} , a generic shock affecting firms' desired price, and ϑ_{it} , which the econometrician uses as the regressor in

the local projection (14). We take z_{it} to follow a two-component Gaussian mixture.¹⁸ ϑ_{it} remains Gaussian so that Proposition 1 applies. Holding the variance of the total innovation fixed, we vary its kurtosis from 3, the Gaussian benchmark, to 9. Figure 7 reports the results. As the tails of the driving shock thicken, the true slope of the Phillips curve and the true Phillips multiplier fall: this is the weakening of the selection effect. Our IRF-implied statistics track the true objects at every kurtosis, with no systematic relation between the kurtosis of the shock and the approximation error. By contrast, the kurtosis ratio-implied Phillips curve rapidly loses informativeness as kurtosis exceeds the Gaussian benchmark. Indeed, the kurtosis of shocks directly affects the kurtosis of price changes, so that the kurtosis-to-frequency ratio does not pin down the policy function, and hence the degree of non-neutrality.

FIGURE 7. Performance of the Robust Statistic with Leptokurtic Shocks



Notes: This figure shows the performance of our approximation of the generalized Phillips curve as we vary the kurtosis of the innovations to firms' desired price. The total innovation to firms' desired price is $z_{it} + \vartheta_{it}$, with the two shocks independent, and where we keep the variance of the total innovation fixed at $\sigma^2 = 0.06^2$. ϑ_{it} is the shock used by the econometrician to estimate the local projection (14); it is Gaussian with variance $\sigma^2/2$. z_{it} is a mixture of two zero-mean Gaussians, drawn with probabilities 0.9 and 0.1, whose variances are set so that the total innovation has variance σ^2 and kurtosis $K \in \{3, 4, 5, 6, 9\}$ (details in Appendix C.3). The x-axis reports K , the kurtosis of the total innovation. We report: the slope of the Phillips curve $J_{\pi,mc^r}^m[0, 0]$ (left), the Phillips multiplier (middle), and the Calvo survival rate ϕ (right). Each panel compares the true menu cost model J_{π,mc^r}^{MC} with three Calvo approximations $J_{\pi,mc^r}^C(\phi)$, which use the IRF-implied survival rate ϕ^{IRF} , the full-information survival rate ϕ^{FI} , and the kurtosis-implied survival rate ϕ^{Kur} . The right panel has no true model object. All other parameters are held at the Nakamura and Steinsson (2010) calibration.

We next turn to the case in which Proposition 1 fails: the data are generated by the menu cost model and the identifying shock ϑ_{it} is not Gaussian. Figure C.5 shows the failure. As the kurtosis of the identifying shock rises above 3, the OLS local projection no longer recovers the correct average of local responses, and the IRF-implied slope of the Phillips

¹⁸Appendix C.3 explains in detail how we build the leptokurtic shocks.

curve and Phillips multiplier drift away from their true values.¹⁹ Appendix B.5.4 details the solution: the correct estimator is an IV regression of the cumulative price change on ϑ_{it} with instrument $s(\vartheta_{it})$ where $s(\vartheta)$ is the score of the shock's density.²⁰ Figure C.7 evaluates the performance of this solution numerically in two cases: the econometrician knows the true distribution and hence the score function, or the econometrician has to estimate the score function from the data. In both cases, the corrected slope and Phillips multiplier are indistinguishable from their true values.

5.3. Measurement Error

Measurement error in prices is a challenge for the measurement of monetary non-neutrality starting from unconditional micro price data. First, measurement error biases frequency of price change measures towards one and affects higher moments of price changes as well, crucially the kurtosis of price changes. Under classical Gaussian measurement error that is sufficiently large with respect to the standard deviation of true prices, the measured frequency of price changes goes to 1, and due to normality in measurement error, the kurtosis goes to 3. The question we answer in this section is the comparative behavior of our statistic and the Alvarez, Le Bihan, and Lippi (2016) statistic under measurement error.

We introduce measurement error as an intermediate step between model simulated data and the data that an econometrician observes. In particular, an econometrician observes $p_{it}^{obs} = p_{it} + me_{it}$, where the measurement error term me_{it} is i.i.d and Gaussian with standard deviation σ_{me} . Since the moments the literature uses are about price changes, then the relevant volatility of the measurement error is that of Δme_{it} which is equal to $\sqrt{2}\sigma_{me}$. To ease the interpretation of measurement error we present it as a ratio of the volatility of productivity shocks ϑ_{it} , which we denote σ_{ϑ} . For a given panel of prices $\{p_{it}\}_{\forall i,t}$, we generate a large number of samples of measurement error of varying volatility, from values ranging $\sqrt{2}\sigma_{me}/\sigma_{\vartheta} \in [0, 50\%]$. We generate 10%-90% confidence bands from the set of samples to generate confidence intervals. We follow the guideline of excluding from the computation of the kurtosis-to-frequency statistic price changes (genuine or not) smaller than 0.1% in absolute value.

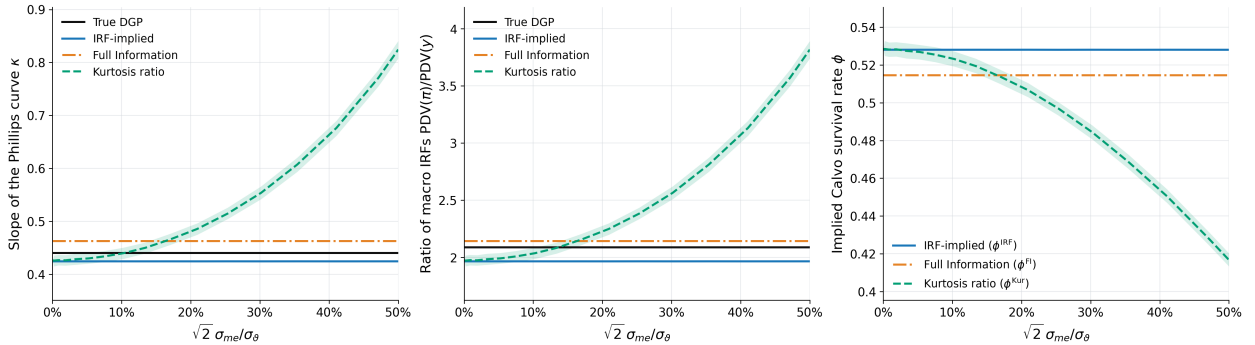
¹⁹Normality is required in the menu cost model only. Figure C.6 shows that when the data are generated by the Calvo model, the IRF-implied statistics track their true counterparts whatever the kurtosis of the identifying shock, as Proposition 1 states.

²⁰With a Gaussian shock the score is proportional to ϑ_{it} and the IV regression is OLS.

Figure 8 shows that measurement error does not bias the inferred slope of the Phillips curve from the IRF-implied statistic $J_{\pi,mc^r}^{\text{IRF}}$, or the inferred Calvo survival rate ϕ^{IRF} . The robust statistic is unaffected by measurement error, an advantage due to the noisy variable being the dependent variable in our regressions. Measurement error increases the noise in the robust statistic, although that is not quantitatively important (the error bands in the robust statistic are not visible in the plot).

By contrast, the effects of measurement error on the kurtosis-to-frequency statistic, and hence on the inferred Phillips curve $J_{\pi,mc^r}^C(\phi^{\text{Kur}})$, are very different. Initially, measurement error increases both kurtosis (there is more mass in small price changes) and frequency which roughly offset. Little by little, measurement error decreases the kurtosis while its effects on the frequency are monotonic. Through the lens of the sufficient statistic that means that monetary non-neutrality decreases, which can only be rationalized in a Calvo model with a steeper slope of the Phillips curve.

FIGURE 8. Price Measurement Error and Inferred Phillips Curve Slopes



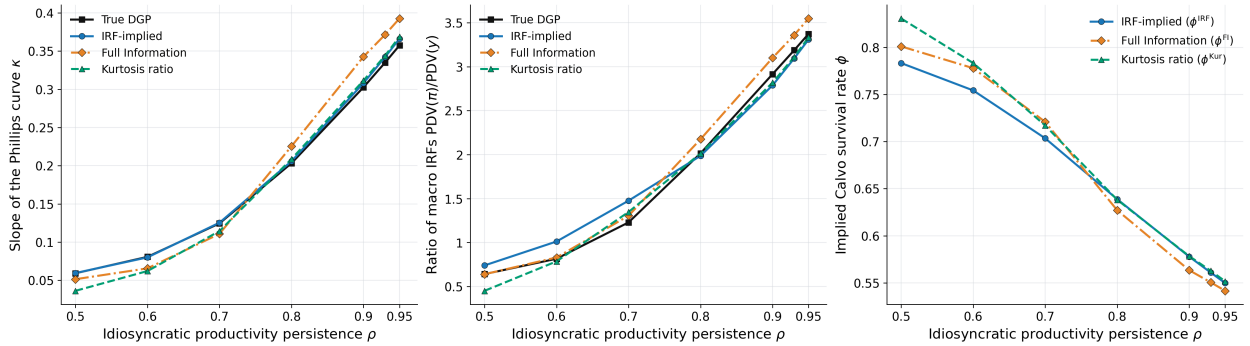
Notes: This figure shows the performance of our approximation of the generalized Phillips curve with measurement error in prices. Observed log prices follow $p_{it}^{\text{obs}} = p_{it} + me_{it}$, with independent Gaussian measurement error in price levels. The x-axis is the standard deviation of measurement error in price changes relative to the cost-innovation standard deviation, $\text{sd}(\Delta me)/\sigma_\theta = \sqrt{2}\sigma_{me}/\sigma_\theta$. We report: the slope of the Phillips curve $J_{\pi,mc^r}^m[0,0]$ (left), the Phillips multiplier (middle), and the Calvo survival rate ϕ (right). The solid blue line is the replication mean of our robust statistic approach $J_{\pi,mc^r}^C(\phi^{\text{IRF}})$, the dashed green line is equivalent object inferred from the Alvarez, Le Bihan, and Lippi (2016) statistic $J_{\pi,mc^r}^C(\phi^{\text{Kur}})$. Shading shows the 10th–90th percentile range across replications; dotted lines show each method’s no-noise value. The ϕ^{Kur} calculation counts observed absolute log price changes above 0.001 when computing frequency and kurtosis. The right panel has no true model object. All other parameters are held at the Nakamura and Steinsson (2010) calibration.

5.4. Non-Random Walk Shocks

Random-walk shocks are appealing due to their tractability. A productivity shock that changes the price gap today will permanently change future expected price gaps in the absence of adjustment. Transitory shocks are different; a firm will consider the persistence of the shock when deciding whether to act on it, changing the size of the inaction region. Sufficient statistics in the literature use the random walk assumption, but may not be reliable when the DGP deviates from this assumption (Dotsey and King 2005).

Our robust statistic works well regardless of the persistence of idiosyncratic shocks. We show that numerically by solving menu cost models with AR(1) productivity shocks with varying persistence. Figure 9 shows that the robust statistic tracks the true slope of the Phillips curve (left panel), the true Phillips multiplier (middle panel), and that we recover the same virtual survival rate as the full information approach (right panel). While we deviate from the random walk assumption in Alvarez, Le Bihan, and Lippi (2016), we find that in practice the kurtosis-implied objects also track their true counterparts well.

FIGURE 9. Performance of the Robust Statistic Across Shock Persistence



Notes: This figure shows the performance of our approximation of the generalized Phillips curve as the persistence of idiosyncratic productivity shocks varies. The idiosyncratic shock follows $\vartheta_{it} = \rho\vartheta_{i,t-1} + \epsilon_{it}$, with ρ the persistence parameter. The innovation standard deviation remains fixed. We report: the slope of the Phillips curve $J_{\pi,mc}^m[0,0]$ (left), the Phillips multiplier (middle), and the Calvo survival rate ϕ (right). Each panel compares the true menu cost model $J_{\pi,mc}^{MC}$ with three Calvo approximations $J_{\pi,mc}^C(\phi)$, which use the IRF-implied survival rate ϕ^{IRF} , the full-information survival rate ϕ^{FI} , and the kurtosis-implied survival rate ϕ^{Kur} . The right panel has no true model object. All other parameters are held at the Nakamura and Steinsson (2010) calibration.

6. Conclusion

In this paper, we propose a robust statistic for monetary non-neutrality: the impact and one-period-ahead responses of a firm’s price to an observed idiosyncratic cost shock, estimated by a local projection and assembled into a Phillips curve as if the economy were Calvo.

Three results support this statistic. First, in the Calvo, Gertler–Leahy, and canonical menu cost models, the average response of firms’ prices to their own cost shock equals the response of the aggregate price level to a permanent nominal cost shock, so that a firm-level regression measures the object the Phillips curve is built from. Second, the two coefficients recover the generalized Phillips curve exactly in the Calvo, Rotemberg, and Gertler–Leahy models, where the survival of a cost shock in firms’ price gaps is geometric. Third, in menu cost models, the two coefficients recover the average one-period-ahead survival rate of a marginal price gap and extrapolating this survival geometrically is accurate across a wide range of parameterizations. The accuracy survives decreasing returns to scale, Kimball demand, leptokurtic shocks, measurement error in prices, and persistent rather than random-walk shocks.

Our approach contrasts with sufficient statistics based on unconditional moments of the distribution of price changes, which tend to quickly lose informativeness as soon as the true data-generating process deviates from the benchmark model under which these statistics are sufficient.

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Appendix

Appendix A. Additional Details of the Model Environments

A.1. The Loss Function from Primitives

Nominal marginal costs of firm i at date t are

$$(A.1) \quad MC_{it} = \frac{W_t}{\gamma A_{it}} \left(\frac{Y_{it}}{A_{it}} \right)^{\frac{1-\gamma}{\gamma}},$$

The desired price. Write the profit (2) as revenue minus cost, $R(P_{it}) = (P_{it}/P_t)^{1-\varepsilon} Y_t$ and $K(P_{it}) = \frac{W_t}{P_t} (Y_t/A_{it})^{1/\gamma} (P_{it}/P_t)^{-\varepsilon/\gamma}$. The first-order condition is

$$(A.2) \quad K(P_{it}^*) = \frac{\gamma(\varepsilon - 1)}{\varepsilon} R(P_{it}^*),$$

the markup condition $P_{it}^* = \frac{\varepsilon}{\varepsilon-1} MC_{it}$ with $MC_{it} = P_t K'(Y)$ from (A.1). Taking logs and using $\frac{\varepsilon}{\gamma} - (\varepsilon - 1) = 1/\omega$,

$$(A.3) \quad \ln P_{it}^* = \omega \left(\ln W_t + \frac{1-\gamma}{\gamma} \ln Y_t - \frac{1}{\gamma} \ln A_{it} \right) + (1-\omega) \ln P_t + \text{const},$$

which is (3) in log deviations, with $mc_t = w_t + \frac{1-\gamma}{\gamma} Y_t$ the nominal marginal cost of a firm with $A_{it} = 1$ producing Y_t .

Gap sufficiency and the curvature. Substituting $P_{it} = P_{it}^* e^x$ and using (A.2),

$$(A.4) \quad \Pi_t(P_{it}^* e^x) = R_{it}^* h_\omega(x), \quad h_\omega(x) \equiv e^{(1-\varepsilon)x} - \frac{\gamma(\varepsilon - 1)}{\varepsilon} e^{-(\varepsilon/\gamma)x},$$

where $R_{it}^* \equiv R(P_{it}^*) = Y_t (P_{it}^*/P_t)^{1-\varepsilon}$ is frictionless revenue. Profit as a function of the log price gap can therefore be written as $R_{it}^* h_\omega(x)$, where h_ω depends only on (ε, γ) and R_{it}^* contains the dependence on firm productivity and aggregate variables. Hence the gap is a sufficient state for the firm's pricing problem exactly, not only to second order.²¹ A second-

²¹It fails under Kimball demand, where the elasticity of demand varies with the relative price so that the shape of profit in the gap depends on the firm's relative cost; there the reduction to a gap problem holds only to second order.

order expansion of (A.4) around $x = 0$ gives (4), with curvature $C_{it} = (\varepsilon - 1)R_{it}^*/\omega$.

From (4) to the objective (5). The curvature factors as $C_{it} = \frac{\varepsilon-1}{\omega} R_{it}^*$, and by (A.3) the desired relative price is

$$\ln \frac{P_{it}^*}{P_t} = \omega \left(\ln \frac{W_t}{P_t} + \frac{1-\gamma}{\gamma} \ln Y_t - \frac{1}{\gamma} \ln A_{it} \right) + \text{const},$$

so

$$(A.5) \quad C_{it} \propto R_{it}^* \propto A_{it}^{\omega(\varepsilon-1)/\gamma} \times \text{aggregate terms},$$

Following Gertler and Leahy (2008), we keep the adjustment problem homogeneous in productivity, and we take the menu cost to be a fixed fraction of the firm's frictionless revenue: menu cost $_{it} = \xi C_{it}$ with ξ a constant. Dividing the date- t objective by C_{it} then gives the per-period loss $\frac{1}{2}x_{it}^2$ and the menu cost ξ , common across firms and dates, for any size of the shocks.

The curvature remains in the continuation value, where the ratio $C_{i,t+s}/C_{it} = R_{i,t+s}^*/R_{it}^*$ weights future losses. The approximation used here is the same as in (4). Consider a joint perturbation in which the innovations to a_{it} , mc_t , and P_t , as well as the band half-width $\bar{x} \propto \sqrt{\xi}$, are all of order v . In that perturbation the gap is $O(v)$ and the ratio $C_{i,t+s}/C_{it}$ is $1 + O(v)$, so its variation multiplies $x^2 = O(v^2)$ and is $O(v^3)$, the order already dropped in (4).²² The household's stochastic discount factor is replaced by β^t for the same reason.

The normalized problem is the canonical menu cost problem: a quadratic loss $\frac{1}{2}x_{it}^2$, an adjustment cost equal to either 0 or the common menu cost ξ , and i.i.d. innovations $\omega(\epsilon_{it}^z + \epsilon_{it})$ to the desired price. The canonical menu cost results used below therefore apply up to terms of order v^3 .

A.2. The Gertler–Leahy Economy

This section restates the parts of Gertler and Leahy (2008) needed for Proposition 3 in the notation of Section 2.1. We maintain the language of the original reference, referring to the

²²This is the term that Gertler and Leahy (2008)'s Proposition 1 shows to be independent of the current price to second order.

large idiosyncratic shock type as *turbulence*.²³

A.2.1. Assumptions and the Firm's Problem

Primitives. A firm is hit by turbulence with probability $1 - \eta$ per period and, independently, draws a free adjustment opportunity with probability λ , both i.i.d.; write $\eta_\lambda \equiv (1 - \lambda)\eta$ for the probability that neither occurs. At a hit $z_{it} = z_{i,t-1} + \epsilon_{it}^z$ with $\omega\epsilon_{it}^z \sim U[-\varsigma/2, \varsigma/2]$ (so that ς is the support width of the turbulence shock in desired-price units), otherwise $z_{it} = z_{i,t-1}$. The firm's desired price is (7). Changing a price costs the menu cost ξ (zero at a free draw). Following GL, a firm knows when it is hit but, to learn the value of the draw and to reconsider its price, it must first pay a small decision cost d ; a firm that does not pay it keeps its price. A free adjustment opportunity waives both ξ and d : the firm observes its state and resets at no cost. GL's assumptions are:

- (A1) the menu cost ξ is second order, so that the band half-width $\bar{x} = O(\sqrt{\xi})$ is first order;
- (A2) innovations to mc_t , P_t and ϑ_{it} are first-order small: ϵ_{it} has bounded support, and the cumulative movement of the non-turbulence part of the desired price, $\omega(mc_t + \vartheta_{it}) + (1 - \omega)P_t$, over any $k(m)$ consecutive periods is bounded by m for every realization, where $k(m)$ is the least integer with $\eta_\lambda^{k(m)} \leq cm$ for a constant $c > 0$;
- (A3) $\varsigma > 4\bar{x} + 2m$;
- (A4) the decision cost d lies in the range for which paying it is optimal after a hit and not after a period with neither a hit nor a free draw.

A.2.2. Characterization of the Optimal Policy

Policy function. By (A4), a firm reconsiders its price only when it is hit or receives a free adjustment opportunity; (A1)–(A3) make the range in (A4) nonempty, since the gain from reconsidering without a hit is of order $m(2\bar{x} + m)$ while the gain after a hit is of order ς^2 .

Under (A1)–(A4), GL's Proposition 1 states that the expected value of an optimal policy after the next hit is independent of the firm's current gap to second order (the same holds after a free draw, which resets the gap); so that in choosing its price the firm only

²³Compared to Gertler and Leahy (2008), (i) we add the free adjustment probability λ of (MC), (ii) we drop the entry–exit process; (iii) the real rigidity is generated by decreasing returns as opposed to island-specific labor markets.

takes into account the states in which neither a further hit nor a free draw has occurred, weighted by $(\eta_\lambda \beta)^s$. With the quadratic loss, the value of a firm that has just reset to p at date t is, up to terms independent of p and third-order terms,

$$(A.6) \quad -\frac{1}{2} \mathbb{E}_t \sum_{s \geq 0} (\eta_\lambda \beta)^s (p - p_{i,t+s}^*)^2,$$

$$p_{i,t+s}^* = \omega(mc_{t+s} + z_{it} + \vartheta_{i,t+s}) + (1 - \omega)P_{t+s} \text{ along the no-hit path.}$$

The first-order condition $\mathbb{E}_t \sum_s (\eta_\lambda \beta)^s (p - p_{i,t+s}^*) = 0$ gives the reset price

$$(A.7) \quad p_{it}^{\text{new}} = \omega z_{it} + (1 - \eta_\lambda \beta) \mathbb{E}_t \sum_{s \geq 0} (\eta_\lambda \beta)^s [\omega(mc_{t+s} + \vartheta_{i,t+s}) + (1 - \omega)P_{t+s}],$$

which is (27). Since z_{it} is frozen along the no-hit path it enters with coefficient ω ; with a random-walk shifter, $\mathbb{E}_t \vartheta_{i,t+s} = \vartheta_{it}$ and $p_{it}^{\text{new}} = \omega(z_{it} + \vartheta_{it}) + (1 - \eta_\lambda \beta) \mathbb{E}_t \sum_s (\eta_\lambda \beta)^s [\omega mc_{t+s} + (1 - \omega)P_{t+s}]$. In the zero-inflation stationary state mc and P are constant, so $p_{it}^{\text{new}} = p_{it}^*$: the firm resets to its current desired price, and the reset gap is zero.

The inaction band. Write the value (A.6) as a function of the gap y between the firm's price and the reset price (A.7); it equals $-\frac{1}{2} y^2 / (1 - \eta_\lambda \beta)$ plus terms independent of y . Value matching equates the value of keeping a price with gap y to the value of resetting net of the menu cost, $\frac{1}{2} y^2 / (1 - \eta_\lambda \beta) = \xi$, so the firm adjusts if and only if $|y| > \bar{x}$ with

$$(A.8) \quad \bar{x} = \sqrt{2(1 - \eta_\lambda \beta) \xi},$$

an expression analogous to Gertler and Leahy (2008).²⁴

Two remarks on what the band is a statement about. First, it is a band on the gap to the *reset* price, $y = p_{i,t-1} - p_{it}^{\text{new}}$, not on the gap $x_{it} = p_{it} - p_{it}^*$ to the flexible-price desired price of Section 2.1: reconsidering means comparing the current price with the price the firm would set, and by GL's Proposition 1 that comparison depends on y alone. The two gaps coincide in the zero-inflation stationary state, where $p_{it}^{\text{new}} = p_{it}^*$, and after an unanticipated permanent shift of the desired price with the price level held fixed, since

²⁴For $\lambda = 0$ and a menu cost b in output units, Gertler and Leahy (2008)'s band is $\bar{x}^2 = 2(1 - \eta\beta) b / ((\varepsilon - 1)\bar{Y})$; here $b = \xi \bar{C}$ with $\bar{C} = (\varepsilon - 1)\bar{Y} / \omega$ at the symmetric steady state $\bar{P}^* = \bar{P}$, so $\bar{x}^2 = 2(1 - \eta_\lambda \beta) \omega b / ((\varepsilon - 1)\bar{Y})$; for the same output-unit menu cost the band is narrower than GL's by $\sqrt{\omega}$, because decreasing returns steepen the firm's profit in its own price whereas GL's island labor market does not.

the problem is stationary again from the shock date on. They differ along anticipated aggregate paths, where the reset price is forward-looking. Second, the band is not just a steady-state object: its half-width (A.8) is constant along every aggregate path admissible under (A2), because the coefficient $1/(1 - \eta_\lambda \beta)$ of the quadratic value does not depend on the state. What moves with the aggregate path is the center of the band, the reset price. The drift in the reset price between two hits is bounded by m and (A2)–(A3) keep the band strictly inside the support of the turbulence draw along every such path.

A.3. Generalized Menu Cost

This subsection allows the menu cost ξ_{it} to be drawn each period from a distribution H with continuous density, independently across firms and time. The canonical model of Section 3.3.3 corresponds instead to the two-point distribution $\xi_{it} \in \{0, \xi\}$. Let $W(y)$ denote the value of carrying a gap of y into the next period without adjusting (Appendix B.4.1); a firm adjusts whenever the gain from resetting exceeds the realized cost, $\xi_{it} < W(y) - W(0)$, so the adjustment probability is the smooth function

$$\Lambda(y) = H(W(y) - W(0)),$$

even and increasing in $|y|$ since W is, in place of the hard band $[-\bar{x}, \bar{x}]$ of the canonical model. The on-impact pass-through b_0 and ϕ^{IRF} admit closed-form extensions to this model.

Policy function. The firm adjusts with probability $\Lambda(y)$ given its pre-adjustment gap y . The frequency of price adjustments is $\text{freq} = \int \Lambda(x) g(x) dx$, where $g(x)$ represents the stationary density of price gaps. A fraction $\Lambda(y)$ of firms at each gap y , reset to $x^* = 0$, and non-adjusters retain their gap with density $g(1 - \Lambda)$.

On-impact pass-through. By Proposition 1 and Appendix B.4.2,

$$(A.9) \quad b_0^{\text{GMC}} = [J_{P,p^*}^{\text{MC}} \mathbf{1}]_0 \omega = \omega \left[\text{freq} + \int y \Lambda'(y) g(y) dy \right],$$

which nests (31) when Λ' reduces to point masses of $(1 - \lambda)$ at $\pm \bar{x}$. As in the canonical model, b_0^{GMC} exceeds ω times the frequency. The shock induces adjustment at every gap

level, with marginal mass proportional to $\Lambda'(y)$, and each marginal adjuster contributes its full price change rather than the displacement that moves it into adjustment.

Term structure of the pass-through matrix. In the generalized model, (33) becomes an intensive-margin component plus a continuum of extensive-margin components indexed by the gap level x :

$$(A.10) \quad J_{P,p^*}^{\text{GMC}} = \int \varrho_e(x) J_{P,p^*}^{\text{TD}}(\Phi^e(x)) dx + \varrho_i J_{P,p^*}^{\text{TD}}(\Phi^i), \quad \int \varrho_e(x) dx + \varrho_i = 1,$$

Here $\Phi^e(x)$ is the survival function associated with the extensive response at gap x , while Φ^i is the intensive-margin survival function.²⁵ Applying Lemma 2 to write $b_h^{\text{GMC}} = \omega[J_{P,p^*}^{\text{GMC}} \mathbf{1}]_h$,

$$(A.11) \quad b_h^{\text{GMC}} = b_0^{\text{GMC}} \sum_{s=0}^h \bar{\Phi}_s, \quad \bar{\Phi}_s \equiv \int m_e(x) \Phi_s^e(x) dx + m_i \Phi_s^i,$$

with weights $m_e(x) \equiv \Lambda'(x)g(x)x/(b_0^{\text{GMC}}/\omega)$ and $m_i \equiv \text{freq}/(b_0^{\text{GMC}}/\omega)$, satisfying $\int m_e(x) dx + m_i = 1$, which reduce to the weights m_e and m_i in the canonical model.

Identification from the term structure. As in the canonical model, (A.11) identifies the average survival function horizon by horizon, $(b_h^{\text{GMC}} - b_{h-1}^{\text{GMC}})/b_0^{\text{GMC}} = \bar{\Phi}_h$ for $h \geq 1$, of which $\phi^{\text{IRF,GMC}}$ is the one-period-ahead value:

$$(A.12) \quad \phi^{\text{IRF,GMC}} = \frac{b_1^{\text{GMC}}}{b_0^{\text{GMC}}} - 1 = \bar{\Phi}_1,$$

as in (34).

Whether the geometric extrapolation of $\phi^{\text{IRF,GMC}}$ to later horizons remains as accurate as in the canonical case, and a formal error bound analogous to Proposition 4, are left for future work.

²⁵ $\Phi_s^e(x) \equiv E^s(x)/x$ is now indexed by the gap level x , and $\Phi_s^i \equiv E^{s'}(0)$ is, as before, a single intensive-margin survival function. The weights

$$\varrho_e(x) \equiv \Lambda'(x)g(x)x \sum_{s \geq 0} \Phi_s^e(x), \quad \varrho_i \equiv \text{freq} \sum_{s \geq 0} \Phi_s^i,$$

generalize ϱ_e, ϱ_i of (33): mass $2(1 - \lambda)g(\bar{x})\bar{x}$ concentrated at the single edge $\bar{x}$ is replaced by a density $\Lambda'(x)g(x)x$ spread over every gap level x .

A.4. General Equilibrium Structure

We solve and simulate a wide range of New Keynesian models whose linearized aggregate block consists of a monetary policy rule, an intertemporal IS curve and a generalized Phillips curve. Let π denote the inflation sequence, $\tilde{y}$ the sequence of log deviations of output from its flexible-price level, i the nominal interest rate and ζ the monetary-policy disturbance. The monetary policy rule is

$$i = \psi_\pi \pi + \zeta,$$

with ψ_π the inflation coefficient. The IS curve is

$$\tilde{y} = -\frac{1}{\sigma_c} (I - F)^{-1} (i - F\pi), \quad \text{i.e.} \quad \tilde{y}_t = -\frac{1}{\sigma_c} \sum_{s \geq 0} (i_{t+s} - \pi_{t+s+1}),$$

with σ_c the inverse intertemporal elasticity of substitution, where $(I - F)^{-1}$ is the upper-triangular matrix of ones and $F(I - F)^{-1}$ its strictly upper-triangular part. The generalized Phillips curve is

$$\pi = J_{\pi, mc}^m \mathbf{mc}^r, \quad \mathbf{mc}^r = \chi \tilde{y}, \quad \text{so that} \quad \pi = J_{\pi, \tilde{y}}^m \tilde{y}, \quad J_{\pi, \tilde{y}}^m \equiv \chi J_{\pi, mc}^m,$$

with χ the elasticity of real marginal cost to output. When the Phillips curve is assembled from the IRF statistics, $J_{\pi, mc}^m$ is replaced by $J_{\pi, mc}^{\text{IRF}} = \kappa^{\text{IRF}} \Theta_\beta$, equation (17).

The monetary-policy disturbance is an AR(1) impulse of size ζ_0 and persistence ρ_ζ ,

$$\zeta = \zeta_0 (1, \rho_\zeta, \rho_\zeta^2, \dots)',$$

truncated at the horizon of the sequence.

Substituting the rule into the IS curve and the result into the Phillips curve gives a linear system for inflation,

$$\left[I + \frac{1}{\sigma_c} J_{\pi, \tilde{y}}^m (I - F)^{-1} (\psi_\pi I - F) \right] \pi = -\frac{1}{\sigma_c} J_{\pi, \tilde{y}}^m (I - F)^{-1} \zeta,$$

which we solve by inverting the bracket on the truncated system. Given π , the output and nominal-rate IRFs follow from the IS curve and the monetary policy rule.

Appendix B. Proofs of the Theoretical Results

B.1. Calvo

B.1.1. Proof of Lemma 1

PROOF. (i) $\Theta(\alpha)\mathbf{1} = (1 - \beta\alpha)^{-1}\mathbf{1}$ and $[\Upsilon(\alpha)\mathbf{1}]_h = \sum_{r=0}^h \alpha^r = (1 - \alpha^{h+1})/(1 - \alpha)$. (ii) Write $c \equiv (1 - \alpha)(1 - \beta\alpha)$ and $J \equiv J_{P,p^*}^C(\alpha)$. Since $(I - J)^{-1}J = (J^{-1} - I)^{-1}$ and $J^{-1} = c^{-1}\Theta(\alpha)^{-1}\Upsilon(\alpha)^{-1} = c^{-1}(I - \beta\alpha F)(I - \alpha L)$, using $FL = I$,

$$J^{-1} - I = c^{-1}[(1 + \beta\alpha^2 - c)I - \alpha L - \beta\alpha F] = \frac{\alpha}{c}[(1 + \beta)I - L - \beta F] = \frac{\alpha}{c}(I - \beta F)(I - L),$$

where the second equality uses $1 + \beta\alpha^2 - (1 - \alpha)(1 - \beta\alpha) = \alpha(1 + \beta)$ and the third again uses $FL = I$. Inverting gives $(I - J)^{-1}J = (c/\alpha)(I - L)^{-1}(I - \beta F)^{-1}$, and premultiplying by $(I - L)$ gives $(I - L)(I - J)^{-1}J = (c/\alpha)(I - \beta F)^{-1} = \kappa(\alpha)\Theta_\beta$. $\square$

B.2. Rotemberg

This section adds a third adjustment technology to the environment of Section 2.1, the convex adjustment cost of Rotemberg (1982), and shows that the IRF statistics are exact there as well. Model (R) keeps the firms, the desired price (7) and the idiosyncratic shocks of (C): ϵ_{it} and ϵ_{it}^z can have any distribution with the moments stated in Section 2.1. The adjustment technology differs. There is no menu cost and no free draw; instead, changing the price by $p_{it} - p_{i,t-1}$ costs $\frac{\nu}{2}(p_{it} - p_{i,t-1})^2$ per period in the curvature units of Appendix A.1, with $\nu > 0$ a constant.²⁶ Every price changes every period, so $\text{freq} = 1$.

The firm's problem. The firm chooses $\{p_{it}\}$ to minimize

$$(B.1) \quad \mathbb{E}_0 \sum_{t \geq 0} \beta^t \left[\frac{1}{2} x_{it}^2 + \frac{\nu}{2} (p_{it} - p_{i,t-1})^2 \right], \quad x_{it} = p_{it} - p_{it}^*.$$

The problem is linear-quadratic, so certainty equivalence holds and the first-order condi-

²⁶As with the menu cost, taking the adjustment cost proportional to frictionless revenue, $\frac{\nu}{2} C_{it}(p_{it} - p_{i,t-1})^2$, makes ν constant across firms and dates once the objective is divided by C_{it} .

tion is

$$(B.2) \quad x_{it} + \nu \left[(p_{it} - p_{i,t-1}) - \beta \mathbb{E}_t(p_{i,t+1} - p_{it}) \right] = 0.$$

LEMMA B.1 (Rotemberg policy). *Let $\alpha_R \in (0, 1)$ be the smaller root of*

$$(B.3) \quad \beta a^2 - (1 + \beta + \frac{1}{\nu})a + 1 = 0.$$

Then $1/\nu = \kappa(\alpha_R)$, with $\kappa(\alpha) \equiv (1 - \alpha)(1 - \beta\alpha)/\alpha$ as in Lemma 1, and the bounded solution of (B.2) is

$$(B.4) \quad p_{it} = \alpha_R p_{i,t-1} + (1 - \alpha_R)(1 - \beta\alpha_R) \mathbb{E}_t \sum_{s \geq 0} (\beta\alpha_R)^s p_{i,t+s}^*, \quad \text{or} \quad \mathbf{p}_i = J_{P,p^*}^C(\alpha_R) \mathbf{p}_i^*,$$

with $J_{P,p^}^C(\alpha) = (1 - \alpha)(1 - \beta\alpha)\Upsilon(\alpha)\Theta(\alpha)$ the Calvo desired-price pass-through of (24).*

PROOF. Substituting $x_{it} = p_{it} - p_{it}^*$ into (B.2) and dividing by ν ,

$$(B.5) \quad \beta \mathbb{E}_t p_{i,t+1} - (1 + \beta + \frac{1}{\nu})p_{it} + p_{i,t-1} = -\frac{1}{\nu} p_{it}^*.$$

The characteristic polynomial $\beta a^2 - (1 + \beta + 1/\nu)a + 1$ equals 1 at $a = 0$ and $-1/\nu < 0$ at $a = 1$, so it has a root $\alpha_R \in (0, 1)$; the product of the roots is $1/\beta$, so the other root is $1/(\beta\alpha_R) > 1$. Evaluating (B.3) at α_R and dividing by α_R gives $1/\nu = (1 - \alpha_R)(1 - \beta\alpha_R)/\alpha_R = \kappa(\alpha_R)$, and also $1 + \beta + 1/\nu = \beta\alpha_R + 1/\alpha_R$. With the latter, (B.5) factors as

$$\beta(\mathbb{E}_t p_{i,t+1} - \alpha_R p_{it}) - \frac{1}{\alpha_R}(p_{it} - \alpha_R p_{i,t-1}) = -\frac{1}{\nu} p_{it}^*.$$

Let $u_{it} \equiv p_{it} - \alpha_R p_{i,t-1}$. Then $u_{it} = \beta\alpha_R \mathbb{E}_t u_{i,t+1} + \frac{\alpha_R}{\nu} p_{it}^*$, whose unique bounded solution is

$$u_{it} = \frac{\alpha_R}{\nu} \mathbb{E}_t \sum_{s \geq 0} (\beta\alpha_R)^s p_{i,t+s}^*;$$

the explosive root is excluded by boundedness. Since $\alpha_R/\nu = \alpha_R \kappa(\alpha_R) = (1 - \alpha_R)(1 - \beta\alpha_R)$, this is (B.4). In sequence space, $\mathbb{E}_t \sum_s (\beta\alpha_R)^s p_{i,t+s}^* = [\Theta(\alpha_R) \mathbf{p}_i^*]_t$, and inverting $p_{it} -$

$\alpha_R p_{i,t-1} = (1 - \alpha_R)(1 - \beta \alpha_R)[\Theta(\alpha_R) \mathbf{p}_i^*]_t$ with $\Upsilon(\alpha_R) = (I - \alpha_R L)^{-1}$ gives

$$\mathbf{p}_i = (1 - \alpha_R)(1 - \beta \alpha_R) \Upsilon(\alpha_R) \Theta(\alpha_R) \mathbf{p}_i^* = J_{P,p^*}^C(\alpha_R) \mathbf{p}_i^*,$$

the map (24) with α_R in place of α . □

The price of a Rotemberg firm is thus the Calvo expected price (23) with α_R in place of α , firm by firm and not only in expectation: the reset formula $\Theta(\alpha_R)$ and the survival weights $\Upsilon(\alpha_R)$ arise here from the two roots of the Euler equation, not from an adjustment lottery.

PROPOSITION B.1 (Rotemberg). *In the (R) economy,*

- (i) *The desired-price pass-through Jacobian is Calvo: $J_{P,p^*}^R = J_{P,p^*}^C(\alpha_R)$, and hence $J_{\pi,mc^r}^R = J_{\pi,mc^r}^C(\alpha_R) = \omega \kappa(\alpha_R) \Theta_\beta = \frac{\omega}{\nu} \Theta_\beta$, the New Keynesian Phillips curve $\pi_t = \frac{\omega}{\nu} mc_t^r + \beta \mathbb{E}_t \pi_{t+1}$;*
- (ii) *Proposition 1 holds for any distribution of ϵ_{it} , and the IRF statistics are exact: $b_h^R = \omega(1 - \alpha_R^{h+1})$; $\phi^{\text{IRF},R} = \alpha_R$; and $J_{\pi,mc^r}^{\text{IRF},R} = \kappa^{\text{IRF},R} \Theta_\beta = J_{\pi,mc^r}^R$;*
- (iii) *The frequency of price changes is one, so $1 - \text{freq} = 0$ while $\phi^{\text{IRF},R} = \alpha_R \in (0, 1)$: the frequency carries no information about the Phillips curve, and the Calvo slope built from it, $\omega \kappa(1 - \text{freq})$, is unbounded.*

PROOF. (i) Integrating (B.4) across firms, the idiosyncratic components of $\mathbf{p}_i^*$ wash out as in Section 3.3.1, so $\mathbf{P} = J_{P,p^*}^C(\alpha_R)[\omega \mathbf{mc} + (1 - \omega) \mathbf{P}]$ and $J_{P,p^*}^R = J_{P,p^*}^C(\alpha_R)$. Lemma 1(ii) then gives $J_{\pi,mc^r}^R = \omega \kappa(\alpha_R) \Theta_\beta$, and $\kappa(\alpha_R) = 1/\nu$ by Lemma B.1.

(ii) Fix a firm, its predetermined state S and the other draws $\mathcal{D}$. By (B.4) the price path is affine in the path of desired prices, and under the random walk a shift δ in ϵ_{it} raises $p_{i,t+s}^*$ by $\omega \delta$ for every $s \geq 0$ and nothing else; hence the cumulative price change $f_h(\epsilon_{it}; S, \mathcal{D})$ is affine in ϵ_{it} with slope $\omega[J_{P,p^*}^C(\alpha_R) \mathbf{1}]_h$, which does not depend on $(S, \mathcal{D})$. Case (ii) of Lemma B.4 therefore applies for any distribution of ϵ_{it} , and with Lemma B.3, $b_h^R = \omega[J_{P,p^*}^C(\alpha_R) \mathbf{1}]_h = \omega(1 - \alpha_R^{h+1})$ by Lemma 1(i). Proposition 2 then applies verbatim with $\alpha = \alpha_R$: $\phi^{\text{IRF},R} = \alpha_R$ and $J_{\pi,mc^r}^{\text{IRF},R} = \kappa^{\text{IRF},R} \Theta_\beta = \omega \kappa(\alpha_R) \Theta_\beta = J_{\pi,mc^r}^R$.

(iii) Under a convex cost the optimal price moves whenever the desired price does, so every price changes every period and $\text{freq} = 1$. Since $\kappa(\alpha) \rightarrow \infty$ as $\alpha \rightarrow 0$, the Calvo slope evaluated at $1 - \text{freq} = 0$ is unbounded. □

The result is analogous to Proposition 3 with α_R in place of η_λ , for the through a dif-

ferent mechanism. In (GL) the shock dissipates from the expected price gap at a geometric rate because reconsideration is a lottery that does not depend on the gap. In (R) the shock dissipates from the gap at a geometric rate because every firm closes a constant fraction $1 - \alpha_R$ of its distance to the forward-looking target each period; there is no selection, and $\phi^{\text{IRF,R}}$ is a pure measure of the survival of the price gap. In both models the statistic identifies the decay of the shock in price gaps, while the frequency of price changes is uninformative: below the relevant hazard in (GL), and equal to one in (R).

B.3. Gertler–Leahy

B.3.1. Selection

Remember that the relevant gap is the one to the reset price, y , on which the band of Appendix A.2 is defined. For a firm hit at t , this gap is:

$$(B.6) \quad y_{it} = p_{i,t-1} - p_{it}^{\text{new}} = \left[(p_{i,t-1} - \omega z_{i,t-1}) - (p_{it}^{\text{new}} - \omega z_{it}) \right] - \omega \epsilon_{it}^z,$$

where, by (27), $p_{it}^{\text{new}} - \omega z_{it}$ does not depend on the current draw. Since z is frozen between hits, the bracket is the gap the firm had after its last hit, in $[-\bar{x}, \bar{x}]$, plus the drift of $p^{\text{new}} - \omega z$ since then, bounded by m under (A2) for every firm whose last hit is within $k(m)$ periods, a set of mass at least $1 - cm$; so it is at most $\bar{x} + m$ in absolute value for those firms, and the complement contributes a second-order error to Lemma B.2, which therefore holds to the order of the approximation. Since $\omega \epsilon_{it}^z$ is uniform on a support of half-width $\varsigma/2 > 2\bar{x} + m$ by (A3), y_{it} is uniform on an interval of width ς that contains $[-\bar{x}, \bar{x}]$.

LEMMA B.2 (Selection). *Conditional on a hit at t , on the aggregate and shifter paths, and on the firm's history: (i) a firm without a free draw adjusts with probability $1 - 2\bar{x}/\varsigma$, and one with a free draw with certainty; (ii) $\mathbb{E}[p_{it} \mid \text{hit}] = \mathbb{E}[p_{it}^{\text{new}} \mid \text{hit}]$, the expectation being over the turbulence draw.*

PROOF. (i) A firm without a free draw keeps its price if $y_{it} \in [-\bar{x}, \bar{x}]$, an event of probability $2\bar{x}/\varsigma$ under the uniform distribution; a free draw makes adjustment costless. (ii) If the firm adjusts, freely or because its gap is outside the band, $p_{it} - p_{it}^{\text{new}} = 0$; if not, $p_{it} - p_{it}^{\text{new}} = y_{it}$ with $|y_{it}| \leq \bar{x}$. Hence $\mathbb{E}[p_{it} - p_{it}^{\text{new}} \mid \text{hit}] = (1 - \lambda) \mathbb{E}[y_{it} \mathbf{1}\{|y_{it}| \leq \bar{x}\}] = (1 - \lambda) \varsigma^{-1} \int_{-\bar{x}}^{\bar{x}} y \, dy = 0$. $\square$

Uniformity makes the conditional density of non-adjusters flat, and symmetry of the band puts the reset price at its midpoint; both are needed. With a non-uniform shock the truncated mass would tilt toward one edge and the conditional mean would miss p_{it}^{new} .

B.3.2. The Calvo Law of Expected Prices and Proof of Proposition 3

The selection property (29) is Lemma B.2(ii). *The Calvo law* (30). Take expectations over the whole adjustment process (hits, free draws and turbulence draws), conditional on the paths of mc , P and ϑ_i . Partition firms at t into free adjusters (mass λ), whose price is the reset price; firms without a free draw that are hit (mass $(1 - \lambda)(1 - \eta)$), whose expected price is the expected reset price by Lemma B.2(ii); and the rest (mass $\eta\lambda$), which keep their price, $p_{it} = p_{i,t-1}$. The expectation of the reset price over the turbulence history is $\bar{p}_{it}^{\text{new}}$, equation (28), because turbulence draws are mean zero, and the event “neither hit nor free” is independent of the firm’s history, so the expected price of the third group is $\bar{p}_{i,t-1}$. Hence

$$(B.7) \quad \bar{p}_{it} = \eta\lambda \bar{p}_{i,t-1} + (1 - \eta\lambda) \bar{p}_{it}^{\text{new}},$$

which is (30).

Proposition 3, macro. By the law of large numbers across firms, $P_t = \int p_{it} di = \int \bar{p}_{it} di$, and since $\int z_{it} di = \int \vartheta_{it} di = 0$, integrating (B.7) across firms gives

$$(B.8) \quad P_t = \eta\lambda P_{t-1} + (1 - \eta\lambda)(1 - \eta\lambda\beta) \sum_{s \geq 0} (\eta\lambda\beta)^s [\omega mc_{t+s} + (1 - \omega)P_{t+s}].$$

In sequence form $\mathbf{P} = (1 - \eta\lambda)\Upsilon(\eta\lambda)(1 - \eta\lambda\beta)\Theta(\eta\lambda)[\omega\mathbf{mc} + (1 - \omega)\mathbf{P}] = J_{P,p^*}^C(\eta\lambda)[\omega\mathbf{mc} + (1 - \omega)\mathbf{P}]$, so the desired-price pass-through is $J_{P,p^*}^{\text{GL}} = J_{P,p^*}^C(\eta\lambda)$ and $J_{P,mc}^{\text{GL}} = \omega J_{P,p^*}^C(\eta\lambda)$. By Lemma 1(ii), $J_{\pi,mc}^{\text{GL}} = \omega\kappa(\eta\lambda)\Theta_\beta$, i.e. $\pi_t = \omega\kappa(\eta\lambda)mc_t^r + \beta\pi_{t+1}$ with $\kappa(\eta\lambda) = (1 - \eta\lambda)(1 - \eta\lambda\beta)/\eta\lambda$.

Proposition 3, micro. By Lemma B.2, which holds for every realization of the shifter path under (A2)–(A3), the conditional expectation of $p_{i,t+h}$ given the shifter path is obtained by iterating (B.7) and is linear in that path. With a random walk, $\partial \bar{p}_{i,t+j}^{\text{new}} / \partial \epsilon_{it} = \omega$ for $j \geq 0$

by (28), so

$$(B.9) \quad \frac{\partial \bar{p}_{i,t+h}}{\partial \epsilon_{it}} = \omega(1 - \eta_\lambda) \sum_{j=0}^h \eta_\lambda^{h-j} = \omega(1 - \eta_\lambda^{h+1}) = \omega[J_{P,p^*}^C(\eta_\lambda) \mathbf{1}]_h,$$

and by linearity the regression coefficient equals this derivative, $b_h^{\text{GL}} = \omega(1 - \eta_\lambda^{h+1})$. Then $\phi^{\text{IRF,GL}} = b_1/b_0 - 1 = (1 - \eta_\lambda^2)/(1 - \eta_\lambda) - 1 = \eta_\lambda$.

Frequency. A firm changes its price if and only if it draws a free adjustment, or is hit without one and adjusts: $\text{freq} = \lambda + (1 - \lambda)(1 - \eta)(1 - 2\bar{x}/\varsigma)$ by Lemma B.2(i), and $1 - \text{freq} = \eta_\lambda + (1 - \lambda)(1 - \eta) 2\bar{x}/\varsigma > \eta_\lambda$. $\square$

B.4. Menu Cost

Throughout this subsection we work in model (MC) and write $\phi^{\text{IRF}}, \kappa^{\text{IRF}}$, and b_h for $\phi^{\text{IRF,MC}}, \kappa^{\text{IRF,MC}}$, and b_h^{MC} .

B.4.1. The Optimal Policy and the Stationary Distribution

This subsection derives the Ss form of the policy stated in Section 2.1, for the normalized problem of Appendix A.1. In gap units the firm's state at the start of a period is its pre-adjustment gap $y = x_- - v$: the gap carried from the previous period, x_- , once the desired price has moved by the innovation $v = \omega(\epsilon_{it}^z + \epsilon_{it})$ and before any adjustment, where v is i.i.d. with a density f that is symmetric and single-peaked; the menu cost draw is $\xi_{it} \in \{0, \xi\}$ with $\Pr(\xi_{it} = 0) = \lambda$, and ξ is the normalized menu cost of Appendix A.1. Let $V(y)$ be the expected discounted loss of a firm with pre-adjustment gap y , before the menu-cost draw, and

$$(B.10) \quad W(x) \equiv \frac{1}{2}x^2 + \beta \mathbb{E}_{v'}[V(x - v')]$$

the value of ending the period with gap x . The firm's problem is

$$(B.11) \quad V(y) = \lambda \min_x W(x) + (1 - \lambda) \min \left\{ W(y), \xi + \min_x W(x) \right\}$$

: with a free draw the firm chooses the best gap; otherwise it keeps y or pays ξ and chooses the best gap.

(i) *Reset gap.* The right-hand side of (B.11) maps even functions into even functions when f is symmetric, since the static loss is even and $\mathbb{E}_{v'} V(x - v') = \mathbb{E}_{v'} V(-x - v')$ for even V ; so V and W are even and the set of minimizers of W is symmetric. Auclert et al. (2024) show that for single-peaked f the minimizer is unique, hence $x^* = 0$: the reset gap is zero, and a firm that adjusts sets its price at the desired price.

(ii) *Band.* A firm without a free draw adjusts if and only if $W(y) - W(0) > \xi$. Since W is even and increasing in $|y|$ (single-peakedness of f propagates through the expectation), the adjustment region is $|y| > \bar{x}$, with the half-width defined by value matching,

$$(B.12) \quad W(\bar{x}) - W(0) = \xi,$$

so the inaction region is the symmetric band $[-\bar{x}, \bar{x}]$; in discrete time there is no smooth-pasting condition. Near zero, $W(y) - W(0) \approx \frac{1}{2}cy^2$, so a second-order menu cost implies a first-order band, $\bar{x} = O(\sqrt{\xi})$.

(iii) *Stationary distribution and frequency.* The post-adjustment gap is 0 for a firm that adjusts and y for one that does not, and the next pre-adjustment gap is the post-adjustment gap minus the next innovation. Writing μ for the stationary law of the post-adjustment gap and g for the stationary density of the pre-adjustment gap,

$$(B.13) \quad g(y) = \int f(y-x) \mu(dx), \quad \mu = \text{freq} \delta_0 + (1-\lambda) g \mathbf{1}_{[-\bar{x}, \bar{x}]}, \quad \text{freq} = 1 - (1-\lambda) \int_{-\bar{x}}^{\bar{x}} g(y) dy$$

: μ has an atom at the reset point and a density on the band, and g is a density (a convolution with f), as smooth as f . Written out, since μ has mass freq at zero and density $(1-\lambda)g$ on the band, the first equation in (B.13) is the integral equation

$$(B.14) \quad g(y) = \text{freq} f(y) + (1-\lambda) \int_{-\bar{x}}^{\bar{x}} f(y-x) g(x) dx.$$

The atom at zero will matter for the cross-sectional law used in Appendix B.5, while convolution with f makes g smooth. Existence and uniqueness of (μ, g) follow from the reinjection at zero, which makes the transition kernel a mixture with a fixed component, as Auclert et al. (2024) show.

(iv) *Smooth hazard.* With a menu cost drawn from a continuous distribution H , the firm

adjusts if and only if $\xi_{it} < W(y) - W(0)$, so the adjustment probability given the gap is

$$(B.15) \quad \Lambda(y) = H(W(y) - W(0)),$$

even, increasing in $|y|$, and as smooth as H and W ; the CalvoPlus case is $H = \lambda\delta_0 + (1-\lambda)\delta_\xi$, for which $\Lambda(y) = \lambda + (1-\lambda)\mathbf{1}\{|y| > \bar{x}\}$, and (B.13) holds with $\int \Lambda dg$ in place of the band integral.

B.4.2. Impact Entry of the Pass-Through Matrix

This subsection computes the first entry of the desired-price pass-through matrix, $[J_{P,p^*}^{\text{MC}} \mathbf{1}]_0$: the date- t response of the price level to a permanent unit shift of every firm's desired price from date t on, with the price level held fixed as in (11), evaluated at the stationary cross-section. Let $y \equiv x_{i,t-1} - v_{it}$ be the pre-adjustment gap, with stationary density g (equation (B.13)), and raise every desired price by δ from date t on. With the price level fixed, the shift leaves the normalized problem of Appendix B.4.1 unchanged: the reset gap remains $x^* = 0$, so an adjuster sets its price at the shifted desired price, and the band $[-\bar{x}, \bar{x}]$ is the same. Its only effect is to lower every pre-adjustment gap by δ , from y to $y - \delta$. A firm adjusts if it draws a free adjustment (probability λ) or if $|y - \delta| > \bar{x}$, and its price change is then $-(y - \delta)$; otherwise its price change is zero. The average price change at date t is therefore

$$(B.16) \quad \Delta \bar{p}_t(\delta) = -\lambda \int (y - \delta) g(y) dy - (1 - \lambda) \int_{|y - \delta| > \bar{x}} (y - \delta) g(y) dy,$$

and $[J_{P,p^*}^{\text{MC}} \mathbf{1}]_0$ is its derivative at $\delta = 0$, since the pre-shift price level does not depend on δ . Differentiating (B.16) (Leibniz's rule, g being continuous) gives two terms. On the adjustment region the price change moves one for one with δ , which contributes the mass of adjusters, $\lambda + (1 - \lambda) \int_{|y| > \bar{x}} g(y) dy = \text{freq}$. At the edges of the band the adjustment decision itself changes: firms with $y \in (\bar{x}, \bar{x} + \delta]$, of mass $(1 - \lambda)g(\bar{x})\delta$, are pulled inside the band and forgo a price change of $-\bar{x}$, and firms with $y \in (-\bar{x}, -\bar{x} + \delta]$, of mass $(1 - \lambda)g(-\bar{x})\delta$, are pushed outside and change their price by $+\bar{x}$; each group contributes $+\bar{x}$ per unit of mass. Hence

$$(B.17) \quad [J_{P,p^*}^{\text{MC}} \mathbf{1}]_0 = \text{freq} + (1 - \lambda) \bar{x} [g(\bar{x}) + g(-\bar{x})] = \text{freq} + 2(1 - \lambda) g(\bar{x}) \bar{x},$$

using the symmetry of g . The first term is the intensive margin and the second the extensive margin. With a continuous menu-cost distribution and adjustment probability $\Lambda(y)$, equation (B.15), averaging over the menu-cost draw gives an expected price change of $-y\Lambda(y)$ for a firm with pre-adjustment gap y , and $-(y - \delta)\Lambda(y - \delta)$ after the shift, so that $\Delta \bar{p}_t(\delta) = -\int (y - \delta)\Lambda(y - \delta)g(y) dy$ and, differentiating under the integral sign,

$$(B.18) \quad [J_{P,p^*}^{MC} \mathbf{1}]_0 = \int [\Lambda(y) + y\Lambda'(y)]g(y) dy = \text{freq} + \int y\Lambda'(y)g(y) dy,$$

which nests (B.17) when Λ' reduces to point masses of $+(1 - \lambda)$ at $y = \bar{x}$ and $-(1 - \lambda)$ at $y = -\bar{x}$. Proposition 1 gives $b_0 = \omega[J_{P,p^*}^{MC} \mathbf{1}]_0$, equation (31). This is the index of macroeconomic flexibility of Caballero and Engel (2007).

Horizon-1 entry. Write $v = \omega(\epsilon_{it}^z + \epsilon_{it})$ for the desired-price innovation, with density f : symmetric and single-peaked as in Appendix B.4.1, but not Gaussian, since only ϵ_{it} is (the convolution of the symmetric single-peaked law of ϵ_{it}^z with a Gaussian is symmetric and single-peaked). Only the symmetry and continuity of f are used below. Let

$$(B.19) \quad m(x) \equiv -\mathbb{E}_v[(x - v)\Lambda(x - v)]$$

be the expected price change next period of a firm that carries the post-adjustment gap x into the period: it adjusts with probability $\Lambda(x - v)$ and then closes the gap $x - v$. In the notation of the footnote to (33), $m(x) = E^1(x) - x$, so the one-period virtual survival values are $\Phi_1^e = E^1(\bar{x})/\bar{x} = 1 + m(\bar{x})/\bar{x}$ and $\Phi_1^i = E^{1'}(0) = 1 + m'(0)$. Since f is symmetric and Λ even, m is odd, $m(0) = 0$, and $m(\bar{x}) < 0$: a firm at the upper edge of the band expects to lower its price. Its derivative is

$$(B.20) \quad -m'(x) = \Lambda_1(x) + (1 - \lambda)\bar{x}[f(x - \bar{x}) + f(x + \bar{x})], \quad \Lambda_1(x) \equiv \mathbb{E}_v[\Lambda(x - v)],$$

the firm-level impact response to a displacement of its gap: its probability of adjusting next period, $\Lambda_1(x)$, plus the extensive margin generated by its own next-period gap density $f(\cdot - x)$ at the edges of the band. Averaging (B.20) over the stationary post-adjustment law μ of (B.13) and using $g(y) = \int f(y - x)\mu(dx)$ recovers (B.17): $\int (-m') d\mu = \text{freq} + (1 - \lambda)\bar{x}[g(\bar{x}) + g(-\bar{x})]$.

Consider the shifted cross-section of date t . The post-adjustment law under the shift, μ_δ , has an atom at zero of mass $\text{freq}(\delta) = \lambda + (1 - \lambda) \Pr(|y - \delta| > \bar{x})$ and, on the band, the density $(1 - \lambda)g(x + \delta)$ of the non-adjusters, whose gap has been displaced by $-\delta$; by the symmetry of g , $\text{freq}'(0) = (1 - \lambda)[g(-\bar{x}) - g(\bar{x})] = 0$, so to first order the shift only reshapes the band density, $\partial_\delta \mu_\delta = (1 - \lambda)g'(x) dx$ on $[-\bar{x}, \bar{x}]$. The desired price is at its shifted level from t on and the problem is stationary again, so the average price change at $t + 1$ is $\int m(x) \mu_\delta(dx)$, and its derivative at $\delta = 0$ is the increment of the pass-through between the two dates:

$$(B.21) \quad \begin{aligned} [J_{P,p^*}^{\text{MC}} \mathbf{1}]_1 - [J_{P,p^*}^{\text{MC}} \mathbf{1}]_0 &= (1 - \lambda) \int_{-\bar{x}}^{\bar{x}} m(x) g'(x) dx \\ &= 2(1 - \lambda) g(\bar{x}) m(\bar{x}) + (1 - \lambda) \int_{-\bar{x}}^{\bar{x}} [-m'(x)] g(x) dx, \end{aligned}$$

by integration by parts, m being odd and g even. Substitute (B.20) and use, with $F_v(\bar{x}) \equiv \Pr(|v| \leq \bar{x})$ and the convolution identity for g ,

$$\begin{aligned} (1 - \lambda) \int_{-\bar{x}}^{\bar{x}} \Lambda_1(x) g(x) dx &= \Pr(\text{keep at } t, \text{ adjust at } t + 1) = \text{freq} (1 - \lambda) F_v(\bar{x}), \\ (1 - \lambda) \int_{-\bar{x}}^{\bar{x}} f(x \mp \bar{x}) g(x) dx &= g(\pm \bar{x}) - \text{freq} f(\pm \bar{x}), \end{aligned}$$

so that

$$(B.22) \quad \begin{aligned} [J_{P,p^*}^{\text{MC}} \mathbf{1}]_1 &= [J_{P,p^*}^{\text{MC}} \mathbf{1}]_0 + \underbrace{\text{freq} (1 - \lambda) F_v(\bar{x})}_{\text{intensive margin at } t+1} + \underbrace{2(1 - \lambda) \bar{x} [g(\bar{x}) - \text{freq} f(\bar{x})]}_{\text{extensive margin at } t+1} \\ &\quad - \underbrace{2(1 - \lambda) g(\bar{x}) |m(\bar{x})|}_{\text{unwinding of the date-}t \text{ extensive margin}}. \end{aligned}$$

At $t + 1$, the response comes from date- t non-adjusters and from the firms that were marginal at the date- t thresholds. Date- t adjusters start from gap zero relative to the shifted desired price and therefore behave as in the steady state, contributing nothing further. Date- t non-adjusters carry a gap displaced by $-\delta$ and generate a new impact response at $t + 1$: their adjustment probability gives the intensive term, while $\bar{x}$ times their mass at the band edges gives the extensive term. Their contribution to the density at $\pm \bar{x}$ is $g(\bar{x}) - \text{freq} f(\bar{x})$, where the second term is the part generated by the atom at zero. The

extensive-margin contribution at $t + 1$ is therefore smaller than at t .

The firms that were marginal at date t generate an additional unwinding term. The firm pulled inside the band at $+\bar{x}$ expects a price change $m(\bar{x}) < 0$ rather than zero, while the firm pushed out at $-\bar{x}$ was reset to zero rather than carrying the expected change $m(-\bar{x}) > 0$. These terms offset part of the date- t extensive response. Hence, in the two-margin representation (33), the survival function associated with the extensive margin is virtual.

Equation (B.22) is the two-margin representation at horizon one. With the virtual survival functions of the footnote to (33), $\Phi_s^i = E^{s'}(0)$ and $\Phi_s^e = E^s(\bar{x})/\bar{x}$, and since $E^1(x) = x + m(x)$ (the expected gap next period is the current gap plus the expected price change, the innovation having mean zero),

$$(B.23) \quad \Phi_1^i = 1 + m'(0) = (1 - \lambda)F_v(\bar{x}) - 2(1 - \lambda)\bar{x}f(\bar{x}), \quad \Phi_1^e = 1 + \frac{m(\bar{x})}{\bar{x}},$$

and regrouping the terms of (B.22) with the help of (B.17),

$$(B.24) \quad [J_{P,p^*}^{\text{MC}} \mathbf{1}]_1 = \text{freq} (1 + \Phi_1^i) + 2(1 - \lambda) g(\bar{x}) \bar{x} (1 + \Phi_1^e),$$

which is Lemma 2 applied to (33) at $h = 1$, i.e. (34) with $b_1/\omega = (b_0/\omega)(1 + \bar{\Phi}_1)$. Two remarks. First, Φ_1^i is not the plain one-period survival probability $(1 - \lambda)F_v(\bar{x})$ of a freshly reset price: it carries the firm-level extensive correction $-2(1 - \lambda)\bar{x}f(\bar{x})$, because the intensive component of the representation is the derivative of the expected gap at the reset point, and that derivative includes the mass of freshly reset firms that the next innovation places at the edge of the band. Equations (B.22) and (B.24) are algebraically equivalent decompositions of the same response. Second, in primitives the statistic of Proposition 4 is

$$(B.25) \quad \phi^{\text{IRF}} = \frac{b_1}{b_0} - 1 = \bar{\Phi}_1 = \frac{\text{freq} \Phi_1^i + 2(1 - \lambda) g(\bar{x}) \bar{x} \Phi_1^e}{\text{freq} + 2(1 - \lambda) g(\bar{x}) \bar{x}},$$

with Φ_1^i and Φ_1^e given by (B.23). The extensive component decays faster in the benchmark calibration, $\Phi_1^e < \Phi_1^i$. The same recursion holds at every horizon, $[J_{P,p^*}^{\text{MC}} \mathbf{1}]_h - [J_{P,p^*}^{\text{MC}} \mathbf{1}]_{h-1} = \text{freq} \Phi_h^i + 2(1 - \lambda)g(\bar{x})\bar{x} \Phi_h^e$. Section 3.3.3 extrapolates from (b_0, b_1) .

Comparison with 1 – freq: selection versus the hazard. Section 3.3.3 states that $\phi^{\text{IRF}} = \bar{\Phi}_1$ falls short of 1 – freq, the survival probability of a price spell. The closed forms above characterize the difference between ϕ^{IRF} and 1 – freq. Let $q(x) \equiv \Pr_v(|x - v| > \bar{x})$ be the probability that a firm carrying the post-adjustment gap x into the next period is pushed out of the band, so that $\Lambda_1(x) = \lambda + (1 - \lambda)q(x)$. Since f is symmetric and decreasing on $\mathbb{R}_+$, q is even and increasing in $|x|$: the threshold firm is the most exposed to exit, the reset firm the least. Averaging Λ_1 over the post-adjustment law μ of (B.13) gives the frequency, $\text{freq} = \lambda + (1 - \lambda) \int q d\mu$, and $F_v(\bar{x}) = 1 - q(0)$. By (B.25),

$$(B.26) \quad [J_{P,p^*}^{\text{MC}} \mathbf{1}]_0 [\phi^{\text{IRF}} - (1 - \text{freq})] = \text{freq} [\Phi_1^i - (1 - \text{freq})] + 2(1 - \lambda) g(\bar{x}) \bar{x} [\Phi_1^e - (1 - \text{freq})],$$

so $\phi^{\text{IRF}} - (1 - \text{freq})$ has the sign of a weighted sum of the two component gaps, weighted by the intensive and extensive masses of (B.17).

The extensive component is always below 1 – freq. From (B.23), $1 - \Phi_1^e = |m(\bar{x})|/\bar{x}$. Splitting the expectation in $m(\bar{x})$ according to whether the innovation pushes the threshold firm out of the band ($v < 0$ or $v > 2\bar{x}$) or leaves it inside, and using the symmetry of f ,

$$(B.27) \quad \begin{aligned} 1 - \Phi_1^e &= \lambda + (1 - \lambda) q(\bar{x}) + \frac{1 - \lambda}{\bar{x}} \int_0^{2\bar{x}} v f(v) dv, \\ \text{hence} \quad \Phi_1^e - (1 - \text{freq}) &= -(1 - \lambda) \left[q(\bar{x}) - \int q d\mu + \frac{1}{\bar{x}} \int_0^{2\bar{x}} v f(v) dv \right] < 0. \end{aligned}$$

The first term reflects the higher exit probability at the threshold, while the second captures mean reversion conditional on remaining inside the band; both are nonnegative. Since $q(\bar{x}) \geq \Pr(v < 0) = \frac{1}{2}$, also $\Phi_1^e < (1 - \lambda)/2$: a marginal gap at the threshold loses more than half of itself in one period, whatever the innovation density.

The intensive component can be on either side. From (B.23), $\Phi_1^i = (1 - \lambda)[1 - q(0)] - 2(1 - \lambda)\bar{x}f(\bar{x})$, so

$$(B.28) \quad \Phi_1^i - (1 - \text{freq}) = (1 - \lambda) \left[\underbrace{\int q d\mu - q(0)}_{\text{increasing hazard}} - \underbrace{2\bar{x}f(\bar{x})}_{\text{reset-point selection}} \right].$$

The first term is positive because a freshly reset price at gap zero exits less often than the average price. The second term captures reset-point selection: among freshly reset firms, a

mass $2\bar{x}f(\bar{x})$ is placed at the band edges by the next innovation. Their relative magnitude depends on the width of the band relative to the innovation dispersion. For narrow and medium bands, the second term dominates and $\Phi_1^i < 1 - \text{freq}$; for sufficiently wide bands, the first term dominates and $\Phi_1^i > 1 - \text{freq}$.

Sufficient conditions. Write $H \equiv (1 - \lambda)[\int q d\mu - q(0)] = (1 - \lambda)^2 \int_{-\bar{x}}^{\bar{x}} [q(x) - q(0)]g(x) dx$ for the hazard term and $w \equiv 2(1 - \lambda)g(\bar{x})\bar{x}$ for the extensive mass. Then $H \leq w$: for $|x| \leq \bar{x}$, $q(x) - q(0) = \int_0^{|x|} [f(\bar{x} - s) - f(\bar{x} + s)] ds \leq \int_0^{|x|} f(\bar{x} - s) ds \leq \bar{x} f(\bar{x} - |x|)$, because f is decreasing on $\mathbb{R}_+$, so that

$$\begin{aligned} H &\leq (1 - \lambda)^2 \bar{x} \int_{-\bar{x}}^{\bar{x}} f(\bar{x} - |x|) g(x) dx = 2(1 - \lambda)^2 \bar{x} \int_0^{\bar{x}} f(\bar{x} - x) g(x) dx \\ &\leq 2(1 - \lambda)^2 \bar{x} \int_{-\bar{x}}^{\bar{x}} f(\bar{x} - x) g(x) dx \leq 2(1 - \lambda) \bar{x} g(\bar{x}) = w, \end{aligned}$$

the last step by the convolution identity $g(\bar{x}) = \text{freq}f(\bar{x}) + (1 - \lambda) \int_{-\bar{x}}^{\bar{x}} f(\bar{x} - x)g(x) dx$. Substituting (B.27) and (B.28) into (B.26), with $2(1 - \lambda)\bar{x}f(\bar{x}) = w f(\bar{x})/g(\bar{x})$,

$$\begin{aligned} (B.29) \quad [J_{P,p^*}^{\text{MC}} \mathbf{1}]_0 [\phi^{\text{IRF}} - (1 - \text{freq})] &= \text{freq} H - \text{freq} w \frac{f(\bar{x})}{g(\bar{x})} - w [1 - \Phi_1^e - \text{freq}] \\ &\leq w \left\{ \text{freq} \left[2 - \frac{f(\bar{x})}{g(\bar{x})} \right] - (1 - \Phi_1^e) \right\}. \end{aligned}$$

Therefore $\phi^{\text{IRF}} < 1 - \text{freq}$ whenever either

- (a) reset-point selection dominates the hazard, $2\bar{x}f(\bar{x}) \geq \int q d\mu - q(0)$: then both component gaps are nonpositive by (B.27)–(B.28), and $\phi^{\text{IRF}} < 1 - \text{freq}$ as a weighted average; or
- (b) $\text{freq} [2 - f(\bar{x})/g(\bar{x})] \leq 1 - \Phi_1^e$, by (B.29); since $1 - \Phi_1^e > (1 + \lambda)/2$ by (B.27) and $f(\bar{x})/g(\bar{x}) \geq 0$, this holds in particular whenever $\text{freq} \leq (1 + \lambda)/4$.

Consistency with the pass-through mixture (33). Equation (B.24) was already identified above as Lemma 2 applied to (33) at $h = 1$; the same argument, run once for general h , recovers both closed forms of this subsection together. Lemma 2 gives $[J_{P,p^*}^{\text{TD}}(\Phi^k) \mathbf{1}]_h = \sum_{s=0}^h \Phi_s^k / \sum_{s \geq 0} \Phi_s^k$ for $k \in \{e, i\}$, so that

$$[J_{P,p^*}^{\text{MC}} \mathbf{1}]_h = \varrho_e \frac{\sum_{s=0}^h \Phi_s^e}{\sum_{s \geq 0} \Phi_s^e} + \varrho_i \frac{\sum_{s=0}^h \Phi_s^i}{\sum_{s \geq 0} \Phi_s^i}.$$

Since $E^0(x) = x$, $\Phi_0^e = \Phi_0^i = 1$, and the mixture weights $\varrho_e = 2(1 - \lambda)g(\bar{x})\bar{x} \sum_s \Phi_s^e$, $\varrho_i = \text{freq} \sum_s \Phi_s^i$ cancel the two denominators at every horizon, leaving

$$[J_{P,p^*}^{\text{MC}} \mathbf{1}]_h = 2(1 - \lambda)g(\bar{x})\bar{x} \sum_{s=0}^h \Phi_s^e + \text{freq} \sum_{s=0}^h \Phi_s^i.$$

At $h = 0$ this is (B.17), and at $h = 1$ it is (B.24).

B.4.3. Proof of Proposition 4

Writing $e \equiv E\mathbf{1}$ and using $b_h = \omega[J_{P,p^*}^{\text{MC}} \mathbf{1}]_h$ (Proposition 1) together with Lemma 1(i),

$$(B.30) \quad \frac{b_h}{\omega} = 1 - (\alpha^*)^{h+1} + e_h, \quad \text{so} \quad \frac{b_0}{\omega} = 1 - \alpha^* + e_0, \quad \frac{b_1}{\omega} = 1 - (\alpha^*)^2 + e_1.$$

The real rigidity cancels from the ratio b_1/b_0 . Substituting into $\phi^{\text{IRF}} = b_1/b_0 - 1$ and subtracting α^* ,

$$(B.31) \quad \phi^{\text{IRF}} - \alpha^* = \frac{e_1 - (1 + \alpha^*)e_0}{1 - \alpha^* + e_0}.$$

Under Assumption 1, $|e_0| \leq \bar{e}$ and $|e_1| \leq \bar{e}$ (since $\|E\mathbf{1}\|_t \leq \|E\|_H$ for $t \leq H$, with $H \geq 1$), and the denominator of (B.31) is at least $1 - \alpha^* - \bar{e} > 0$, so

$$(B.32) \quad |\phi^{\text{IRF}} - \alpha^*| \leq \frac{(2 + \alpha^*)\bar{e}}{1 - \alpha^* - \bar{e}}.$$

The interval $I_\alpha = [\min\{\alpha^*, \phi^{\text{IRF}}\}, \max\{\alpha^*, \phi^{\text{IRF}}\}]$ satisfies $I_\alpha \subset (0, 1)$ as long as $\bar{\Phi}_1 \in (0, 1)$, as in every calibration we consider, since $\phi^{\text{IRF}} = \bar{\Phi}_1$; on it, $M_H(I_\alpha) = \sup_{\alpha \in I_\alpha} \|\partial J_{P,p^*}^{\text{C}}(\alpha)/\partial \alpha\|_H$ is finite. Writing the difference of the two Calvo Jacobians as an integral and taking norms (the mean-value inequality),

$$\|J_{P,p^*}^{\text{C}}(\phi^{\text{IRF}}) - J_{P,p^*}^{\text{C}}(\alpha^*)\|_H = \left\| \int_{\alpha^*}^{\phi^{\text{IRF}}} \frac{\partial J_{P,p^*}^{\text{C}}(\alpha)}{\partial \alpha} d\alpha \right\|_H \leq M_H(I_\alpha) |\phi^{\text{IRF}} - \alpha^*|.$$

Combining with (B.32) and the triangle inequality applied to $J_{P,p^*}^{\text{MC}} - J_{P,p^*}^{\text{C}}(\phi^{\text{IRF}}) = E + J_{P,p^*}^{\text{C}}(\alpha^*) - J_{P,p^*}^{\text{C}}(\phi^{\text{IRF}})$,

$$\|J_{P,p^*}^{\text{MC}} - J_{P,p^*}^{\text{C}}(\phi^{\text{IRF}})\|_H \leq \|E\|_H + M_H(I_\alpha) |\phi^{\text{IRF}} - \alpha^*| \leq \bar{e} + M_H(I_\alpha) \frac{(2 + \alpha^*) \bar{e}}{1 - \alpha^* - \bar{e}},$$

which is (38).

Part (ii). Write $J \equiv J_{P,p^*}^{\text{MC}}$ and $\tilde{J} \equiv J_{P,p^*}^{\text{C}}(\phi^{\text{IRF}})$, both truncated at the reporting horizon H . On $(H+1) \times (H+1)$ matrices the maximum-absolute-row-sum norm is the operator norm induced by the ℓ^∞ vector norm, hence submultiplicative, and $\|I - L\| \leq 2$. For the truncated Calvo matrix, $\|\tilde{J}\| = \max_{h \leq H} (1 - (\phi^{\text{IRF}})^{h+1}) < 1$ (its entries are positive with row sums $1 - (\phi^{\text{IRF}})^{h+1}$), so $I - \tilde{J}$ is invertible; invertibility of the truncated $I - J$ is assumed in the statement (it holds in all our computations, and whenever $\|J\| < 1$). Using $(I - A)^{-1}A = (I - A)^{-1} - I$ for both matrices and differencing,

$$(B.33) \quad (I - \tilde{J})^{-1}\tilde{J} - (I - J)^{-1}J = (I - \tilde{J})^{-1}(\tilde{J} - J)(I - J)^{-1}.$$

By (12)–(13), $J_{\pi,mc^r}^{\text{MC}} - J_{\pi,mc^r}^{\text{C}}(\phi^{\text{IRF}}) = -\omega(I - L)[(I - \tilde{J})^{-1}\tilde{J} - (I - J)^{-1}J]$, so taking norms and using submultiplicativity,

$$\|J_{\pi,mc^r}^{\text{MC}} - J_{\pi,mc^r}^{\text{C}}(\phi^{\text{IRF}})\| \leq \omega \|I - L\| \|(I - \tilde{J})^{-1}\| \|\tilde{J} - J\| \|(I - J)^{-1}\| \leq 2\omega \|(I - \tilde{J})^{-1}\| \|(I - J)^{-1}\| B_H,$$

where the last step uses that the truncated difference obeys $\|\tilde{J} - J\| \leq \|J_{P,p^*}^{\text{C}}(\phi^{\text{IRF}}) - J_{P,p^*}^{\text{MC}}\|_H \leq B_H$: its row sums over $s \leq H$ are bounded by the corresponding row sums over all s , and part (i) applies. This is (39).

B.5. Proof of Proposition 1

Expectations. Fix a date t and a firm i . Everything random about the firm's prices from t to $t+h$ is one of three objects: its predetermined state S (the gap $x_{i,t-1}$, the predetermined price $p_{i,t-1}$, and $\vartheta_{i,t-1}$ in the AR(1) case); its identifying innovation ϵ_{it} ; and $\mathcal{D}$, all other random draws affecting its prices from t to $t+h$ (the innovations $\epsilon_{i,t+j}^z$ and $\epsilon_{i,t+j}$ for $j \geq 1$, the date- t innovation ϵ_{it}^z , and the menu-cost or lottery draws). Their laws are $S \sim \mu_S$, the cross-sectional law of the state at date t , stationary in its gap component (price levels

have no stationary law, but $p_{i,t-1}$ enters every object below only additively and cancels from every cumulative price change); $\epsilon_{it} \sim \mu_\epsilon$, with density φ_σ in the Gaussian case; and $\mathcal{D} \sim \mu_{\mathcal{D}}$. The three are mutually independent: ϵ_{it} is independent of $(S, \mathcal{D})$ by the maintained assumption of (8) (i.i.d. over time, independent of the z process and of all adjustment draws), and $\mathcal{D}$ consists of date- t draws other than ϵ_{it} and of later innovations and draws, all independent of S . We denote expectations with a subscript naming the law it integrates over, the remaining arguments being held fixed: $\mathbb{E}_\epsilon$ integrates against μ_ϵ alone, $\mathbb{E}_{S,\mathcal{D}}$ against $\mu_S \otimes \mu_{\mathcal{D}}$, $\mathbb{E}_S$ against μ_S , and

$$(B.34) \quad \mathbb{E}_{S,\epsilon,\mathcal{D}} \equiv \mathbb{E}_{S,\mathcal{D}} \mathbb{E}_\epsilon = \mathbb{E}_\epsilon \mathbb{E}_{S,\mathcal{D}}$$

integrates against the product law $\mu_S \otimes \mu_\epsilon \otimes \mu_{\mathcal{D}}$. This last one is the *cross-sectional* expectation: the cross-section of firms at date t is the ergodic law of a single firm's $(S, \epsilon_{it}, \mathcal{D})$, so averaging over the distribution of firms means integrating against $\mu_S \otimes \mu_\epsilon \otimes \mu_{\mathcal{D}}$. Variances, covariances and probabilities carry the same subscripts.

Road map. The proof has two steps, stated as two lemmas. Conditional on $(S, \mathcal{D})$, the firm's cumulative price change is a deterministic function of its innovation, $p_{i,t+h} - p_{i,t-1} = f_h(\epsilon_{it}; S, \mathcal{D})$. Define the cross-sectional average derivative of the price path with respect to the firm's own shifter,

$$(B.35) \quad A_h \equiv \frac{d}{d\delta} \mathbb{E}_{S,\epsilon,\mathcal{D}} [f_h(\epsilon_{it} + \delta; S, \mathcal{D})] \Big|_{\delta=0},$$

the derivative of the average cumulative price change when every firm's identifying innovation is raised by δ . Step 1 (Section B.5.1, Lemma B.3) shows $[J_{P,mc}^m \mathbf{1}]_h = A_h$ in every model, using only the structure of the firm's problem, the random-walk structure of the shifter, and no property of the marginal law of ϵ_{it} . Step 2 (Section B.5.2, Lemma B.4) shows $A_h = b_h^m$: the regression coefficient equals the average derivative. This is where the distribution of the identifying shock matters, and where Gaussianity enters for the menu cost model. Proposition 1 is the two lemmas combined.

B.5.1. Step 1: The Aggregate Response Is the Average Derivative

Fix a date t and a firm i , with S , ϵ_{it} , $\mathcal{D}$ and f_h as defined above. A shift of ϵ_{it} by δ is the experiment behind A_h in (B.35).

LEMMA B.3 (Aggregation). *In each model of Section 2.1, with a random-walk shifter, let $R_h(\delta)$ be the mean price at $t + h$ when nominal marginal cost is shifted by δ from date t on, so that every firm's desired price is shifted by $\omega\delta$, with the price level and all other aggregates at their steady-state values and no feedback from prices to costs, as in (11), starting from the stationary cross-section. Then, for every δ (in (GL)), for every δ within the slack of the small-shock bounds (A2)–(A3)),*

$$(B.36) \quad R_h(\delta) = \mathbb{E}_S[p_{i,t-1}] + \mathbb{E}_{S,\epsilon,\mathcal{D}}[f_h(\epsilon_{it} + \delta; S, \mathcal{D})],$$

and, whenever R_h is differentiable at 0 (equivalently, by (B.36), whenever A_h exists),

$$(B.37) \quad [J_{P,mc}^m \mathbf{1}]_h = R'_h(0) = A_h.$$

Beyond the random-walk structure of ϑ and the independence of ϵ_{it} from later innovations and adjustment draws, which keep the continuation problem stationary, no property of the law of ϵ_{it} is used: its shape, moments and support, and its dependence on the predetermined state, play no role.

PROOF. Fix $(S, \epsilon_{it}, \mathcal{D})$ and a real δ , and compare two experiments, both started from the same state and driven by the same draws:

- (E1) the firm's own innovation is $\epsilon_{it} + \delta$ instead of ϵ_{it} ; nominal marginal cost stays at its steady-state value;
- (E2) the firm's innovation is ϵ_{it} ; nominal marginal cost rises permanently and unexpectedly by δ at date t , with the price level and all other aggregates held at their steady-state values as in (11).

Write $p_{i,t+h}^{(1)}(\delta)$ and $p_{i,t+h}^{(2)}(\delta)$ for the resulting prices at $t+h$; both are deterministic functions of $(S, \epsilon_{it}, \mathcal{D}, \delta)$.

(a) *Pathwise equivalence:* $p_{i,t+h}^{(1)}(\delta) = p_{i,t+h}^{(2)}(\delta)$ for every $(S, \epsilon_{it}, \mathcal{D})$ and every δ . By (5)—the normalized problem of Appendix A.1, in which the menu cost is proportional to frictionless revenue, so that a common cost shift leaves the normalized menu cost unchanged—the firm's objective and its adjustment technology depend on $(mc_\tau, z_{i\tau}, \vartheta_{i\tau}, P_\tau)$ only through the desired price $p_{i\tau}^*$, and on $p_{i\tau}^*$ only through the gap $x_{i\tau} = p_{i\tau} - p_{i\tau}^*$; its optimal policy at

any date $\tau \geq t$ is therefore a function of the pre-adjustment gap at τ and of the conditional law of the future increments $\{p_{i,\tau+s}^* - p_{i\tau}^*\}_{s \geq 1}$. In (E1), $p_{i,t+s}^*$ rises by $\omega\delta$ for every $s \geq 0$ because ϑ is a random walk and enters (7) with weight ω , and the law of the future increments is unchanged because ϵ_{it} is independent of all later innovations and draws. In (E2), $p_{i,t+s}^*$ rises by the same $\omega\delta$ for every $s \geq 0$, since mc enters (7) with the same weight and the price level is held fixed, and the law of the future increments is unchanged because the shift is permanent and not revised. Hence in both experiments the pre-adjustment gap at t is $x_{i,t-1} - v_{it} - \omega\delta$, the desired price at $t+h$ is $p_{i,t+h}^* + \omega\delta$, and the continuation from t is solved by the stationary policies (bands, reset gap $x^* = 0$, hazard) applied to the same draws in $\mathcal{D}$. The realized price is therefore, in both experiments,

$$(B.38) \quad p_{i,t+h} = p_{i,t+h}^* + \omega\delta + \mathcal{X}_h(x_{i,t-1} - v_{it} - \omega\delta; v_{i,t+1}, \dots, v_{i,t+h}; \text{adjustment draws}),$$

where $v_{i\tau} = \omega(\epsilon_{i\tau}^z + \epsilon_{i\tau})$ is the date- τ innovation to the desired price (with $z_{i\tau} - z_{i,\tau-1}$ in place of $\epsilon_{i\tau}^z$ when z is not a random walk) and $\mathcal{X}_h$ is the gap h periods later under the stationary policies; when those policies condition on further state variables (z_{it} in (C) if z is not a random walk, ϑ_{it} in the AR(1) case), these enter $\mathcal{X}_h$ as additional arguments, none of which depends on δ . In (GL) the argument holds for $|\delta|$ within the slack of the small-shock bounds (A2)–(A3), under which the policies of Appendix A.2 remain the firm's policies.

In the firm's problem, the two shocks enter identically: both shift $p_{it}^* = \omega(mc_t + z_{it} + \vartheta_{it}) + (1-\omega)P_t$ by $\omega\delta$. Hence the two pass-throughs are instances of the same desired-price pass-through, scaled by ω .

(b) *Aggregation.* The cross-section at date t is the law $\mu_S \otimes \mu_\epsilon \otimes \mu_{\mathcal{D}}$ of $(S, \epsilon_{it}, \mathcal{D})$ in both experiments: a common cost shift alters no idiosyncratic draw. Since the price level and aggregate output (P_t, Y_t) are held fixed, the shifted cost mc_t is common to all firms, and, given these, no firm's problem involves any other firm's price—with decreasing returns the firm's own output enters its own marginal cost, but that is a function of its own price and of the aggregates, and Appendix A.1 shows it is absorbed in the profit function $R_{it}^* h_\omega(x_{it})$, i.e. in the weight ω of the desired price and in the curvature—firms respond independently, and the mean price in (E2) is the integral of $p_{i,t+h}^{(2)}(\delta)$ against that law.

Therefore

$$R_h(\delta) = \mathbb{E}_{S,\epsilon,\mathcal{D}}[p_{i,t+h}^{(2)}(\delta)] = \mathbb{E}_{S,\epsilon,\mathcal{D}}[p_{i,t+h}^{(1)}(\delta)] = \mathbb{E}_S[p_{i,t-1}] + \mathbb{E}_{S,\epsilon,\mathcal{D}}[f_h(\epsilon_{it} + \delta; S, \mathcal{D})],$$

the middle equality by (a) and the last by the definition of f_h , since $p_{i,t+h}^{(1)}(\delta) = p_{i,t-1} + f_h(\epsilon_{it} + \delta; S, \mathcal{D})$; this is (B.36). Subtracting $\mathbb{E}_S[p_{i,t-1}]$, which does not depend on δ , and differentiating at $\delta = 0$ gives $R'_h(0) = A_h$ whenever either side exists. Finally, $[J_{P,mc}^m \mathbf{1}]_h$ is by definition the derivative of the aggregate price at horizon h with respect to a permanent unit shift of nominal marginal cost at date 0, unanticipated and with aggregates fixed, starting from the stationary cross-section; relabelling date 0 as t is admissible because the cross-section is the same at every date. Hence $[J_{P,mc}^m \mathbf{1}]_h = R'_h(0) = A_h$, which is (B.37). $\square$

When the derivative can be moved inside the expectation over $(S, \mathcal{D})$, equation (B.37) reads $[J_{P,mc}^m \mathbf{1}]_h = \mathbb{E}_{S,\mathcal{D}}[\frac{d}{d\delta} \mathbb{E}_\epsilon[f_h(\epsilon + \delta; S, \mathcal{D})] \big|_{\delta=0}]$. This is the aggregation logic of Caballero and Engel (2007): the cross-section over which the aggregate response averages is the ergodic distribution of a single firm's states, so the macro response is the average of micro marginal responses.

Note that Lemma B.3 requires no continuity of f_h in ϵ and no density for $S, \mathcal{D}$ or ϵ_{it} .²⁷

Step (a) is also where the limits of the identification appear: the regression generates the displacement whose profile is the impulse response of p^* to the shifter innovation, a permanent step for a random-walk shifter. An mc path whose profile differs from the shifter's own impulse response—a news shock, or a transitory shift when the shifter is a random walk—is not that displacement, the policies respond to the anticipated path, and no regression on innovations to that shifter can reproduce it. This is why the bridge delivers exactly $J_{P,mc}^m \mathbf{1}$ and no other feature of the matrix.

²⁷This is because it differentiates nothing inside an expectation: it identifies R_h with the cross-sectional average of the shifted price path and then differentiates that average. Whether R_h is differentiable at 0 is a property of the model's aggregate response, satisfied under the assumptions of Section 2.1 on the idiosyncratic shocks. Take the case (MC): conditional on $(S, \epsilon_{it}, \mathcal{D})$, a firm's price at $t + h$ is a step function of the common shift: it equals the old price until δ tips the firm across the band and the reset price afterwards. Its pointwise derivative exists for almost every realization and takes values in $\{0, \omega\}$ (ω if the firm has adjusted since t , zero otherwise), but its expectation is only the intensive margin: the integrand jumps at the values of δ at which the firm's adjustment decision flips, and the mass swept across those jumps, the extensive margin, is missed by any argument that differentiates realization by realization. A derivative of the *aggregate* exists because, across firms, the pre-adjustment gap $x_{i,t-1} - \omega(\epsilon_{it}^z + \epsilon_{it})$ has a continuous density, so that the mass tipped over by a small δ is proportional to δ .

Persistent shifters. If $\vartheta_{it} = \rho\vartheta_{i,t-1} + \epsilon_{it}$, a shift δ in ϵ_{it} raises $p_{i,t+s}^*$ by $\omega\rho^s\delta$, other innovations held fixed, exactly as an unanticipated cost path $(\delta, \rho\delta, \rho^2\delta, \dots)$ does with the price level fixed. The firm's state now includes ϑ_{it} ; along the shifted path the pair (gap, ϑ) evolves exactly as it would under an unanticipated cost path $(\delta, \rho\delta, \rho^2\delta, \dots)$ starting from the same state, with identical policy functions (the policy under the anticipated cost path is formally time-varying but coincides at $t+j$ with the stationary policy evaluated at $\vartheta_{i,t+j} + \rho^j\delta$). Lemma B.3, combined with Lemma B.4, then gives $b_h^m = [J_{P,mc}^m \tilde{\vartheta}]_h$ with $\tilde{\vartheta} = (1, \rho, \rho^2, \dots)'$, provided ϵ_{it} is independent of $(x_{i,t-1}, \vartheta_{i,t-1})$, the domination (B.46) still holds (it does, since $\rho \leq 1$), and the same ρ is used for the shifter and for the cost path.

B.5.2. Step 2: The Average Derivative Is the Regression Coefficient

Since ϵ_{it} is independent of $(S, \mathcal{D})$, by definition of the OLS coefficient,

$$(B.39) \quad b_h^m = \frac{\mathbb{E}_{S,\epsilon,\mathcal{D}}[f_h(\epsilon_{it}; S, \mathcal{D}) \epsilon_{it}]}{\sigma_{\vartheta}^2} = \mathbb{E}_{S,\mathcal{D}} \left[\frac{\mathbb{E}_{\epsilon}[f_h(\epsilon; S, \mathcal{D}) \epsilon]}{\sigma_{\vartheta}^2} \right].$$

The lemma gives two sufficient conditions under which this regression coefficient equals A_h .

LEMMA B.4 (Regression coefficient equals average derivative). *Let $(S, \epsilon_{it}, \mathcal{D})$ be mutually independent, with ϵ_{it} of mean zero and variance σ_{ϑ}^2 , and let $|f_h(\epsilon; S, \mathcal{D})| \leq M_h(S, \mathcal{D}) + |\epsilon|$ for all ϵ , with $\mathbb{E}_{S,\mathcal{D}}[M_h] < \infty$, so that b_h^m is well defined. Suppose that either*

(G) $\epsilon_{it} \sim N(0, \sigma_{\vartheta}^2)$; or

(L) $F_h(\epsilon; S) \equiv \mathbb{E}_{\mathcal{D}}[f_h(\epsilon; S, \mathcal{D})]$ is affine in ϵ on an open set containing the support of ϵ_{it} , for μ_S -almost every S : $F_h(\epsilon; S) = a_h(S) + s_h(S)\epsilon$ there, with $\mathbb{E}_S|a_h|, \mathbb{E}_S|s_h| < \infty$.

Then A_h exists and $A_h = b_h^m$. Under (G), moreover, $A_h = \mathbb{E}_{S,\mathcal{D}} \left[\frac{d}{d\delta} \mathbb{E}_{\epsilon}[f_h(\epsilon + \delta; S, \mathcal{D})] \right]_{\delta=0}$.

PROOF. Case (L). Because $(S, \epsilon_{it}, \mathcal{D})$ are mutually independent and f_h is dominated by $M_h + |\epsilon|$, Fubini's theorem allows integrating $\mathcal{D}$ out first in both objects, for $|\delta|$ small enough that $\epsilon + \delta$ stays in the set where F_h is affine. For the regression coefficient, $\mathbb{E}_{\epsilon}[F_h(\epsilon; S)\epsilon] = \mathbb{E}_{\epsilon}[(a_h + s_h\epsilon)\epsilon] = s_h(S)\sigma_{\vartheta}^2$, so $b_h^m = \mathbb{E}_S[s_h]$. For the average derivative, $\mathbb{E}_{S,\epsilon,\mathcal{D}}[f_h(\epsilon + \delta)] = \mathbb{E}_S[\mathbb{E}_{\epsilon}[a_h + s_h(\epsilon + \delta)]] = \mathbb{E}_S[a_h] + \delta \mathbb{E}_S[s_h]$, which is affine in δ , so $A_h = \mathbb{E}_S[s_h]$. No exchange of limit and expectation is needed, and no property of the law of ϵ_{it} beyond a finite variance

and the domination above.

Case (G). The argument has three parts: Stein's identity for a single Gaussian variable, its application conditionally on $(S, \mathcal{D})$, and the exchange of $d/d\delta$ with $\mathbb{E}_{S, \mathcal{D}}$.

Stein's identity. Let $\epsilon \sim N(0, \sigma^2)$ with density φ_σ , let f be measurable with polynomial growth, $|f(u)| \leq C_f(1 + |u|^q)$ for some constants C_f, q , and fix $c > 0$ (constants local to this appendix, unrelated to the bounds m, c of (A2)). Then

$$(B.40) \quad \frac{d}{d\delta} \mathbb{E}_\epsilon[f(\epsilon + \delta)] \Big|_{\delta=0} = \frac{\mathbb{E}_\epsilon[f(\epsilon)\epsilon]}{\sigma^2}.$$

This is Stein's lemma (Stein 1981, Lemma 1) in location-derivative form. The classical statement, $\mathbb{E}_\epsilon[f(\epsilon)\epsilon] = \sigma^2 \mathbb{E}_\epsilon[f'(\epsilon)]$, assumes f absolutely continuous, which the menu cost response is not; the form (B.40) needs no regularity of f because the derivative falls on the density. Writing $\mathbb{E}_\epsilon f(\epsilon + \delta) = \int f(u) \varphi_\sigma(u - \delta) du$, the integrand is C^∞ in δ with derivative $-f(u) \varphi'_\sigma(u - \delta) = f(u) \frac{u - \delta}{\sigma^2} \varphi_\sigma(u - \delta)$, which for $|\delta| \leq c$ is dominated, uniformly in δ , by

$$(B.41) \quad G(u) \equiv \frac{C_f(1 + |u|^q)(|u| + c)}{\sigma^3 \sqrt{2\pi}} e^{-[(|u| - c)_+]^2/2\sigma^2}, \quad \int G(u) du < \infty,$$

so differentiation under the integral sign is licensed (mean value theorem on the difference quotients, then dominated convergence), and the change of variable $u = \epsilon + \delta$ gives $\frac{d}{d\delta} \mathbb{E}_\epsilon f(\epsilon + \delta) = \mathbb{E}_\epsilon[f(\epsilon + \delta)\epsilon]/\sigma^2$ for all $|\delta| < c$. When f is piecewise C^1 with finitely many jumps $\Delta f_j \equiv f(u_j^+) - f(u_j^-)$ at points u_j and $|f'| \leq C_f(1 + |u|^q)$ on the smooth stretches, integrating by parts stretch by stretch gives the “intensive plus extensive” form $\mathbb{E}_\epsilon[f'(\epsilon)] + \sum_j \Delta f_j \varphi_\sigma(u_j)$, which is the classical statement when there are no jumps. Conversely, (B.40) characterizes the Gaussian (Stein 1972): it holds for every f only if $\varphi' = -(u/\sigma^2)\varphi$; for linear f it holds for any distribution.

Conditional application. The domination $|f_h(u; S, \mathcal{D})| \leq M_h(S, \mathcal{D}) + |u|$ gives $f_h(\cdot; S, \mathcal{D})$ polynomial growth of degree one for every $(S, \mathcal{D})$, so under (G) (B.40) applies conditionally on $(S, \mathcal{D})$:

$$(B.42) \quad b_h^m = \mathbb{E}_{S, \mathcal{D}} \left[\frac{d}{d\delta} \mathbb{E}_\epsilon[f_h(\epsilon + \delta; S, \mathcal{D})] \Big|_{\delta=0} \right].$$

Exchanging the derivative and the expectation over $(S, \mathcal{D})$. Write $\gamma_h(\delta; S, \mathcal{D}) \equiv \mathbb{E}_\epsilon[f_h(\epsilon + \delta; S, \mathcal{D})]$, so that (B.42) reads $b_h^m = \mathbb{E}_{S, \mathcal{D}}[\gamma_h'(\delta)]$, while the claim is $b_h^m = \frac{d}{d\delta} \mathbb{E}_{S, \mathcal{D}}[\gamma_h(\delta)]|_{\delta=0}$. Three facts let the derivative move outside the expectation.

- (1) For every $(S, \mathcal{D})$, γ_h is a smooth function of δ , because δ enters $\gamma_h(\delta) = \int f_h(u; S, \mathcal{D}) \varphi_\sigma(u - \delta) du$ only through the Gaussian density (by independence of ϵ_{it} from $(S, \mathcal{D})$); by the computation behind (B.40), $\gamma_h'(\delta) = \mathbb{E}_\epsilon[f_h(\epsilon + \delta; S, \mathcal{D}) \epsilon] / \sigma_\theta^2$.
- (2) Its derivative is bounded, uniformly over $|\delta| \leq c$, by an integrable function G_h of $(S, \mathcal{D})$ alone.²⁸
- (3) Dominated convergence then applies to the difference quotients: by the mean value theorem, $\{\gamma_h(\delta + r) - \gamma_h(\delta)\} / r = \gamma_h'(\tilde{\delta})$ for some $\tilde{\delta}$ between δ and $\delta + r$, so the quotients are bounded by G_h whenever $|\delta| + |r| \leq c$, and letting $r \rightarrow 0$ inside the expectation gives $\frac{d}{d\delta} \mathbb{E}_{S, \mathcal{D}}[\gamma_h(\delta)] = \mathbb{E}_{S, \mathcal{D}}[\gamma_h'(\delta)]$ on $(-c, c)$.

At $\delta = 0$, (B.42) therefore becomes

$$(B.44) \quad b_h^m = \frac{d}{d\delta} \mathbb{E}_{S, \epsilon, \mathcal{D}}[f_h(\epsilon_{it} + \delta; S, \mathcal{D})] \Big|_{\delta=0}.$$

The right-hand side is A_h , so $b_h^m = A_h$; and (B.42) together with (B.44) is the second claim of the lemma. $\square$

Note that no density, continuity or stationarity of μ_S or $\mu_{\mathcal{D}}$ is used here. In particular, μ_S has an atom at the reset gap $x^* = 0$, where every adjuster sits, and this is harmless because the smoothing in δ is supplied by ϵ_{it} alone.

The three models. *Model (C).* Conditional on the lottery draws, which belong to $\mathcal{D}$, f_h is affine in ϵ with slope $\omega \mathbf{1}\{\text{a reset occurs in } [t, t + h]\}$: under the random walk every reset price (22) moves by ω per unit of ϵ_{it} , and a price that has not been reset does not move. Averaging over the draws, $F_h(\epsilon; S) = \mathbb{E}_{\mathcal{D}}[f_h]$ is affine in ϵ with slope $\omega \Pr(\text{reset in } [t, t + h]) =$

²⁸By the domination hypothesis, $|f_h(\epsilon + \delta; S, \mathcal{D})| \leq M_h + |\epsilon| + c$ for $|\delta| \leq c$, so

$$(B.43) \quad |\gamma_h'(\delta; S, \mathcal{D})| \leq \frac{\mathbb{E}_\epsilon[(M_h + |\epsilon| + c) |\epsilon|]}{\sigma_\theta^2} = \frac{(M_h + c) \sqrt{2/\pi}}{\sigma_\theta} + 1 \equiv G_h(S, \mathcal{D}),$$

and $\mathbb{E}_{S, \mathcal{D}}[G_h] < \infty$ because $\mathbb{E}_{S, \mathcal{D}}[M_h] < \infty$. The same domination gives $\mathbb{E}_{S, \mathcal{D}}|\gamma_h(\delta)| \leq \mathbb{E}_{S, \mathcal{D}}[M_h] + \mathbb{E}_\epsilon|\epsilon| + c < \infty$, so $\mathbb{E}_{S, \mathcal{D}}[\gamma_h(\delta)]$ is well defined for $|\delta| \leq c$. Note that $\mathbb{E}_{S, \mathcal{D}}[G_h] < \infty$ requires finite first moments under $\mu_S \otimes \mu_{\mathcal{D}}$: $\mathbb{E}_S|x_{i,t-1}| < \infty$ and finite $\mathbb{E}_{\mathcal{D}}|\epsilon_{it}^z|$, $\mathbb{E}_{\mathcal{D}}|\epsilon_{i,t+j}^z|$, $\mathbb{E}_{\mathcal{D}}|\epsilon_{i,t+j}|$ for $j \leq h$ but the menu-cost and lottery draws do not enter M_h .

$\omega(1 - \alpha^{h+1})$ (equation (23)), and the domination $|f_h| \leq M_h + |\epsilon|$ holds by the representation (B.45) below with the reset price in place of p_{i,τ^*}^* (plus the forecast terms of z , of finite mean, when z is not a random walk): condition (L) holds for any distribution of ϵ , and Lemma B.4 gives $A_h = b_h^C = \omega(1 - \alpha^{h+1})$.

Model (GL). Conditional on hits and turbulence draws, f_h is not linear in ϵ (whether the firm adjusts depends on ϵ through its gap), but after integrating over the turbulence draws it is: Lemma B.2 holds for every shifter path under (A2)–(A3), so $F_h(\epsilon; S) = \mathbb{E}_{\mathcal{D}}[f_h]$ is affine in ϵ with slope $\omega(1 - \eta_\lambda^{h+1})$ (Appendix B.3.2): condition (L) holds to the order of the GL approximation, for any distribution of ϵ with support compatible with (A2). Affinity fails only on the set of histories whose last hit is older than the horizon over which (A2) bounds the cumulated shifter, a set of μ_S -mass $O(m)$ on which the conditional mean of $x_{i,t+h}$ is bounded by $\bar{x}$, so the error in b_h is $O(m\bar{x})$, the order of the approximation in the reset rule (A.6); the domination $|f_h| \leq M_h + |\epsilon|$ holds as in (C). Lemma B.4 then gives $A_h = b_h^{\text{GL}} = \omega(1 - \eta_\lambda^{h+1}) = \omega[J_{P,p^*}^C(\eta_\lambda)\mathbf{1}]_h$ to that order. Note that (A2) is a bounded-support condition on ϵ , incompatible with the Gaussian shifter of (MC): the two models place different conditions on the identifying shock, and thin-tailed shifters make the GL error exponentially small (see Section B.5.4, Scope).

Model (MC), Gaussian ϵ . Here f_h is not differentiable in ϵ : it jumps whenever ϵ pushes the pre-adjustment gap $x_{i,t-1} - \omega(\epsilon_{it}^z + \epsilon)$ across the band at t or, for a firm that has not yet adjusted, at a later date. Condition (G) is verified from the policy. If the firm does not adjust between t and $t + h$ then $f_h = 0$. Otherwise let $\tau^* \in \{t, \dots, t + h\}$ be its last adjustment date; since the reset gap is $x^* = 0$, $p_{i,t+h} = p_{i,\tau^*}^*$ and, with the price level fixed,

$$(B.45) \quad f_h(\epsilon; S, \mathcal{D}) = p_{i,\tau^*}^* - p_{i,t-1} = -x_{i,t-1} + \omega(\epsilon_{it}^z + \epsilon) + \sum_{j=1}^{\tau^*-t} v_{i,t+j}, \quad v_{i\tau} \equiv \omega(\epsilon_{i\tau}^z + \epsilon_{i\tau}).$$

Hence, for every ϵ and every $|\delta| \leq c$, using $\omega \leq 1$,

$$(B.46) \quad \begin{aligned} |f_h(\epsilon + \delta; S, \mathcal{D})| &\leq M_h(S, \mathcal{D}) + \omega(|\epsilon| + c) \leq M_h(S, \mathcal{D}) + |\epsilon| + c, \\ M_h &\equiv |x_{i,t-1}| + \omega|\epsilon_{it}^z| + \sum_{j=1}^h |v_{i,t+j}|, \end{aligned}$$

and $\mathbb{E}_{S,\mathcal{D}}[M_h] < \infty$: post-adjustment gaps lie in the band, so $|x_{i,t-1}| \leq \bar{x}$ (with a menu-cost

distribution H of unbounded support, $\mathbb{E}_S|x_{i,t-1}| < \infty$ is the assumption of Proposition 1; it is automatic when H has bounded support), and the innovations are Gaussian, so that the domination hypothesis holds and (G) holds.

Proof of Proposition 1. Under the conditions of the proposition the hypotheses of Lemma B.4 hold: the mutual independence of $(S, \epsilon_{it}, \mathcal{D})$ and the domination of f_h in every model, and (L) in (C), (L) to the order of the approximation in (GL) under (A2)–(A3), (G) in (MC) with Gaussian ϵ_{it} . Lemma B.4 therefore shows that A_h exists and equals b_h^m . By (B.36), the existence of A_h is the differentiability of R_h at 0 required in Lemma B.3, which then gives $[J_{P,mc}^m \mathbf{1}]_h = A_h = b_h^m$. $\square$

B.5.3. Intuition for Step 2, Case (G): The Regression as a Weighted Average Derivative

This section rereads the proof of case (G) of Lemma B.4. The regression coefficient and the average derivative are both averages of the same *local responses* of the price change to the shock, under two different weightings; they agree for every response function exactly when the two weightings coincide, and that happens only for the Gaussian. The statements below are written for a discontinuous response, since that is the menu cost case. Throughout, $(S, \mathcal{D})$ is fixed, $f \equiv f_h(\cdot; S, \mathcal{D})$, and ϵ is a single random variable with law μ , mean zero and variance $\sigma^2 \in (0, \infty)$; every expectation is $\mathbb{E}_\epsilon$. The shock is measured in desired-price units, $\omega\epsilon$, so that $\omega = 1$ throughout this subsection without loss of generality.

Local responses. In model (MC), f is affine on each of finitely many intervals, with slopes in $\{0, 1\}$ and at most $2(h + 1)$ jump points, and $|f(u)| \leq M_h + |u|$ by (B.46). Under the Ss policy, given the draws in $\mathcal{D}$, the set of shocks u under which the firm has not adjusted by a given date is an intersection of intervals, hence an interval, and these intervals shrink with the date; so the first-adjustment date is piecewise constant in u , with at most two boundary points per date, and once the firm has adjusted its gap is reset to $x^* = 0$ and nothing later depends on u , so that f is affine with slope 1 on each piece where the firm adjusts by $t + h$ and equal to 0 where it does not. We take f right-continuous at its jump points $u_1 < \dots < u_J$, a normalization that is immaterial when μ has no atom there, and write $\Delta f_j \equiv f(u_j^+) - f(u_j^-)$. The *local response* at u is the slope $f'(u)$ off the jump points and the jump Δf_j at u_j ; to average local responses against a weight k we use the

generalized derivative $Df \equiv f'(u) du + \sum_j \Delta f_j \delta_{u_j}$, i.e.

$$(B.47) \quad \int k dDf \equiv \int f'(u) k(u) du + \sum_{j=1}^J \Delta f_j k(u_j),$$

so that jumps count as point masses, and the fundamental theorem of calculus reads

$$(B.48) \quad f(b) - f(a) = Df((a, b]) = \int_a^b f'(u) du + \sum_{j: a < u_j \leq b} \Delta f_j, \quad a < b.$$

For the impact response of a firm that starts at a zero gap, $f(u) = u \mathbf{1}\{|u| > \bar{x}\}$, one has $f' = \mathbf{1}\{|u| > \bar{x}\}$ and $\Delta f = \bar{x}$ at both $u = \pm \bar{x}$, so

$$(B.49) \quad \int k dDf = \int_{|u| > \bar{x}} k(u) du + \bar{x} [k(\bar{x}) + k(-\bar{x})] :$$

with k the density of the shock, the intensive margin (the adjusters move one for one with the desired price) plus the extensive margin (the firms swept across the thresholds, each moving by $\bar{x}$), the structure of the impact entry (B.17) and of b_0^{MC} .

The two weightings. The following proposition shows that the regression coefficient and the average derivative are both averages of the same local responses $f'(u)$, under two different weightings. They agree for every response function exactly when the two weightings coincide, and that happens only for the Gaussian.

PROPOSITION B.2 (Two weightings of the local responses). *Let f be piecewise C^1 with finitely many jumps and $|f(u)| + |f'(u)| \leq C_f(1 + |u|)$ off the jump points, and let μ have mean zero, variance σ^2 , $\mathbb{E}_\epsilon |\epsilon|^3 < \infty$, and no atom at the jump points of f .*

(i) *Regression weighting. Define*

$$(B.50) \quad w(u) \equiv \frac{1}{\sigma^2} \mathbb{E}_\epsilon [\epsilon \mathbf{1}\{\epsilon > u\}] = \frac{1}{\sigma^2} \mathbb{E}_\epsilon [(-\epsilon) \mathbf{1}\{\epsilon \leq u\}], \quad u \in \mathbb{R},$$

the two forms being equal because $\mathbb{E}_\epsilon \epsilon = 0$. Then w is a probability density, without atoms,

continuous except at an atom of μ located at $u \neq 0$, where it jumps by $u \mu(\{u\})/\sigma^2$, and

$$(B.51) \quad \frac{\mathbb{E}_\epsilon[f(\epsilon)\epsilon]}{\sigma^2} = \int w \, dDf = \int f'(u) w(u) \, du + \sum_{j=1}^J \Delta f_j w(u_j).$$

(ii) Shift weighting. If μ has a continuously differentiable density φ with $|u|\varphi(u) \rightarrow 0$ as $|u| \rightarrow \infty$ and $\int \sup_{|\delta| \leq c} (1 + |u|) |\varphi'(u - \delta)| \, du < \infty$ for some $c > 0$, then

$$(B.52) \quad \frac{d}{d\delta} \mathbb{E}_\epsilon[f(\epsilon + \delta)] \Big|_{\delta=0} = - \int f(u) \varphi'(u) \, du = \int \varphi \, dDf.$$

(iii) When they coincide. If μ has a continuous density φ , then w is continuously differentiable with

$$(B.53) \quad w'(u) = -\frac{u \varphi(u)}{\sigma^2},$$

and $w = \varphi$ on $\mathbb{R}$ if and only if $\varphi = \varphi_\sigma$, the $N(0, \sigma^2)$ density.

PROOF. (i) Fix a continuity point c of f . Since $\mathbb{E}_\epsilon \epsilon = 0$, $\mathbb{E}_\epsilon[f(\epsilon)\epsilon] = \mathbb{E}_\epsilon[(f(\epsilon) - f(c))\epsilon]$, and by (B.48)

$$(B.54) \quad f(\epsilon) - f(c) = \begin{cases} \int_c^\infty f'(u) \mathbf{1}\{u < \epsilon\} \, du + \sum_j \Delta f_j \mathbf{1}\{c < u_j \leq \epsilon\}, & \epsilon > c, \\ - \int_{-\infty}^c f'(u) \mathbf{1}\{\epsilon < u\} \, du - \sum_j \Delta f_j \mathbf{1}\{\epsilon < u_j \leq c\}, & \epsilon \leq c. \end{cases}$$

Multiply by ϵ and integrate against μ . Fubini's theorem allows the order of integration in the du terms to be exchanged, the integrand being bounded by $C_f(1 + |u|)|\epsilon|$ on a u -range of length at most $|\epsilon| + |c|$ on which $|u| \leq |\epsilon| + |c|$, which is integrable by the third moment; the jump terms are finite sums. Reading off $\mathbb{E}_\epsilon[\epsilon \mathbf{1}\{\epsilon > u\}] = \sigma^2 w(u)$ and $\mathbb{E}_\epsilon[(-\epsilon) \mathbf{1}\{\epsilon \leq u\}] = \sigma^2 w(u)$ from (B.50), and using $\mu(\{u_j\}) = 0$ so that $\mathbf{1}\{u_j \leq \epsilon\} = \mathbf{1}\{u_j < \epsilon\}$ almost surely,

$$(B.55) \quad \mathbb{E}_\epsilon[f(\epsilon)\epsilon] = \sigma^2 \left[\int_{-\infty}^\infty f'(u) w(u) \, du + \sum_{j=1}^J \Delta f_j w(u_j) \right],$$

which is (B.51); the base point c drops out, as it must. As for w : the first form in (B.50) shows $w \geq 0$ for $u \geq 0$ and the second for $u < 0$; Tonelli's theorem gives $\int w = \sigma^{-2} \mathbb{E}_\epsilon[\epsilon^2] =$

1; and dominated convergence on the indicators gives right-continuity and $w(u^-) - w(u) = u \mu(\{u\})/\sigma^2$.

(ii) The first equality is the argument behind (B.40)–(B.41) with φ in place of φ_σ : the integrand $f(u)\varphi(u-\delta)$ of $\mathbb{E}_\epsilon f(\epsilon+\delta)$ is C^1 in δ , its derivative is dominated for $|\delta| \leq c$ by $C_f(1+|u|) \sup_{|\delta'| \leq c} |\varphi'(u-\delta')|$, which the stated condition makes integrable (it is what (B.41) delivers in the Gaussian case), and the mean value theorem with dominated convergence lets the derivative pass under the integral. For the second, integrate $-\int f\varphi'$ by parts on each stretch between consecutive jump points: the two boundary terms at a jump point combine, by continuity of φ , into $f(u_j^+)\varphi(u_j) - f(u_j^-)\varphi(u_j) = \Delta f_j \varphi(u_j)$, the terms at $\pm\infty$ vanish because $|f\varphi| \leq C_f(1+|u|)\varphi \rightarrow 0$, and $\int |f'\varphi| < \infty$ by the first moment.

(iii) (B.53) is the fundamental theorem of calculus applied to $w(u) = \sigma^{-2} \int_u^\infty v\varphi(v) dv$. If $w = \varphi$ then $\varphi' = -u\varphi/\sigma^2$, a linear first-order equation whose solutions are $\varphi(0)e^{-u^2/2\sigma^2}$, and $\int \varphi = 1$ fixes $\varphi(0) = (\sigma\sqrt{2\pi})^{-1}$. Conversely $v\varphi_\sigma(v) = -\sigma^2\varphi'_\sigma(v)$ gives $\int_u^\infty v\varphi_\sigma = \sigma^2\varphi_\sigma(u)$. $\square$

Interpretation of the two weightings: Chords and tangents. The proof of Proposition B.2(i) reads the regression through the total displacement $f(\epsilon) - f(c)$, the sum of the local responses along the path from c to ϵ ; the local response at u is credited to every firm whose path crosses u , each carrying its regression weight ϵ , which is why the weight at u is the shock-weighted mass of firms beyond u , (B.50). Part (ii) of the proposition instead credits the local response at u to the firms located at u , with weight $\varphi(u)$. The regression uses chord slopes between c and ϵ , while the economy responds to the tangent slope at ϵ . The weight w determines how the tangent slopes are averaged in the regression. Equivalently, by the Tonelli computation in that proof with $c = 0$, w is the density of a point drawn uniformly on the segment between 0 and ϵ in a sample in which firms are drawn with probability proportional to ϵ^2 : once because longer paths contain more points, once because the regression weights each firm by ϵ . Part (iii) says that for the bell curve, and only for it, the shock-weighted mass beyond u equals σ^2 times the density at u , so chord sampling and endpoint sampling agree at every u ; this is the same property that makes the score linear, $\varphi'_\sigma = -(u/\sigma^2)\varphi_\sigma$.

If f' is the same number everywhere and f has no jumps, $\int w dDf = \int \varphi dDf$ whatever w and φ are: the weighting is irrelevant and any law works. Applied to $F_h = \mathbb{E}_{\mathcal{D}} f_h$, this is

the content of case (L) of Lemma B.4, which covers the Calvo model and, to the order of its approximation, the GL model.

Note: A general fact about regressions on a randomized continuous treatment. Nothing in Proposition B.2 is specific to price setting. Consider any outcome $y = g(\epsilon; \zeta)$ of a continuous treatment ϵ that is randomly assigned, i.e. independent of the unit's characteristics ζ , with response functions $g(\cdot; \zeta)$ that may be heterogeneous across units and nonlinear in the treatment, and let $m(u) \equiv \mathbb{E}_\zeta[g(u; \zeta)]$ be the average response, assumed piecewise C^1 with finitely many jumps and linear growth. The average marginal treatment effect is $\int \varphi dDm$, which equals $\mathbb{E}_\epsilon[m'(\epsilon)]$ when m is smooth and includes the mass swept across the jumps when individual responses are discontinuous, as in part (ii). The OLS coefficient of y on ϵ is $\mathbb{E}[y\epsilon]/\sigma^2 = \mathbb{E}_\epsilon[m(\epsilon)\epsilon]/\sigma^2 = \int w dDm$ by independence and part (i): heterogeneity enters only through m . The two coincide for a given m when m is affine, which covers heterogeneous but linear responses, and for every m if and only if the treatment is Gaussian (part (iii)). Otherwise OLS is a weighted average of marginal effects with the weights w , a result due to Yitzhaki (1996) and discussed by Angrist and Krueger (1999).

B.5.4. Non-Gaussian Identifying Shocks

Let $\bar{F}_h(\epsilon) \equiv \mathbb{E}_{S, \mathcal{D}}[f_h(\epsilon; S, \mathcal{D})]$ be the response averaged over the state and the other draws. It is smooth even though each realization f_h jumps, because the driving innovations ϵ^z have smooth densities and spread the jump points out; this is the only place where a property of ϵ^z beyond finite first moments is used. By Proposition B.2, the regression averages the local responses of $\bar{F}_h$ under the weight w and the shift experiment averages them under the density of the shock, and the two agree for every response only when the shock is Gaussian. Their moments differ otherwise: $\int u w(u) du = \mu_3/(2\sigma_y^2)$ and $\int u^2 w(u) du = \sigma_y^2 \text{kurt}/3$, so a skewed shock tilts the regression weight and a fat-tailed one spreads it beyond the population density.

The bias. Suppose ϵ has mean zero, variance σ_y^2 and bounded absolute standardized moments up to order five, and that $\bar{F}_h$ is C^4 near zero with linear growth. Expanding $\bar{F}_h$

inside the two averages,

$$(B.56) \quad \frac{\mathbb{E}_\epsilon[\bar{F}_h(\epsilon)\epsilon]}{\sigma_\vartheta^2} - \frac{d}{d\delta} \mathbb{E}_\epsilon[\bar{F}_h(\epsilon+\delta)] \Big|_{\delta=0} = \frac{\bar{F}_h''(0)}{2} \text{skew } \sigma_\vartheta + \frac{\bar{F}_h'''(0)}{6} (\text{kurt} - 3) \sigma_\vartheta^2 + O(\sigma_\vartheta^3),$$

which vanishes at all orders in the Gaussian case by (B.40). At the zero-inflation steady state the model is invariant to the sign flip of all gaps and innovations, so $\bar{F}_h$ is odd up to $O(\sigma_\vartheta^3)$ so the leading bias is $\frac{1}{6} \bar{F}_h'''(0)(\text{kurt} - 3)\sigma_\vartheta^2$, proportional to excess kurtosis.

Solution: regress on the score. Let p be the density of ϵ , absolutely continuous with $\int \sup_{|\delta| \leq c} (1 + |u|) |p'(u - \delta)| du < \infty$ for some $c > 0$ (Gaussian, Student- t and Laplace densities qualify), and let $s(\epsilon) \equiv -p'(\epsilon)/p(\epsilon)$ be its score. Integration by parts, with jumps counted as in (B.47), gives $\mathbb{E}_\epsilon[f(\epsilon)s(\epsilon)] = \frac{d}{d\delta} \mathbb{E}_\epsilon[f(\epsilon + \delta)] \Big|_{\delta=0}$ for every piecewise- C^1 f of linear growth; the Gaussian score is $\epsilon/\sigma_\vartheta^2$, which recovers (B.40). Repeating the two steps of the proof with $s(\epsilon_{it})$ in place of $\epsilon_{it}/\sigma_\vartheta^2$,

$$(B.57) \quad \mathbb{E}_{S,\epsilon,\mathcal{D}}[(p_{i,t+h} - p_{i,t-1}) s(\epsilon_{it})] = [J_{P,mc}^m \mathbf{1}]_h$$

exactly, for any such density. Since $\mathbb{E}_\epsilon[\epsilon s(\epsilon)] = 1$ and $\mathbb{E}_\epsilon[s(\epsilon)] = 0$, this is the coefficient of an IV regression of price changes on ϵ_{it} with instrument $s(\epsilon_{it})$: the average-derivative estimator of Stoker (1986) and Härdle and Stoker (1989). p can be estimated from the observed panel of ϵ_{it} .

B.5.5. The Turbulence Regression in the Gertler–Leahy Model

Suppose the econometrician regresses price changes on the turbulence innovation ϵ_{it}^z itself (equal to zero for firms not hit), a violation of the conditions of Proposition 1 in (GL). This section shows that the resulting coefficients are $b_h^z = \omega(1 - \iota^{\text{GL}} \eta_\lambda^h)$ with $\iota^{\text{GL}} \equiv (1 - \lambda)(2\bar{x}/\varsigma)^3 < 1/8$: the impact pass-through is close to its long-run value ω and $b_1^z/b_0^z - 1$ does not recover η_λ . Set mc and P constant; the shifter innovations are independent of ϵ_{it}^z and drop out of every covariance below, so we also set ϑ constant, and $p_{it}^* = \omega z_{it}$ up to a constant and $p_{i,t+h}^* - p_{i,t-1}^* = \omega(z_{i,t+h} - z_{i,t-1})$ is a random walk in the turbulence innovations. Define

$$b_h^z \equiv \frac{\text{Cov}(p_{i,t+h} - p_{i,t-1}, \epsilon_{it}^z)}{\text{Var}(\epsilon_{it}^z)}$$

over all firms, with $\epsilon_{it}^z = 0$ for firms not hit at t ; since those firms contribute zero to both the covariance and the variance, b_h^z equals the same ratio computed among hit firms. Corollary 1 applies with ϵ^z in place of ϵ : $b_h^z = \omega + \text{Cov}(x_{i,t+h}, \epsilon_{it}^z) / \text{Var}(\epsilon_{it}^z)$. It is convenient to work with the turbulence innovation in desired-price units, $u_{it}^z \equiv \omega \epsilon_{it}^z$, uniform on $[-\varsigma/2, \varsigma/2]$ at a hit; since $\epsilon^z = u^z / \omega$,

$$\frac{\text{Cov}(x_{i,t+h}, \epsilon_{it}^z)}{\text{Var}(\epsilon_{it}^z)} = \omega \frac{\text{Cov}(x_{i,t+h}, u_{it}^z)}{\text{Var}(u_{it}^z)}.$$

With mc , P and ϑ constant the reset price net of the turbulence component, $p_{it}^{\text{new}} - \omega z_{it}$, is a constant, normalized to zero. Consider a firm hit at t with pre-hit gap $x_- \equiv p_{i,t-1} - \omega z_{i,t-1} \in [-\bar{x}, \bar{x}]$, independent of u_{it}^z . Its would-be gap is $y = x_- - u_{it}^z$, and its gap after the hit is $x_{it} = y \mathbf{1}\{|y| \leq \bar{x}\}$ if it has no free draw (probability $1 - \lambda$) and 0 if it has one. Substituting $u = -y$, which ranges over an interval of width ς containing $[-\bar{x}, \bar{x}]$ by (A3),

$$(B.58) \quad \mathbb{E}[x_{it} u_{it}^z \mid x_-] = \mathbb{E}[y (x_- - y) \mathbf{1}\{|y| \leq \bar{x}\} \mid x_-] = -\frac{1}{\varsigma} \int_{-\bar{x}}^{\bar{x}} u (u + x_-) du = -\frac{2\bar{x}^3}{3\varsigma},$$

independent of x_- , while $\text{Var}(u_{it}^z \mid \text{hit}) = \varsigma^2/12$. Multiplying by the probability $1 - \lambda$ of no free draw, $\text{Cov}(x_{it}, u_{it}^z) / \text{Var}(u_{it}^z) = -(1 - \lambda) 8\bar{x}^3 / \varsigma^3 = -\iota^{\text{GL}}$ and $b_0^z = \omega(1 - \iota^{\text{GL}})$. For later horizons, $x_{i,t+1} = x_{it}$ with probability η_λ ; with probability λ the firm resets freely, so $x_{i,t+1} = 0$; and with the remaining probability it is hit again with would-be gap $x_{it} - u_{i,t+1}^z$, uniform on an interval containing the band, so that $\mathbb{E}[x_{i,t+1} \mid x_{it}, \text{hit}] = \varsigma^{-1} \int_{-\bar{x}}^{\bar{x}} u du = 0$. Therefore $\mathbb{E}[x_{i,t+h} \mid x_{it}] = \eta_\lambda^h x_{it}$, $\text{Cov}(x_{i,t+h}, u_{it}^z) = \eta_\lambda^h \text{Cov}(x_{it}, u_{it}^z)$, and

$$(B.59) \quad b_h^z = \omega(1 - \iota^{\text{GL}} \eta_\lambda^h),$$

the result stated above. In particular $b_h^z \rightarrow \omega$ (full long-run pass-through of the desired price) and $b_1^z / b_0^z - 1 = \iota^{\text{GL}}(1 - \eta_\lambda) / (1 - \iota^{\text{GL}})$. Contrast with the shifter: there $\text{Cov}(x_{i,t+h}, \epsilon_{it}) / \sigma_\vartheta^2 = -\omega \eta_\lambda^{h+1}$, because a firm that has neither been hit nor freely reset since the shock carries all of it in its gap; here a firm hit at t carries, on average across the hit population, only the part ι^{GL} of the innovation that its non-adjusters absorb.

The weighting representation (B.51) of Section B.5.3 explains this result. Take ϵ_{it}^z as the regressor, so that it plays the role of ϵ_{it} and $\mathcal{D}^z$ collects every other draw: with probability η it equals zero (no hit), and with probability $1 - \eta$ it is uniform on $[-\varsigma/(2\omega), \varsigma/(2\omega)]$ and triggers reconsideration; conditional on a hit the firm adjusts with probability $1 -$

$2\bar{x}/\varsigma$ and its expected price equals the reset price (Lemma B.2). The average derivative is A_h^z , the derivative of the cross-sectional average price with respect to a shift δ of every firm's ϵ_{it}^z , i.e. a shift $\omega\delta$ of every desired price in (B.38), computed in expectation over $\mathcal{D}^z$ through Lemma B.2. A firm hit at t , or one with a free draw, passes $\omega\delta$ through fully at every horizon in that expectation. A firm that is neither is displaced by $\omega\delta$ inside its band, does not reconsider its price at t (assumption (A4)), and passes $\omega\delta$ through once it is hit or free, which by $t + h$ has probability $1 - \eta_\lambda^h$. Hence

$$(B.60) \quad A_h^z = \omega[(1 - \eta_\lambda) \cdot 1 + \eta_\lambda(1 - \eta_\lambda^h)] = \omega(1 - \eta_\lambda^{h+1}) = [J_{P,mc}^{\text{GL}} \mathbf{1}]_h,$$

the pass-through of mc , as step (a) requires. The regression instead computes (B.51) with the weight ϵ_{it}^z : the atom at zero receives no weight, so the non-hit firms, which carry the mass η_λ of zero impact responses in (B.60) (the free adjusters among them respond, but are discarded too), drop out, and the coefficient is the pass-through among hit firms, $b_h^z = \omega(1 - \iota^{\text{GL}} \eta_\lambda^h)$ derived above. The discrepancy $b_h^z - [J_{P,mc}^{\text{GL}} \mathbf{1}]_h = \omega \eta_\lambda^h (\eta_\lambda - \iota^{\text{GL}})$ is, up to the small selection term ι^{GL} , ω times the mass η_λ^{h+1} of firms neither hit nor freely reset through $t + h$; at $h = 0$ it is, net of ι^{GL} , ω times the discarded mass η_λ . The identifying shock must therefore be absorbed the way mc is: it must enter p^* with the same weight ω as mc and leave the adjustment decision to other forces, so that adjusters and non-adjusters enter the regression in their population proportions. The small shifter ϑ does this in (GL) for any distribution because it never triggers adjustment. In the language of Lemma B.4, the turbulence regressor violates both conditions: it has an atom and is not Gaussian, so (G) fails, and the $\mathcal{D}$ -averaged response is not affine in it, so (L) fails.

B.6. Non-Random-Walk Shocks

The main text takes z_{it} and ϑ_{it} to be driftless random walks in every model. This section investigates the case of an AR(1) shifter, $\vartheta_{it} = \rho \vartheta_{i,t-1} + \epsilon_{it}$ with $0 \leq \rho < 1$. A persistent component of the z_{it} shock, with persistence ρ_z , is treated in parallel where it matters. With the price level fixed,

$$(B.61) \quad \mathbb{E}_t[p_{i,t+s}^*] - p_{it}^* = -\omega(1 - \rho^s) \vartheta_{it},$$

so two things change in general: the reset price is no longer the current desired price, and the conditional law of future desired-price increments depends on the current level ϑ_{it} . On the econometric side the regression is unchanged, and Remark 1 gives the formulas that replace (25)–(26): $\phi^{\text{IRF}} = b_1/b_0 - \rho$ and $\kappa^{\text{IRF}} = b_0(1 - \beta\phi^{\text{IRF}}\rho)/\phi^{\text{IRF}}$, with ρ estimated from the autocorrelation of the observed shifter.

A benchmark for time-dependent models. The following lemma states what the geometric extrapolation recovers under any time-dependent technology, and hence what the canonical menu cost policy would predict.

LEMMA B.5 (TD response to a persistent shifter). *In a time-dependent model with survival function Φ and the reset rule underlying (32), the response of prices to the shifter innovation is*

$$(B.62) \quad b_h = \omega c(\rho) \frac{\sum_{j=0}^h \rho^{h-j} \Phi_j}{\sum_{s \geq 0} \Phi_s}, \quad c(\rho) \equiv \frac{\sum_{s \geq 0} \beta^s \Phi_s \rho^s}{\sum_{s \geq 0} \beta^s \Phi_s},$$

so that $b_1/b_0 - \rho = \Phi_1$ exactly. For a mixture $\sum_k \varrho_k J_{P,p^*}^{\text{TD}}(\Phi^k)$, $b_1/b_0 - \rho$ is a weighted average of the Φ_1^k with weights proportional to $\varrho_k c_k(\rho)/\sum_s \Phi_s^k$, which reduces to $\bar{\Phi}_1$ at $\rho = 1$.

PROOF. A firm resetting at $t + j$ sets

$$p_{i,t+j}^{\text{new}} = \frac{\sum_{s \geq 0} \beta^s \Phi_s \mathbb{E}_{t+j} p_{i,t+j+s}^*}{\sum_{s \geq 0} \beta^s \Phi_s},$$

and $\partial \mathbb{E}_{t+j} p_{i,t+j+s}^* / \partial \epsilon_{it} = \omega \rho^{j+s}$, so $\partial p_{i,t+j}^{\text{new}} / \partial \epsilon_{it} = \omega \rho^j c(\rho)$. The expected price at $t + h$ averages reset prices over the age distribution of prices, with weight $\Phi_j / \sum_s \Phi_s$ on the reset price set j periods earlier, and only resets at dates $t + h - j \geq t$ respond; differentiating gives (B.62), and $b_1/b_0 = (\rho\Phi_0 + \Phi_1)/\Phi_0$ with $\Phi_0 = 1$. For a mixture, b_h is the ϱ_k -weighted sum of (B.62) across components, and the ratio follows. $\square$

(C) **Calvo.** Nothing changes in the structure: adjustment is exogenous and the reset price is a linear forecast of desired prices,

$$\bar{p}_i = J_{P,p^*}^{\text{C}}(\alpha) [\omega(\mathbf{mc} + \mathbb{E}\mathbf{z}_i + \mathbb{E}\vartheta_i) + (1 - \omega)\mathbf{P}],$$

in which ϑ_{it} enters with the weight $w_\rho \equiv (1 - \beta\alpha)/(1 - \beta\alpha\rho)$ and the law of z is irrelevant. Lemma B.5 with $\Phi_j = \alpha^j$ gives $c(\rho) = w_\rho$ and

$$(B.63) \quad b_h = \omega w_\rho (1 - \alpha) \sum_{j=0}^h \rho^{h-j} \alpha^j, \quad b_0 = \omega(1 - \alpha)w_\rho, \quad \frac{b_1}{b_0} = \alpha + \rho,$$

so $\phi^{\text{IRF,C}} = \alpha$ and $\kappa^{\text{IRF,C}} = b_0(1 - \beta\alpha\rho)/\alpha = \omega\kappa(\alpha)$, the formulas of Remark 1. They hold exactly and for any law of ϵ_{it} , since the draw-averaged response is affine (case (L) of Lemma B.4). The same holds for any process for ϑ : $\tilde{\vartheta}$ is then its impulse response and $b_h = \omega[J_{P,p^*}^C(\alpha)\tilde{\vartheta}]_h$.

(GL) Gertler–Leahy. Neither component needs to be a random walk. Let turbulence mean-revert between hits, $z_{it} = \rho_z z_{i,t-1}$ with the uniform jump added at a hit, and let the shifter be the AR(1) above. A firm that reconsiders its price at t still chooses p to minimize $\mathbb{E}_t \sum_{s \geq 0} (\eta_\lambda \beta)^s (p - p_{i,t+s}^*)^2$ along the path with no further reconsideration, so the reset price is the same linear forecast with the persistence-dependent weights

$$(B.64) \quad p_{it}^{\text{new}} = \omega w_{\rho_z}^{\text{GL}} z_{it} + \omega w_\rho^{\text{GL}} \vartheta_{it} + (1 - \eta_\lambda \beta) \mathbb{E}_t \sum_{s \geq 0} (\eta_\lambda \beta)^s [\omega m c_{t+s} + (1 - \omega) P_{t+s}],$$

with $w_r^{\text{GL}} \equiv (1 - \eta_\lambda \beta)/(1 - \eta_\lambda \beta r)$. The adjustment decision still depends on the would-be gap $y = p_{i,t-1} - p_{it}^{\text{new}}$ alone. Write $d_s \equiv p_{it}^{\text{new}} - \mathbb{E}_t p_{i,t+s}^*$ for the expected drift of the desired price away from the reset price, nonzero now. The value of keeping the old price is $-\frac{1}{2} \sum_s (\eta_\lambda \beta)^s \mathbb{E}_t (y + d_s + u_s)^2$ with u_s the unexpected part, and expanding the square, the cross term $2y \sum_s (\eta_\lambda \beta)^s d_s$ vanishes by the first-order condition that defines p_{it}^{new} ; what is left is $-\frac{1}{2} y^2 / (1 - \eta_\lambda \beta)$ plus terms that do not involve y and are common to the keep and reset branches. Value matching therefore gives the same band (A.8), and the levels of z_{it} and ϑ_{it} enter the decision only through p_{it}^{new} : the firm's state remains one-dimensional because it does not reconsider between hits and free draws, and because the loss is quadratic. The rest of Appendix B.3 goes through with two changes. In the selection lemma, the drift of $p^{\text{new}} - \omega z$ between two hits now includes the expected reversion of the shifter, which (A2) bounds, and the turbulence draw enters the would-be gap with the weight $\omega w_{\rho_z}^{\text{GL}}$, so the uniform support in (A3) is scaled by $w_{\rho_z}^{\text{GL}}$ (with $\rho_z = 1$ nothing changes). In the

aggregation step, $\partial \bar{p}_{i,t+j}^{\text{new}} / \partial \epsilon_{it} = \omega w_{\rho}^{\text{GL}} \rho^j$, and Lemma B.5 with $\Phi_j = \eta_{\lambda}^j$ gives

$$(B.65) \quad b_h = \omega w_{\rho}^{\text{GL}} (1 - \eta_{\lambda}) \sum_{j=0}^h \rho^{h-j} \eta_{\lambda}^j, \quad \phi^{\text{IRF,GL}} = \frac{b_1}{b_0} - \rho = \eta_{\lambda}, \quad \kappa^{\text{IRF,GL}} = \omega \kappa(\eta_{\lambda}),$$

to the order of the approximation.

(MC) Canonical menu cost. The firm reconsiders its price every period and the value of waiting depends on where the desired price is expected to go relative to the band. The state is the pair (y, ϑ) and the Bellman equation (B.11) becomes

$$(B.66) \quad \begin{aligned} V(y, \vartheta) &= \lambda \min_x W(x, \vartheta) + (1 - \lambda) \min \left\{ W(y, \vartheta), \xi + \min_x W(x, \vartheta) \right\}, \\ W(x, \vartheta) &= \frac{1}{2} x^2 + \beta \mathbb{E}[V(x - v', \rho\vartheta + \epsilon') \mid \vartheta], \quad v' = \omega(\epsilon^{z'} + \epsilon' - (1 - \rho)\vartheta), \end{aligned}$$

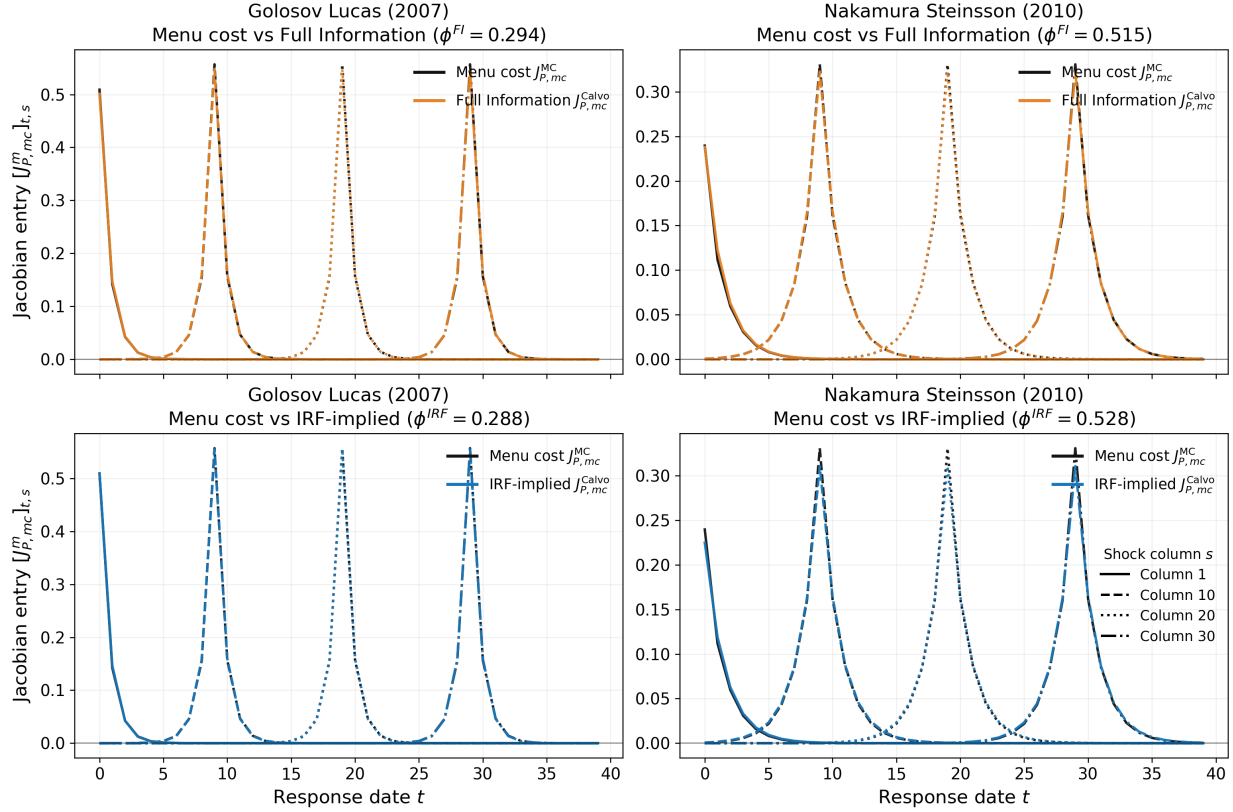
where the desired-price innovation v' has conditional mean $-\omega(1 - \rho)\vartheta$. The reset gap $x^*(\vartheta)$ and the band $[\underline{x}(\vartheta), \bar{x}(\vartheta)]$ depend on ϑ : a firm with a high shifter expects its desired price to fall, so it resets below the current desired price, $x^*(\vartheta) < 0$, and tolerates larger negative than positive gaps, since a negative gap is expected to close on its own. The stationary law is a joint law of (y, ϑ) .

Proposition 1 still applies and implies $b_h = [J_{P,mc}^m \tilde{\vartheta}]_h$, $\tilde{\vartheta} = (1, \rho, \rho^2, \dots)'$. However, the analytical results on the entries of $J_{P,mc}^m$ in Section 3.3.3 rest on the one-dimensional state and do not apply. We provide numerical results for this case in Section 5.4.

Appendix C. Additional Numerical Results

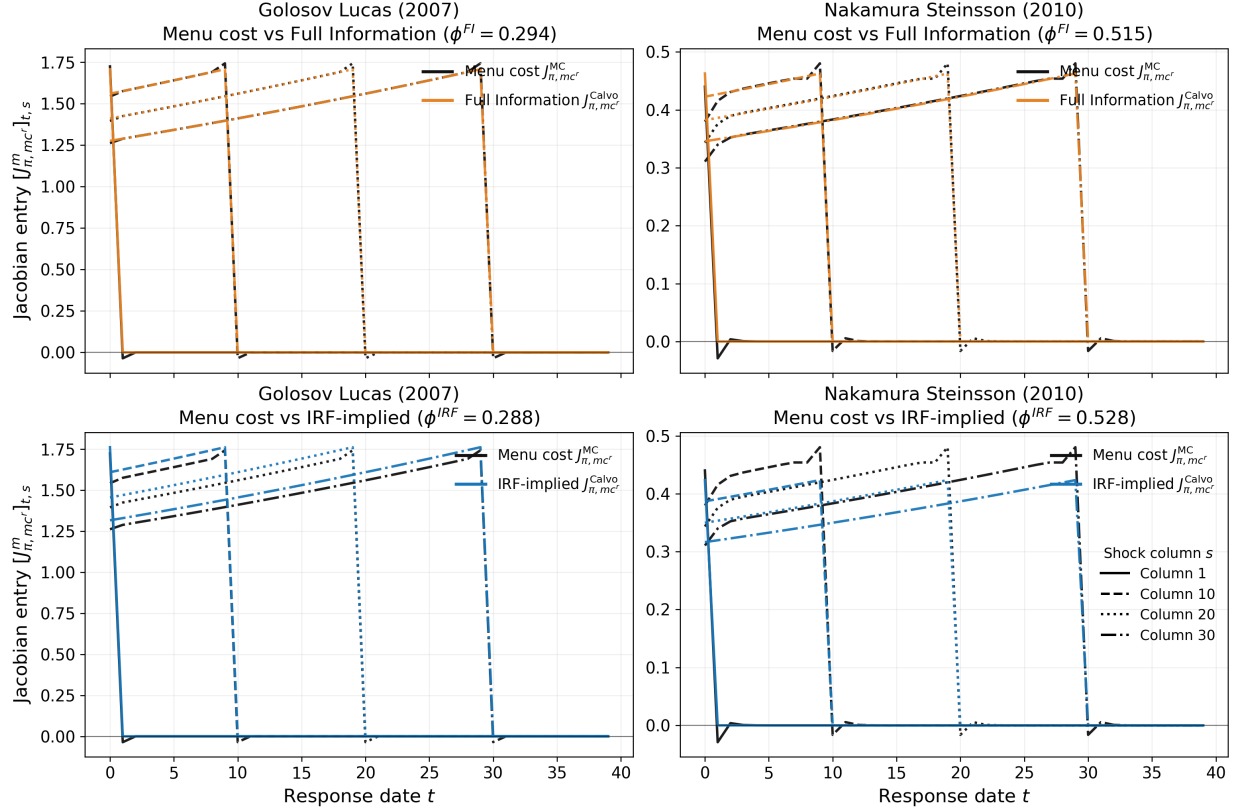
C.1. Selected columns of the nominal passthrough and Phillips curve Jacobian

FIGURE C.1. Nominal price pass-through: selected Jacobian columns



Note: The panels compare columns of the menu-cost nominal pass-through Jacobian $J_{P,mc}^{MC}$ with their Calvo approximation counterparts. The left column uses the Golosov–Lucas (2007) calibration and the right column the Nakamura–Steinsson (2010) calibration. Each curve gives the response of the aggregate log price level at date t to a unit nominal marginal-cost disturbance at the indicated date $s \in \{1, 10, 20, 30\}$. Line styles distinguish shock dates. The black line is the true nominal pass-through Jacobian $J_{P,mc}^{MC}$. The orange line is the Calvo approximation assembled from the full-information survival rate ϕ^{FI} ; the blue line is the Calvo approximation assembled from IRF-implied survival rate ϕ^{IRF} .

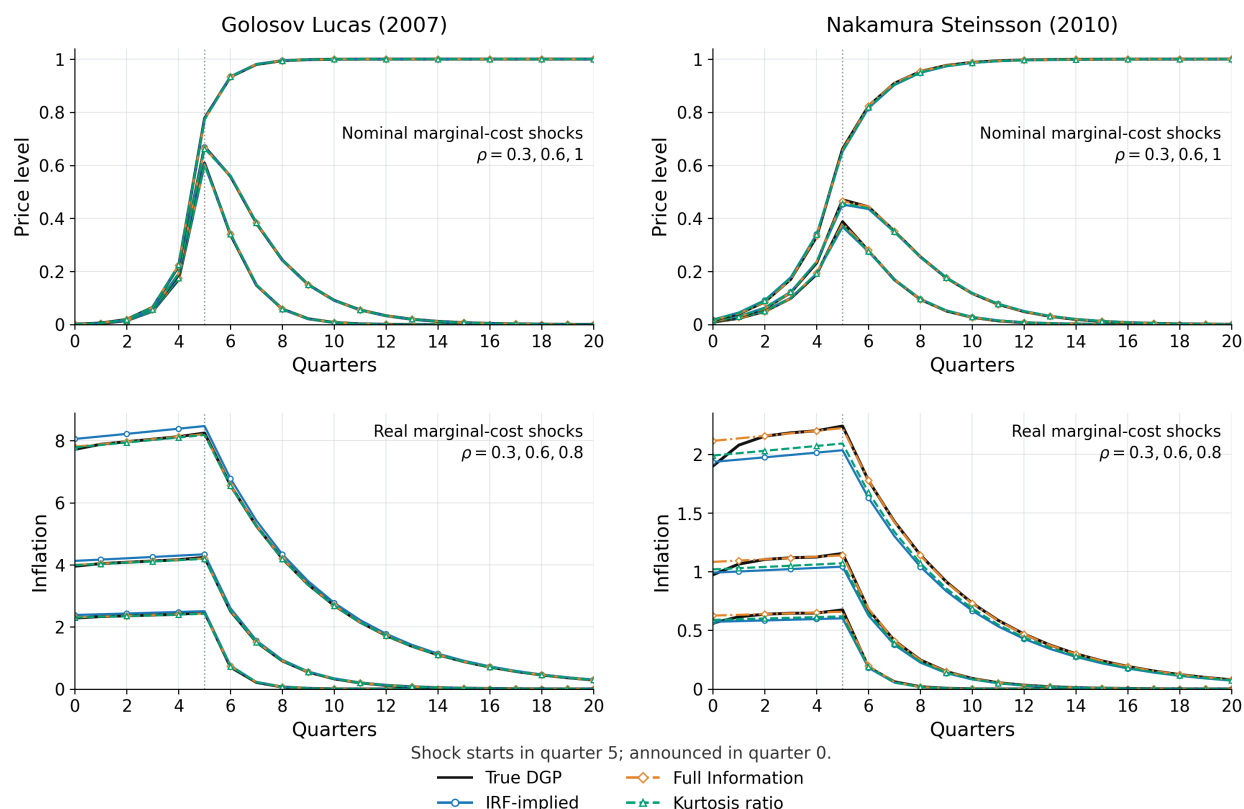
FIGURE C.2. Generalized Phillips curve: selected Jacobian columns



Note: The panels compare columns of J_{π, mc^r}^{MC} , the true generalized Phillips curve Jacobian with their Calvo approximation counterparts. The left column uses the Golosov–Lucas (2007) calibration and the right column the Nakamura–Steinsson (2010) calibration. The horizontal axis is response date t ; line styles identify the date of the unit cost disturbance, $s \in \{1, 10, 20, 30\}$. The black line is the true Phillips curve Jacobian J_{π, mc^r}^{MC} . The orange line is the Calvo approximation assembled from the full-information survival rate ϕ^{FI} ; the blue line is the Calvo approximation assembled from IRF-implied survival rate ϕ^{IRF} .

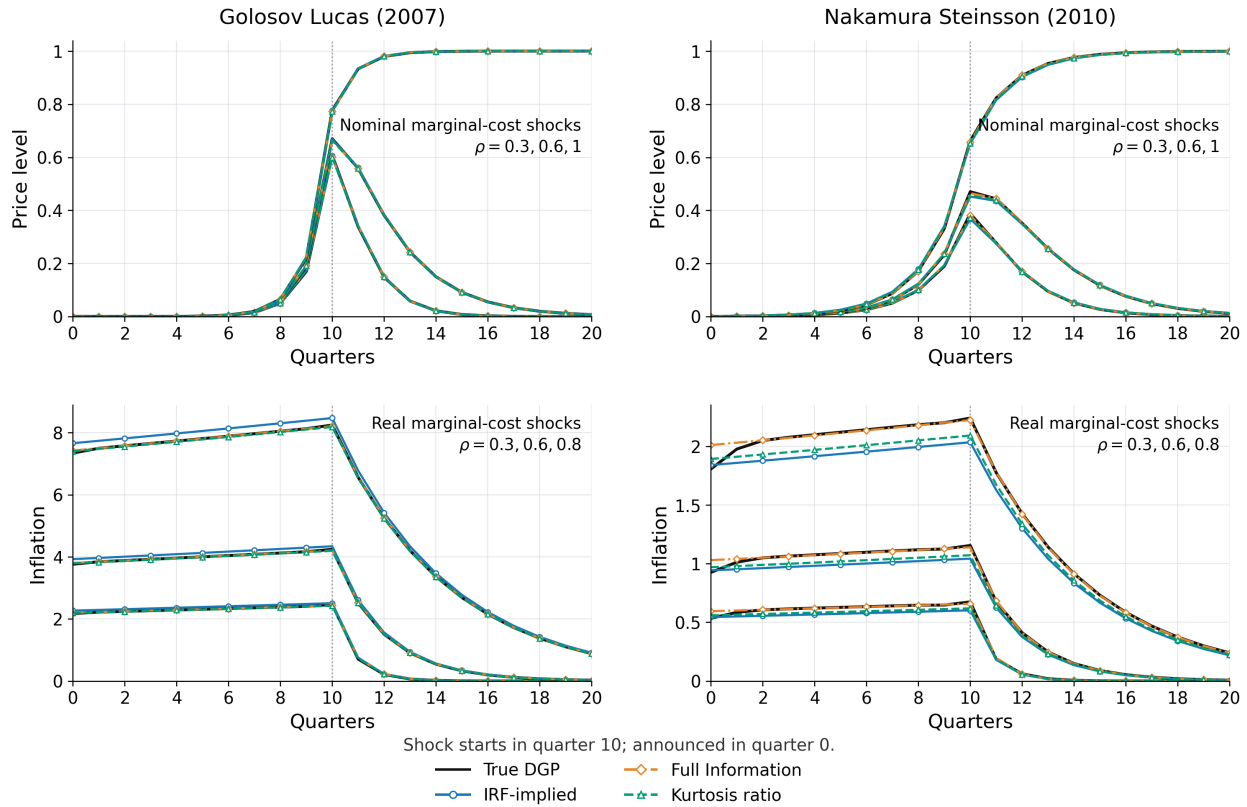
C.2. Impulse responses to nominal and real marginal cost shocks

FIGURE C.3. Responses to Anticipated Nominal and Real Marginal-Cost Innovations (5 quarters ahead)



Notes: The left column uses the Golosov–Lucas (2007) calibration and the right column the Nakamura–Steinsson (2010) calibration. The top row reports price-level responses to nominal marginal-cost paths $(1, \rho, \rho^2, \dots)$ for $\rho \in \{0.3, 0.6, 1\}$; $\rho = 1$ is a permanent nominal innovation. The bottom row reports inflation responses to real marginal-cost paths with $\rho \in \{0.3, 0.6, 0.8\}$. Innovations are announced in quarter 0 but beginning in quarter 5: they are zero before quarter 5 and follow ρ^{t-5} thereafter. Black denotes the true DGP, blue the IRF-implied approximation, orange Full Information, and green the Kurtosis ratio. Within each color, the three paths correspond to the persistence values listed inside the panel.

FIGURE C.4. Responses to Anticipated Nominal and Real Marginal-Cost Innovations (10 quarters ahead)



Notes: The left column uses the Golosov–Lucas (2007) calibration and the right column the Nakamura–Steinsson (2010) calibration. The top row reports price-level responses to nominal marginal-cost paths $(1, \rho, \rho^2, \dots)$ for $\rho \in \{0.3, 0.6, 1\}$; $\rho = 1$ is a permanent nominal innovation. The bottom row reports inflation responses to real marginal-cost paths with $\rho \in \{0.3, 0.6, 0.8\}$. Innovations are announced in quarter 0 but beginning in quarter 10: they are zero before quarter 10 and follow ρ^{t-10} thereafter. Black denotes the true DGP, blue the IRF-implied approximation, orange Full Information, and green the Kurtosis ratio. Within each color, the three paths correspond to the persistence values listed inside the panel.

C.3. Leptokurtic Shocks

Construction of leptokurtic shocks. We first discuss the case where the only shock z_{it} is a leptokurtic shock. z_{it} a mixture of two zero-mean Gaussian distributions. Each firm in each period draws the low-variance Gaussian shock with probability 0.9 and a high-variance Gaussian shock with probability 0.1. We generate

$$z_{it} \sim 0.9 \mathcal{N}(0, \sigma_L^2) + 0.1 \mathcal{N}(0, \sigma_H^2).$$

The Gaussian draws are independent across firms and periods. We hold the unconditional standard deviation fixed at $\sigma = 0.06$ and vary the target kurtosis over $K \in \{3, 4, 5, 6, 9\}$.

Choosing the component variances σ_L^2 and σ_H^2 . Let $p = 0.1$ denote the probability of the high-variance component and define

$$d = \sqrt{\frac{K/3 - 1}{p(1-p)}}, \quad \sigma_L^2 = \sigma^2(1 - pd), \quad \sigma_H^2 = \sigma^2(1 + (1-p)d).$$

These choices give zero mean and satisfy

$$\text{Var}(z_{it}) = (1-p)\sigma_L^2 + p\sigma_H^2 = \sigma^2, \quad \frac{\mathbb{E}[z_{it}^4]}{\sigma^4} = \frac{3[(1-p)\sigma_L^4 + p\sigma_H^4]}{\sigma^4} = K.$$

That is, we are changing the kurtosis of the mixture while keeping both the total variance and the mixture weights fixed. The resulting component standard deviations are

Kurtosis K	σ_L (probability 0.9)	σ_H (probability 0.1)
3	0.060	0.060
4	0.054	0.099
5	0.051	0.111
6	0.049	0.120
9	0.044	0.137

For $K = 3$, the two components coincide and the code draws a single Gaussian with standard deviation 0.06.

Gaussian shifter used as independent variable in the local projection. Now we generate $\varepsilon_{it} = \vartheta_{it} + z_{it}$, where the Gaussian shock ϑ_{it} and the potentially leptokurtic shock z_{it}

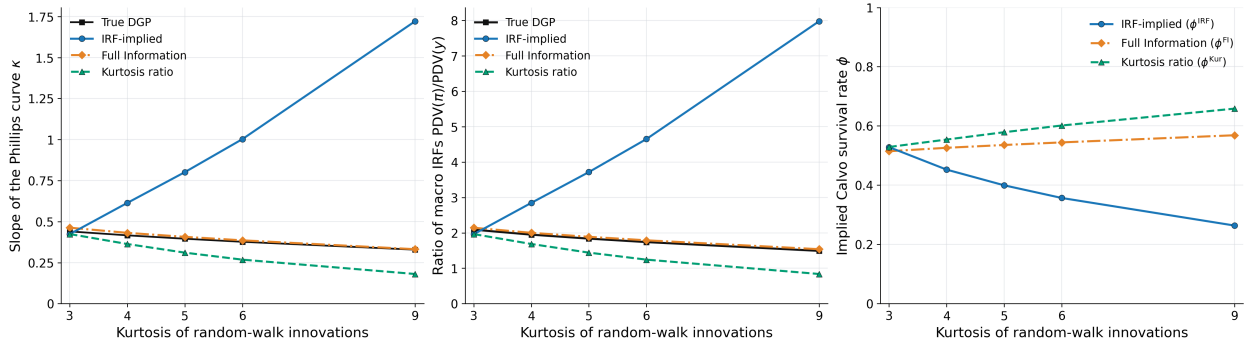
are independent and

$$\begin{aligned}\vartheta_{it} &\sim \mathcal{N}(0, \sigma^2/2), \\ z_{it} &\sim 0.9 \mathcal{N}(0, \sigma_L^2 - \sigma^2/2) + 0.1 \mathcal{N}(0, \sigma_H^2 - \sigma^2/2).\end{aligned}$$

We adjust the variance of ϑ_{it} and z_{it} so each accounts for 50% of the variance in idiosyncratic productivity and the overall shifter has a constant standard deviation 0.06 and kurtosis K . The regression uses the observed Gaussian shock ϑ_{it} .

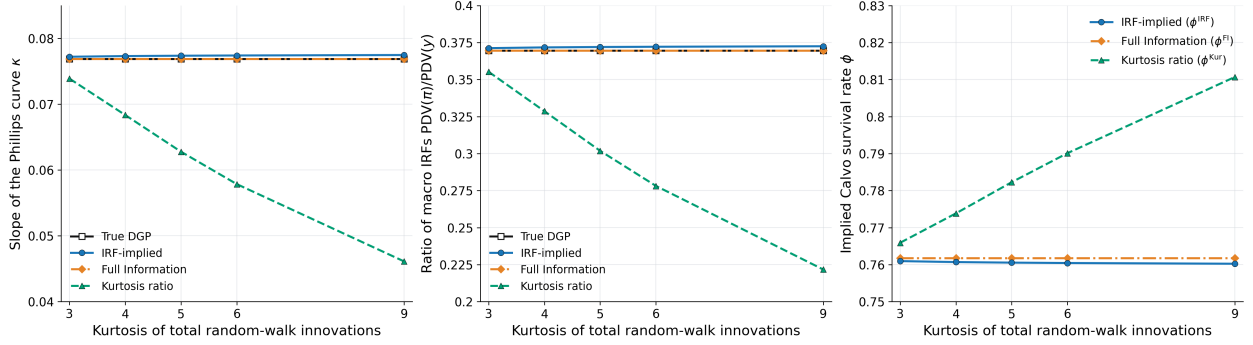
Results. The results are displayed below.

FIGURE C.5. Micro IRF-implied Statistics with a Leptokurtic Identifying Shock: Menu Cost Model



Notes: This figure shows the performance of our approximation of the generalized Phillips curve as we vary the kurtosis of the innovations to firms' desired price. The total innovation to firms' desired price is z_{it} , a mixture of two zero-mean Gaussians, drawn with probabilities 0.9 and 0.1, whose variances are set so that the total innovation has variance $\sigma^2 = 0.06^2$ and kurtosis $K \in \{3, 4, 5, 6, 9\}$ (details above). z_{it} is used by the econometrician to estimate the local projection (14). The x-axis reports K , the kurtosis of the total innovation. We report: the slope of the Phillips curve $J_{\pi, mc^r}^m[0, 0]$ (left), the Phillips multiplier (middle), and the Calvo survival rate ϕ (right). Each panel compares the true menu cost model J_{π, mc^r}^{MC} with three Calvo approximations $J_{\pi, mc^r}^C(\phi)$, which use the IRF-implied survival rate ϕ^{IRF} , the full-information survival rate ϕ^{FI} , and the kurtosis-implied survival rate ϕ^{Kur} . The right panel has no true model object. All other parameters are held at the Nakamura and Steinsson (2010) calibration.

FIGURE C.6. Micro IRF-implied Statistics with a Leptokurtic Identifying Shock: Calvo Model



Notes: This figure shows the performance of our approximation of the generalized Phillips curve as we vary the kurtosis of the innovations to firms' desired price. The total innovation to firms' desired price is z_{it} , a mixture of two zero-mean Gaussians, drawn with probabilities 0.9 and 0.1, whose variances are set so that the total innovation has variance $\sigma^2 = 0.06^2$ and kurtosis $K \in \{3, 4, 5, 6, 9\}$ (details above). z_{it} is used by the econometrician to estimate the local projection (14). The x-axis reports K , the kurtosis of the total innovation. We report: the slope of the Phillips curve $J_{\pi,mc}^m[0, 0]$ (left), the Phillips multiplier (middle), and the Calvo survival rate ϕ (right). Each panel compares the true model $J_{\pi,mc}^C$ with three Calvo approximations $J_{\pi,mc}^C(\phi)$, which use the IRF-implied survival rate ϕ^{IRF} , the full-information survival rate ϕ^{FI} , and the kurtosis-implied survival rate ϕ^{Kur} . The DGP is a Calvo model whose quarterly adjustment probability is fixed at the stationary adjustment frequency of the Gaussian Nakamura and Steinsson (2010) menu cost benchmark.

Implementation of the correction in the case of leptokurtic identifying innovations.

We implement the solution presented in Appendix B.5.4.

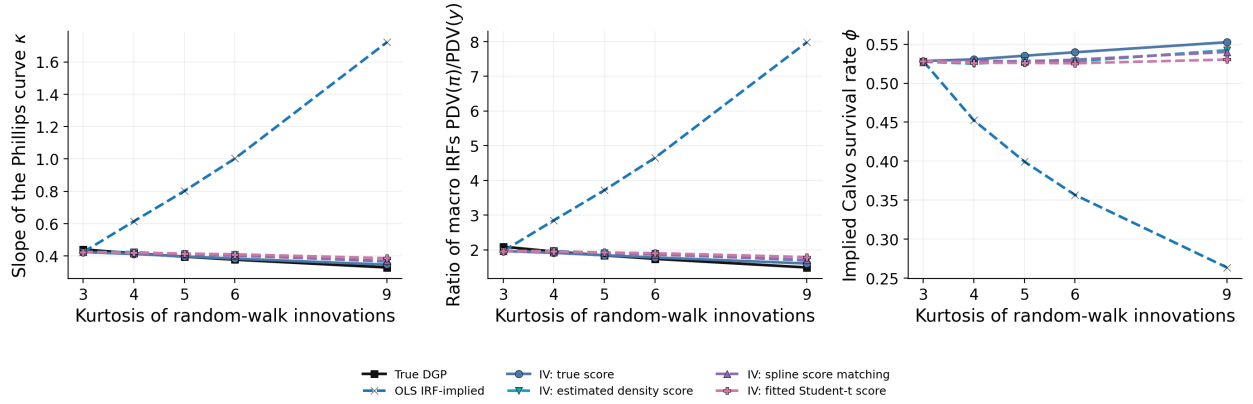
Step 1. Estimate the score function of the identifying innovations. Denote the potentially non-normal innovation in the identifying shock as ε_{it} . Let p be the true density of ε_{it} . The score function is defined as $s(\varepsilon_{it}) = -p'(\varepsilon_{it})/p(\varepsilon_{it})$. We take the point of view of an econometrician who does not know the true distribution p and estimate s in three ways: a score obtained from an estimated density, a spline score-matching estimate, and a fitted Student- t score. As a benchmark, we provide results for the case where the true score function is known.

Step 2. Estimate the IV local projection with the score as instrument. We estimate the local projection

$$(C.1) \quad p_{it+h} - p_{i,t-1} = a_h + b_h \varepsilon_{it} + u_{it}$$

where ε_{it} is instrumented by $\hat{s}(\varepsilon_{it})$. We use the estimated coefficients b_0 and b_1 to construct our statistic.

FIGURE C.7. Correcting the IRF-Implied Statistics for Leptokurtic Identifying Innovations



Notes: This figure shows the performance of our IV-IRF approximation of the generalized Phillips curve in the case where the identifying shock is not Gaussian. The total innovation to firms' desired price is z_{it} , a mixture of two zero-mean Gaussians, drawn with probabilities 0.9 and 0.1, whose variances are set so that the total innovation has variance $\sigma^2 = 0.06^2$ and kurtosis $K \in \{3, 4, 5, 6, 9\}$ (details above). z_{it} is used by the econometrician to estimate the local projection (C.1), where z_{it} is instrumented by its score $\hat{s}(z_{it})$. The score function is either the true score of the known shock distribution, or an estimate of the score function from the observable data obtained from an estimated density, a spline score-matching estimate, and a fitted Student- t score. As a benchmark, we also show the case where z_{it} is not-instrumented. The x-axis reports K , the kurtosis of the total innovation. The y-axis reports: the slope of the Phillips curve $J_{\pi, mcr}^m[0, 0]$ (left), the Phillips multiplier (middle), and the Calvo survival rate ϕ (right). All other parameters are held at the Nakamura and Steinsson (2010) calibration.